Supplementary Material for "Reciprocal Validation on DAG-Based IIoT Ledgers: Security, Liveness, and Validation-Work Bounds"

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This supplementary material extends the main manuscript. Appendix I broadens the related-work discussion; Appendices II–III specify field-level protocol details, edge cases, and deployment scope; Appendix IV presents the complete proofs and numerical details; and Appendix V reports additional evaluation settings, results, and cost accounting. Appendix VI collects the message fields, detailed invocation and verification algorithms, and architectural comparison. Appendix VII adds the four-method comparison with complete cost and coverage results. Appendices VIII and IX add the parent-count parameter grid and the observation-time sensitivity study.

Appendix I
Extended Related Work

This appendix expands the mechanism comparisons in main-manuscript Section II. The relevant distinction is what a method validates or optimizes, which participants perform that work, and which assumptions connect their actions to the reported guarantee.

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A. Ledger Structure, Dynamics, and Storage

CoDAG organizes blocks into levels and limits the number of blocks in each level. Its placement rules connect new blocks to an earlier level, and its analysis addresses confirmation and attacks within that ledger design [1]. Guo *et al.* model the degree evolution of the Tangle under batch vertex arrivals and attachment dynamics. They derive a double Pareto lognormal distribution and use an expectation–maximization procedure to estimate its parameters from ledger snapshots [2]. GpDB takes a storage perspective: it uses transaction freshness to partition the DAG held by an edge server into retained and removed portions [3]. These are specific mechanisms for ledger organization, topology modeling, and storage. MBSC instead needs an event model for accepting a computed output, together with an accounting rule for the re-execution of earlier invocations. A topology model or storage reduction does not directly supply that application-level event model.

Recent IoT ledger designs provide further examples of these distinctions. Double-DAG couples communication and transaction-set DAGs, LNLCA uses transaction sets and sampled parent nodes for batch verification, and TidyBlock organizes transactions before block formation and selects blocks for verification [4]–[6]. MBSC specifically studies deterministic function re-execution and the accounting of metered verification work.

B. DAG Ordering and Execution Correctness

Mysticeti-C uses an uncertified DAG for commit decisions, Shoal++ changes anchor commitment and scheduling, and Sailfish combines reliable broadcast with a leader vertex in each round [7]–[9]. Ladder uses a twin-chain proof-of-work structure with committee-based Byzantine fault-tolerant recovery [10]. Their DAG structures serve different commitment rules and fault models. MBSC assumes an underlying ledger service and separately defines when an application may use an invocation's output. Its eligible-approval threshold is not the quorum or commit rule of any of these consensus protocols.

ZKWASM and Jolt illustrate cryptographic verification of execution correctness [11], [12]. A computation proof and a signed parent reference serve different verification purposes: the former is verified against a specified computation relation, whereas the latter authenticates a message's signer and

contents. MBSC does not add a succinct-proof system. The analysis here concerns what can be inferred from MBSC's signed messages, as explained in main-manuscript Section III.

C. DAG-Supported Edge Computation

The FETA architecture of Mazzocca *et al.* combines the IOTA Tangle with IPFS for federated learning. A participant publishes a signed reference to a locally trained partial model; other participants retrieve models and aggregate them for the relevant training round. Authorization and model provenance form part of the protocol, while resource-allocation algorithms and incentive mechanisms are explicitly outside its scope [13]. Its shared objects are learned models, whereas MBSC treats a bounded function, its explicit inputs, and a claimed output as a unit of re-executable work. The verification objects and operations therefore differ between the two architectures.

Jiang *et al.* couple federated learning with a sharded DAG blockchain and resource scheduling in digital-twin-driven IIoT [14]. Their framework includes verification of local and global model updates; that fact alone does not identify the verification operation with MBSC's re-execution of a bounded function. Yang *et al.* combine DAG-aided trust management for edge task execution with bandwidth and computation-resource allocation [15]. These works address application-specific trust and resource allocation, whereas the present analysis relates an invocation-level work obligation to finite-time endorsement.

Other edge studies address complementary decisions. Priyadarshni *et al.* formulate task offloading as a Markov decision process, combine deep Q-learning with a long short-term memory model for the offloading decision, and record processed results through IOTA [16]. Li *et al.* jointly analyze expected job completion time and processing cost for fault tolerance in a hybrid edge–cloud system [17]. Zhang *et al.* combine a cryptographic accumulator and a stateless blockchain structure for IIoT identity management [18]. Here, "stateless" refers to the identity-management architecture; MBSC uses the term to restrict a function's dependence on mutable shared application state. Yang *et al.* couple trust-aware pricing and DAG consensus with resource trading modeled as a Stackelberg game [19]. These studies address task allocation, identity state, and resource trading under models distinct from MBSC's stateless invocation model.

D. ISC Committees and Their Security Semantics

The 2021 ISC white paper describes multiple Layer-2 chains anchored to the Layer-1 ledger. A designated committee executes a chain's state transitions and produces threshold signatures. The committee can rotate through the chain's governance mechanism; a corporate or private chain can operate without monetary fees [20]. Committee governance and fee policy are configurable aspects of that architecture.

For a committee with N validators, where N denotes the committee size, the white paper gives the usual signature threshold $q_{\mathrm{ISC}} = \lfloor 2N/3 \rfloor + 1$; q_{ISC} is the number of co-operating validators needed to form the threshold signature, and $\lfloor x \rfloor$ denotes the greatest integer not exceeding a real number x. This rule gives 15 for $N = 21$ [20]. It describes a committee's agreement on a state transition, subject to the protocol's fault and communication assumptions. It is not interchangeable with MBSC's application threshold, which counts eligible approvals of one output during a finite interval. An availability calculation that assigns every node the same online probability therefore compares a single assumption; the matched experiments of main-manuscript Section V compare end-to-end performance under common conditions.

E. Validator-Based Layer-1 Programmability

The IOTA Rebased technical proposal describes an object-based ledger with MoveVM execution, delegated proof of stake, transaction fees, and sponsored transactions [21]. Its upgrade documentation describes the transition from the earlier network and the validator-based genesis procedure [22]. These sources establish the architectural setting compared in Table XIII.

F. Verification Allocation and Observable Evidence

The question studied here is how to relate a compliant issuer's submitted workload to its checking of earlier deterministic outputs. The work rule is analyzed for issuers that follow it. A claimed parent list or a valid output does not establish that its sender expended the prescribed CPU work: observable message validity and past execution are different properties. Unsafe approvals are therefore included explicitly in the security model. The finite-time analysis asks which bounds on unsafe and honest approval arrivals permit an application threshold to satisfy the specified risk and deadline targets; workload accounting separately constrains sustainable completion. These are the mechanism-specific questions addressed in main-manuscript Section IV.

APPENDIX II
EXTENDED PROTOCOL SPECIFICATION

This appendix expands main-manuscript Section III. The workflow in main-manuscript Fig. 1 and parent selection in Algorithm S3 (Appendix VI-E) distinguish public checks from private computation. Each element specified below serves one of the design goals G1–G3 of main-manuscript Section III. Appendix VI supplies the payload fields in Tables XI–XII and the invocation and verification procedures in Algorithms S1 and S2.

A. Payload Semantics and Serialization

A Deployment stores immutable WASM code together with an execution-profile identifier. The profile specifies the runtime version, metering convention, integer semantics, memory bound, permitted imports, and input/output encoding. Its identifier must distinguish profiles with different work-count semantics. The illustrative payload type identifiers 70 and 71 are local to MBSC-capable peers, rather than assignments in the public IOTA registry.

An Invocation identifies the Deployment and exported function, supplies explicit inputs, and declares a serialized result,

digest, work count, and issuance timestamp. The complete message additionally commits the issuer identity, duplicate-free parent identifiers, admission evidence, and signature. Parent references are covered by that signature. The timestamp is checked against the underlying protocol-time rule; signing a timestamp alone does not establish freshness. The finite eligibility condition for child x and parent y is $0 \leq ts(x) - ts(y) \leq T_e$, where ts is validated protocol issuance time. For the counting certificates, these accepted timestamps index issuance events on the same time axis as the intensities and observation deadlines. Every countable reference must be included at elapsed time $ts(x) - ts(y)$ in the target's issuance window, including a message observed later. Time admission must not allow backdating to add endorsements absent from that count. A deployment must justify this interface and its mapping to physical clocks; neither a signature nor the propagation bound supplies a clock-error guarantee.

The `byte[]` payload fields use a deterministic serialization profile. For example, an input schema may contain a repeated list of argument byte strings, and an output schema may contain separate result and digest byte strings. Field order, integer representation, optional values, and handling of unknown fields must be canonicalized. Hash equality is checked over those canonical result bytes, not over an implementation-dependent in-memory object. A work count remains a declaration until the verifying node reproduces it under the committed profile.

B. Attempt State and Time

Prerequisite: immutable deployment. A Deployment becomes a stable code reference under the underlying ledger's own acceptance conditions. Its availability is a prerequisite for later execution. The analysis does not infer code availability from a message hash alone.

Own execution. At the beginning of an attempt the issuer sets a local monotonic deadline D by adding $t_{attempt}$ to the current local time. Own execution and code retrieval are bounded by D, the work limit, and the memory limit. The issuer computes the proposed result and work but does not yet fix the child's protocol issuance timestamp. This avoids treating parents generated during selection as future messages relative to an old child timestamp.

Parent selection and reciprocal work. The issuer draws without replacement from its currently available local pool, subject to d_{max}. Every drawn candidate is permanently removed from that node's local pool, including if the attempt is deferred; it is not restored for later attempts. This removal does not declare the candidate invalid or remove it from other nodes' pools. A candidate can contribute only once and only after its own result has been reproduced. The sum of reproduced counts must cover the issuer's own count and at least k_{min} distinct parents must be accepted. Newly visible candidates may enter the capped pool. The pool-wait horizon starts on entry to selection and ends no later than D. A nonempty pool does not bypass the whole-attempt deadline.

Candidate verification within parent selection. Admission checks precede bounded re-execution of the candidate's invocation. Matching output bytes, digest, and work count establish this candidate's local correctness. They do not establish whether its issuer re-executed its own parents. The routine does not recursively traverse and execute all ancestors. An objective invalidity may be cached by message identifier. A retrieval timeout or exhausted local attempt deadline is not evidence that an otherwise unknown candidate is invalid, so it does not create a permanent invalidity record.

Signing and issuance. After successful selection, message preparation, admission-work preparation, signing, and local submission remain subject to D. The issuer fixes the timestamp and checks all selected parents against it. It then canonically encodes the ordered parent references, issuer public key, and complete invocation payload, including that timestamp, as the unsigned body. The abstract helper $rateControlNonce(body, Target, D)$ must return admission evidence valid for this exact body, or fail by D. The subsequent signature covers the body and nonce, and neither is modified afterward. A concrete ledger adapter must realize this binding contract; no ledger-specific nonce implementation or measured cryptographic cost is supplied here. An expired parent makes the attempt deferred. No automatic unbounded retry is assumed. The work and delay of any explicit later retry belong to the same application request's accounting. The helper $submit(m, D)$ returns success only after the local ledger interface accepts the signed message, including its time and admission evidence, for dissemination before D. If its fixed timestamp has become inadmissible, it fails without retiming the signed body. Changing a bound field in a later attempt requires preparing both the nonce and signature again. On failure, $submit$ returns by D and the attempt is deferred. Successful local submission is not synonymous with network delivery, threshold endorsement, or ledger finality.

C. Public Checks and Private Execution

A receiving peer rejects malformed messages, invalid signatures, duplicates, unsupported profiles, missing work declarations or values outside the permitted range, insufficient parent counts, and references that fail the eligibility rule. The declared parent-work sum can also be compared with the child's declared work. This check rejects an internally inconsistent declaration, not every false declaration. If locally verified cached counts contradict a declaration, the receiver has additional evidence for rejection; such a cache is not assumed to be globally complete.

Actual re-execution detects an incorrect output, digest, or work count. A signed reference to such an incorrect result is attributable to its sender. MBSC leaves the penalty policy, such as economic slashing, reputation, or identity revocation, to an external layer that fixes its own identity, evidence-distribution, and adversarial assumptions.

The information needed by such a mechanism is nevertheless specified. Consider a signed child message x that directly references a target y whose declared output, digest, or work count differs from deterministic re-execution. The evidence record consists of x, y, the immutable Deployment code, and the versioned execution profile; y already contains the explicit

inputs. Any peer can re-execute y under that profile, observe the mismatch, and verify the signature of x over its parent set. Because a compliant verifier rejects y before referencing it, the record shows that the signer of x did not follow the compliant verification rule, whether through fault or intent. The record does not rely on declared work and remains checkable after the eligibility window closes, provided that the Deployment and both messages remain retrievable. An external penalty layer can consume such records. Whether abandoning an identity avoids a penalty depends on that layer's identity-persistence and enforcement rules.

The evidence has an asymmetric scope. An issuer that skips verification but references only correct targets produces no such record. Its references add nothing to the invalid-target endorsement event bounded in main-manuscript Theorem 1, but that issuer falls outside the compliant-work guarantee of Lemma 1. A targeted blind approval of an incorrect result, as in the attack study of Appendix V-D, does produce such a record once the target is re-executed.

Consider two executions that produce the same public message. In the first, the issuer computes the parent results; in the second, it obtains the same correct values without doing that work. The receiver's observation is identical. Consequently, a decision based only on this observation cannot certify which private execution history occurred. The compliant-work theorem is deliberately restricted to issuers following the algorithm. Unsafe endorsement traffic includes selfish skipping even when the selfish issuer's own output is correct. A minimum declared work unit cannot be used to prove a physical attacker-rate limit when that attacker may skip execution. A duplicate-free parent set limits a message to one approval of a target; it does not prevent the same identity from approving that target in several distinct messages. Local pool removal constrains compliant issuers only. A count cap against unsafe issuers requires enforcement by the admission layer.

D. Resource and Completion Controls

The local pool cap N_{max} limits storage and selection scope, d_{max} limits candidate draws, I_{max} limits metered work per execution, and $t_{attempt}$ limits local attempt duration when enforced by the execution and I/O scheduler. These limits are configured explicitly. An average observed pool size is not a substitute for a hard cap, and a deterministic metered-work limit is not by itself a bound on operating-system scheduling or network retrieval latency. The positive lower bound I_{min} excludes zero-work invocations and supplies a minimum metered-work contribution for each accepted parent.

Lemma 1 bounds own plus accepted-parent work by $\max\{3, k_{min}+1\}I_{max}$. At most rI_{max} is added for r rejected candidates whose execution is attempted. Structural checking, network traffic, virtual-machine setup, and queueing are accounted for separately. The same costs must be included for both methods in a performance comparison. Work consumed by a failed or deferred request is not discarded from the total.

An attempt deadline guarantees a finite local outcome only if all helper calls respect it. It does not guarantee that the outcome is SUCCESS. Appendix IV supplies a separate sufficient condition for success under a fixed eligible candidate set. Applying that condition to a changing pool requires justification of the candidate-count and validity bounds throughout the attempt, including final timestamp rechecking.

E. Execution Profile and Measurement Boundary

The execution profile must exclude mutable ledger state and nondeterministic host dependencies from the target functions. The evaluation uses integer-only WASM modules with explicit inputs and no filesystem, clock, or network imports. The recorded engine version and its fuel counter define the tested work unit. Fuel is an engine-defined metering quantity, not a measured CPU-instruction count or a proof of physical effort. Cross-engine equivalence requires an additional canonical instrumentation specification.

Each invocation is executed using the same immutable module and profile as its verifier. Output agreement and repeated fuel-count agreement are checked, as are execution-budget violations. Runtime, memory, compilation, and instance setup are different quantities and are reported separately where measured. The measurement procedure characterizes execution time on the specified platform.

F. Ledger Integration Boundary

The validation mechanism is evaluated in a node-level model of an MBSC-capable DAG. In a different ledger architecture, signed parent endorsements could be represented by explicit application receipt objects, but object admission, ordering, fees, and finality would need a separate integration analysis. A receipt containing a signature and claimed work carries the same evidence semantics as a signed parent reference, so the enforcement analysis above applies unchanged.

For a deployment whose application requires irrevocable results, the target and the counted references must satisfy a precisely defined stable-ledger-view condition. Pending references are tentative and must be withdrawn if that view changes. This study separates the application threshold from that additional ledger condition, which the chosen consensus protocol provides.

APPENDIX III
PROTOCOL EDGE CASES AND DEPLOYMENT CONDITIONS

A. Concurrent Selection and Approval Supply

Several issuers may independently select the same parent before learning of one another's messages. Each compliant issuer still checks its own coverage condition. The resulting direct references are distinct endorsement messages, not necessarily endorsements from independent identities. Local removal of a selected candidate does not imply removal from all other peers' pools.

Aggregate work coverage does not imply that every target receives c endorsements. If E eligible direct-reference edges have been produced in an observation interval and count toward the targets under consideration, at most $\lfloor E/c \rfloor$ of those targets can each obtain c edges from that set. Here E

counts edges, rather than messages, because one message can reference several parents. For a long homogeneous stream with a fixed k parents per message, the supply is approximately k edges per new target. Choosing $c > k$ therefore creates an approval-supply shortfall for near-universal completion, even with correct outputs and no communication failures. Initial seed targets or a finite observation boundary do not remove this long-run accounting constraint. Heterogeneous work may increase MBSC's parent count, but the resulting supply must be measured rather than assumed.

B. Congestion and Eligibility

An offered request stream passes through own execution, parent selection, queueing, and possible deferral before it emits endorsements. Even if request arrivals are Poisson, successful endorsement opportunities need not be a Poisson process with the same rate. The liveness theorem assumes its stated conditional opportunity and visibility bounds.

A finite pool limits resource use but can also evict a valid target before it collects sufficient endorsements. Final timestamp checking may reject parents that were eligible when initially selected. Thus visibility, selection, expiration, and capacity constraints jointly influence completion.

C. Stateless Task Scope

Identical immutable code and declared inputs must give identical output bytes and work counts under the same execution profile. Suitable illustrative tasks include bounded integer sensor rules, fixed-table access decisions, and bounded integer computation. These examples motivate the workload construction. Applications requiring mutable shared balances, long-lived auctions, or cross-request state transitions need additional state and ordering machinery.

D. Hardware, Connectivity, and Finality

Applying the execution profile to another deployment platform requires characterizing its processor, memory, scheduling, and communications. Lightweight sensors that cannot execute the profile can delegate computation to a capable device under additional trust or verification assumptions.

The analytical opportunity model applies only where the target remains eligible and sufficiently visible. A partition or loss of future issuers can remove that premise. Reconnection alone does not guarantee completion before a fixed deadline. Furthermore, reaching the application threshold does not make a tentative ledger record irrevocable. An application that requires both properties must budget and justify each stage separately.

E. Bootstrap and Finite Observation

A system starting with an empty candidate pool needs an explicit initialization policy. Prevalidated seed invocations provide initial candidates but introduce a stated initial condition and initialization cost. They must be the same in paired comparisons and cannot silently supply extra approvals to only one method. The security claims for new targets do not derive the validity of trusted initialization data.

Performance measurement follows a predefined cohort of submitted requests. Background arrivals continue under the same workload policy until every cohort request has reached its deadline. Requests that remain unconfirmed, defer, or expire count toward unsuccessful completion rather than disappearing from the denominator. A conditional delay average among successful requests is reported alongside the full-cohort failure fraction. Retries, if permitted, require an explicit policy and add their resource use and elapsed time to the original request.

APPENDIX IV
PROOFS AND NUMERICAL DETAILS

The effective-approval definition, process assumptions, and principal theorem statements are given in main-manuscript Section IV. This appendix also states Corollary 2 and Proposition 2 alongside their proofs. Throughout this appendix, "confirmation" means attaining the application-level semantic endorsement threshold from distinct signed messages. It does not mean DAG finality or independently verified execution by distinct identities. The subscript m includes malicious, selfish, and other unsafe approval sources, including additional identities under their control. We use $\mathbb{E}$ for expectation and $\mathbf{1}\{B\}$ for the indicator of event B, equal to one when B occurs and zero otherwise. The notation $X \sim \mathrm{Poisson}(\eta)$ means that the nonnegative integer count X has a Poisson distribution with mean η.

A. Invalid-Output Endorsement Bound

This subsection expands main-manuscript Sections IV-A and IV-B, including counting-process terminology and interpretation of the security tail. A simple process has one arrival at a time; non-explosion means finitely many arrivals in every finite interval. Conditional intensity is the instantaneous arrival rate given the preceding history. Predictability restricts it to information available just before that time. An almost-sure bound excludes only a probability-zero event, rather than merely bounding a long-run average.

The mean $\eta_m = \lambda_m^{max} q_m T_e$ combines the maximum unsafe message rate, its maximum targeting probability, and the eligibility duration. For the dominating Poisson count, $e^{-\eta_m}\eta_m^j/j!$ is the probability of exactly j approvals. Summing over $j < c$ gives its lower tail; the complement is the upper tail in main-manuscript Eq. (2).

1) Proof of Theorem 1: Fix one incorrect target and set time zero to its accepted protocol issuance time. Honest issuers re-execute it and reject its incorrect output. Therefore, all effective approvals it acquires must come from unsafe sources. Let $Y(t)$ count those approvals issued by elapsed time t, with $Y(0) = 0$. Each unsafe message counts at most once, regardless of repeated references inside it. An arrival enters Y only when its mark is one. Conditional on the preceding history, this probability is at most q_m, so the approval intensity is at most $q_m\lambda_m(t) \le \lambda_m^{max} q_m = b$. This bound uses the conditional mark assumption and permits history-dependent targeting.

The following calculation proves the tail comparison directly within the conditional-intensity framework [23]. Write $T = T_e$ for this proof and let $\mu_m(t) \le b$ be the predictable intensity of Y. For time $0 \le t \le T$ and integer current count $n \ge 0$, define

$$g(t,n) = \Pr\{n + Z_t \ge c\}, \qquad Z_t \sim \text{Poisson}(b(T-t)). \quad (1)$$

Thus $g(t,n)$ is the chance of reaching c if an ideal Poisson process of rate b supplies the remaining approvals. It is one when $n \ge c$. For $n < c$, its explicit value is

$$g(t,n) = 1 - e^{-b(T-t)} \sum_{j=0}^{c-n-1} \frac{[b(T-t)]^j}{j!}. \quad (2)$$

Let $Dg(t,n) = g(t,n+1) - g(t,n) \ge 0$ be the effect of one additional approval. Differentiating the finite sum yields $\partial_t g(t,n) = -bDg(t,n)$; the identity also holds for $n \ge c$, where both terms are zero.

Between successive arrivals, $g(t, Y(t))$ changes according to its time derivative; at an arrival its increment is $Dg(t, Y(t-))$, where $Y(t-)$ is the count immediately before time t. The defining compensator property of the predictable intensity states that the expected sum of such bounded jump increments equals the expected integral of their size times $\mu_m(t)$. Taking expectations therefore gives

$$\begin{aligned}
\mathbb{E}g(T, Y(T)) &= g(0,0) + \mathbb{E}\int_0^T \big[\partial_t g(t, Y(t-)) \\
&\qquad + \mu_m(t) Dg(t, Y(t-))\big]\, dt \\
&= g(0,0) + \mathbb{E}\int_0^T [\mu_m(t) - b] Dg(t, Y(t-))\, dt \\
&\le g(0,0). \qquad (3)
\end{aligned}$$

The integrands are bounded on this finite interval, so the expectations are well defined. At T, no future Poisson arrivals remain, and $g(T, Y(T)) = \mathbf{1}\{Y(T) \ge c\}$. At time zero, $g(0,0)$ is the upper tail of a Poisson count with mean bT. It follows that

$$\begin{aligned}
P_{invalid} &\le \Pr\{Y(T_e) \ge c\} \\
&\le 1 - e^{-\eta_m} \sum_{j=0}^{c-1} \frac{\eta_m^j}{j!}, \qquad \eta_m = bT_e. \quad (4)
\end{aligned}$$

For $b = 0$, the compensator is zero and $Y(T_e) = 0$ almost surely, giving the same conclusion. This proves main-manuscript Eq. (2) without assuming that Y itself is Poisson. Messages received later cannot increase the total beyond the count issued inside the allowed window. The argument bounds the endorsement event even if a local verifier already knows that the target is incorrect and rejects it; it does not require that such a verifier accept the target. ■

Scope of the guarantee. The argument uses only the predictable bound $\mu_m(t) \le b$ and the conditional mark bound, so Theorem 1 covers coordinated and adaptive unsafe issuers, including those that choose targets from the observed history; no independence or honest-majority assumption enters.

2) Proof of Corollary 1: Fix a finite threshold c, the window $T_e > 0$, and a risk target $\varepsilon \in (0,1)$. Without a finite intensity bound λ_m^{max}, an adversary may issue effective approvals of one fixed incorrect target as a Poisson process of rate r. Its count A_r in the window has mean rT_e. For every fixed $j \in \{0, \dots, c-1\}$, $e^{-rT_e}(rT_e)^j/j! \to 0$ as $r \to \infty$. Consequently,

$$\Pr\{A_r \ge c\} = 1 - e^{-rT_e} \sum_{j=0}^{c-1} \frac{(rT_e)^j}{j!} \longrightarrow 1. \quad (5)$$

Honest issuers reject the incorrect target, so $P_{invalid} = \Pr\{A_r \ge c\}$ for this adversary, and some finite r gives $P_{invalid} > \varepsilon$. A threshold certificate that holds against all adversaries therefore requires a bound on the aggregate unsafe exposure. The constructions below show how an admission layer supplies it, including hard quotas, for which a deterministic count replaces the intensity condition. ■

3) Supplying the exposure bound: The exposure bound of main-manuscript Theorem 1 cannot be derived from MBSC's own records. By the indistinguishability argument of Appendix II-C, declared work and signatures do not reveal how an issuer behaves. The bound must therefore be supplied by the deployment's admission layer. MBSC's compliant re-execution ensures that honest issuers reject an incorrect target; admission limits the remaining unsafe approvals. Theorem 1 converts that limit into a threshold-risk bound. Fixed-k can use the same argument under the same checks and exposure assumptions; the certificate is therefore a property of the checks and the window, and work coverage adds the compliant-work guarantee of main-manuscript Lemma 1. Committee and validator designs meet the same requirement in another form: their guarantees presuppose a bounded number of faulty members, which membership or stake rules supply (main-manuscript Section II). The constructions below specify admission layers that supply this bound.

From risk target to budget. For a chosen threshold c and risk target ε, the security certificate holds whenever $\lambda_m^{max} q_m T_e \le \eta_m^{max}(c;\varepsilon)$, the budget of Eq. (9). An unsafe issuer chooses its own parents, so $q_m = 1$ is the conservative value, and the budget caps the total unsafe approval rate of any one target. For $c = 4$ and $\varepsilon = 10^{-3}$, Appendix IV-D gives $\eta_m^{max} = 0.428552$. With $T_e = 300\,\text{ms}$, the admission layer must keep that rate at or below $1.4285\,\text{s}^{-1}$.

Enrolled identities with a hard issuance cap. Suppose that every admissible message carries an enrolled identity, that at most f enrolled identities are unsafe, including any additional identities controlled by an adversary, and that each identity has at most m admissible messages with validated timestamps in any interval of length T_e. Honest issuers reject an incorrect target, and a message approves a given target at most once (main-manuscript Definition 1). The target can therefore obtain at most fm effective approvals in its window, so $P_{invalid} = 0$ for every $c \ge fm+1$. The same count holds under the weaker rule that each identity contributes at most m approvals of any one target within its window; honest MBSC issuers satisfy it with $m = 1$, because Algorithm S3 removes each drawn candidate permanently from the drawing node's pool. An unsafe issuer can ignore this local removal rule; applying $m =$

1 to it requires an enforced admission cap. This deterministic count replaces the conditional-intensity condition and needs no Poisson comparison. The cap must hold for the admitted message set, not only in one receiver's view: an identity must not gain additional admissible messages by sending different messages to different peers. A per-identity sequence number or time slot that the underlying ledger orders and checks before a message becomes countable is one realization.

Example. A per-identity token bucket with rate ρ and burst β admits at most $\lfloor \beta + \rho T_e \rfloor$ messages in any interval of length T_e. With $\rho = 2\,\mathrm{s}^{-1}$, $\beta = 1$, and $T_e = 300\,\mathrm{ms}$, this gives $m = 1$, so $c \geq f+1$ makes an incorrect target unable to reach the threshold. This cap also shapes honest traffic and does not establish the opportunity process required by Theorem 2. An average honest issuance rate cannot supply that process assumption. Similarly, a token-bucket rate is not a predictable conditional-intensity bound, so the deterministic count is the appropriate security argument for this example.

Randomized body-bound admission. Algorithm S1 fixes the canonical body, including the ordered parent references, before $rateControlNonce$ prepares admission evidence for that exact body (Appendix II-B). Suppose that valid evidence requires a randomized search in which each attempt succeeds with probability p, independently of the history, and that the aggregate attempt process of all unsafe sources has predictable conditional intensity at most H_m. The admitted unsafe messages then have conditional intensity at most $H_m p$, which is the condition of main-manuscript Theorem 1 with $\lambda_m^{max} = H_m p$. Because the body contains the target's identifier, no approval of the target can be prepared before the target exists. The justification thus reduces to a bound on the unsafe share of attempt capacity.

Concentration. In both constructions, unsafe approvals can concentrate on one target ($q_m = 1$), whereas an honest opportunity approves a given valid target only with probability q_h, which is $4/21$ in Appendix IV-F. Equal per-identity allowances therefore favor the adversary. If honest issuers supply the opportunity rate $350\,\mathrm{s}^{-1}$ of Appendix IV-F through the same puzzle, the budget above permits an unsafe share of attempt capacity of at most about 0.41% for $c = 4$ and 0.70% for $c = 5$; the example value $\lambda_m^{max} = 2/3\,\mathrm{s}^{-1}$ corresponds to about 0.19%. The attack study of Appendix V-D runs without these budgets. Its input requests are assigned to issuers independently of their behavior, giving each of the four unsafe nodes an offered rate of about $0.70 C_{ref}/21 \approx 730$ requests/s. Its invalid-target endorsements therefore show the uncapped exposure that an admission budget removes.

B. Finite-Time Semantic Liveness

This subsection expands main-manuscript Sections IV-A and IV-C. The mean $\eta_h(T)$ belongs to an auxiliary honest approval count supported by the lower rate guarantee. Its probability of fewer than c approvals bounds the deadline-miss probability. Both η_m and $\eta_h(T)$ are dimensionless, each combining a message rate, probability, and time. A larger defensible target-selection probability improves the honest lower-rate certificate, as does a higher countable-opportunity rate.

1) Proof of Theorem 2: Consider the assumed Poisson process of locally countable honest message opportunities. Let p_i be the conditional probability that its ith opportunity approves the valid target, given the history just before that opportunity. Until the target first reaches c or the deadline expires, $p_i \geq q_h$. On an enlarged probability space, the approval indicator can be represented as $\mathbf{1}\{U_i \leq p_i\}$, where U_i is a fresh uniform random variable on $[0,1]$, independent of the opportunity process and earlier randomness. Retain the same opportunity in an auxiliary process whenever $U_i \leq q_h$. Before the target reaches c, each auxiliary approval is therefore an actual honest approval.

Each opportunity is retained independently in the auxiliary process with fixed probability q_h. Conditional on n opportunities, the retained count is binomial with parameters n and q_h. Averaging over the Poisson opportunity count gives a Poisson count with its mean multiplied by q_h. This independent-retention operation is called thinning. Its count $A_h(T)$ satisfies

$$A_h(T) \sim \mathrm{Poisson}(\eta_h(T)), \qquad \eta_h(T) = \lambda_h^{loc} q_h T. \tag{6}$$

After the actual threshold is reached, the auxiliary process may continue using fresh marks even if the target leaves the local pool. This continuation is only a device in the proof; it asserts no post-confirmation eligibility. If $T_{conf} > T$, the actual threshold has not been reached anywhere in the interval, so every auxiliary approval up to T is included in the honest count and $A_h(T) < c$. Thus

$$\begin{aligned} \Pr\{T_{conf} > T\} &\leq \Pr\{A_h(T) < c\} \\ &= e^{-\eta_h(T)} \sum_{j=0}^{c-1} \frac{\eta_h(T)^j}{j!}. \end{aligned} \tag{7}$$

Any additional unsafe approvals can only cause the endorsement threshold to be reached earlier for this valid target. The proof requires a uniform lower bound on countable honest opportunities until the threshold is reached; an observed request rate or average selection frequency does not establish that premise. ■

2) Pool-based selection probability: A geometric expression for q_h is available under stronger pool conditions. Suppose, conditional on the history, that the valid target belongs to an opportunity's eligible pool with probability at least $v_{min} \in (0,1]$, that the pool contains at most N_{max} entries, and that at least $k_{min} \geq 1$ distinct candidates are sampled uniformly without replacement. Suppose also that a sampled valid target is retained, verified, and transmitted as a countable approval. With enough candidates to perform these draws,

$$q_h = \min\left\{1, \frac{v_{min} k_{min}}{N_{max}}\right\} \tag{8}$$

is a conservative per-opportunity lower bound, valid while the eligible population supports these draws until the threshold is reached.

Conditional on a valid target being present in a pool of n candidates, with $k_{min} \leq n \leq N_{max}$, the probability that it appears among the first k_{min} draws of a uniform random permutation is k_{min}/n. Indeed, every one of the n positions is equally likely for this target, and exactly k_{min} positions lie in

those draws. Multiplying by the conditional pool-membership floor v_{min} gives at least $v_{min}k_{min}/N_{max}$. Additional distinct draws cannot reduce inclusion. This yields Eq. (8) if a sampled valid target is verified, retained, and delivered as a countable approval. Failed completion, expiry before the final timestamp recheck, or conditioning on a subset of successful attempts can alter the selection law. The geometric calculation does not prove those additional properties.

C. Joint Security–Liveness Certificate

This subsection expands main-manuscript Section IV-D. The endpoint c_{sec} is the smallest threshold passing the security certificate, and c_{live} is the largest passing the liveness certificate. Their definitions describe analytical certificates, not all deployable systems.

1) Proof of Proposition 1: For fixed η_m, $\mathcal{U}(c;\eta_m)$ is nonincreasing in the positive integer c: increasing the threshold removes possible counts from the upper-tail event. Therefore, the security certificate holds precisely for $c \geq c_{sec}$. For fixed $\eta_h(T_{max})$, $\mathcal{L}(c;\eta_h(T_{max}))$ is nondecreasing in c, so its certificate holds precisely for $c \leq c_{live}$, or for no positive integer if $c_{live} = 0$. The two certified integer sets intersect exactly when $c_{sec} \leq c_{live}$, yielding the interval in main-manuscript Eq. (7). These are statements about the specified bounds. Failure of a bound to certify a target is not a proof that the actual target probability is too large. ■

2) Rate and time budgets: For a fixed c, define the corresponding exposure budgets

$$\eta_m^{max}(c;\varepsilon) = \sup\{\eta \geq 0 : \mathcal{U}(c;\eta) \leq \varepsilon\}, \tag{9}$$
$$\eta_h^{min}(c;\delta) = \inf\{\eta \geq 0 : \mathcal{L}(c;\eta) \leq \delta\}. \tag{10}$$

These are the largest permitted unsafe and smallest sufficient honest count means. The proof below derives their monotonicity and boundary properties. Write $a = \lambda_h^{loc} q_h > 0$ and $b = \lambda_m^{max} q_m \geq 0$ for the two effective rates. They describe approved-message opportunities and are not interchangeable with the input request rate.

Corollary 2 (Rate and Time Budgets): *At a fixed threshold c, the two certificates hold exactly when*

$$bT_e \leq \eta_m^{max}(c;\varepsilon), \qquad aT_{max} \geq \eta_h^{min}(c;\delta). \tag{11}$$

For $b > 0$, they imply the necessary condition

$$\frac{a}{b} \geq \frac{\eta_h^{min}(c;\delta)}{\eta_m^{max}(c;\varepsilon)} \frac{T_e}{T_{max}}. \tag{12}$$

The ratio condition alone does not certify fixed time windows. Both inequalities in Eq. (11) *must still be checked. For $b = 0$, the unsafe exposure bT_e is zero, so the security certificate holds for every positive c and only the honest budget needs testing.*

Proof: The derivation follows below.

The two budgets are separately tunable: shortening T_e reduces unsafe exposure, and lengthening an admissible T_{max} adds honest opportunities, even at fixed a/b. Appendix IV-D gives an analytical example in which changing the common window from 10 to 50 s changes an empty certificate interval to $[5, 5]$ at unchanged rates.

TABLE I

EXACT-TAIL BUDGETS FOR $\varepsilon = 10^{-3}$ AND $\delta = 0.05$. THE UNSAFE BUDGET IS ROUNDED DOWN, THE HONEST BUDGET UP, AND THEIR PRINTED RATIO UPWARD. THESE ARE ANALYTICAL CALCULATIONS.

c	η_m^{max}	η_h^{min}	$\eta_h^{min}/\eta_m^{max}$	$\mathcal{U}(c;0.6)$
2	0.045	4.744	105.43	0.121901
3	0.190	6.296	33.14	0.023115
4	0.428	7.754	18.12	0.003358
5	0.739	9.154	12.39	0.000394

c is the endorsement threshold; η_m^{max} and η_h^{min} are the maximum unsafe and minimum honest approval-count means satisfying the stated risk and deadline-miss targets. $\mathcal{U}(c;0.6)$ is the probability that a Poisson count of mean 0.6 reaches c.

3) Proof of Corollary 2: For fixed $c \geq 1$, differentiating the Poisson lower tail with respect to $\eta > 0$ gives

$$\frac{\partial \mathcal{L}(c;\eta)}{\partial \eta} = -e^{-\eta} \frac{\eta^{c-1}}{(c-1)!} < 0. \tag{13}$$

The derivative of $\mathcal{U} = 1 - \mathcal{L}$ has the opposite sign. Continuity and the limits at zero and infinity imply unique positive budget boundaries for $\varepsilon, \delta \in (0, 1)$. Using their definitions in Eqs. (9)–(10),

$$\mathcal{U}(c;bT_e) \leq \varepsilon \iff bT_e \leq \eta_m^{max}(c;\varepsilon), \tag{14}$$
$$\mathcal{L}(c;aT_{max}) \leq \delta \iff aT_{max} \geq \eta_h^{min}(c;\delta). \tag{15}$$

Here $a = \lambda_h^{loc} q_h > 0$ and $b = \lambda_m^{max} q_m \geq 0$. For $b > 0$, dividing these separate rate bounds gives

$$\frac{a}{b} \geq \frac{\eta_h^{min}(c;\delta)/T_{max}}{\eta_m^{max}(c;\varepsilon)/T_e} = \frac{\eta_h^{min}(c;\delta)}{\eta_m^{max}(c;\varepsilon)} \frac{T_e}{T_{max}}. \tag{16}$$

For fixed a, b, c, the full conditions, including the window relation, can also be written as

$$\frac{\eta_h^{min}(c;\delta)}{a} \leq T_{max} \leq T_e \leq \frac{\eta_m^{max}(c;\varepsilon)}{b}. \tag{17}$$

The ratio inequality expresses an ordering between the outer limits but does not put specified time windows inside them. It is therefore necessary, not sufficient, for fixed windows. Dropping $T_e/T_{max} \geq 1$ weakens the necessary condition further. When $b = 0$, the effective unsafe approval count is zero almost surely under the intensity and marking assumptions, so the security certificate holds for every positive c and only the honest budget remains. ■

D. Numerical Inversion of the Certification Bounds

This subsection expands the numerical interpretation in main-manuscript Section IV-D, including the complete common-window example and tail-budget calculations.

For each c, the budget boundaries solve $\mathcal{U}(c;\eta) = \varepsilon$ and $\mathcal{L}(c;\eta) = \delta$, respectively. Bracket a root by starting with $[0, 1]$ and doubling its upper endpoint until the target is crossed. Bisection then reduces the bracket width below 10^{-12}. For $c = 2, 3, 4, 5$, the respective roots, rounded to six decimal places, are

$$\eta_m^{max} = (0.045402,\ 0.190533,\ 0.428552,\ 0.739372), \tag{18}$$
$$\eta_h^{min} = (4.743865,\ 6.295794,\ 7.753657,\ 9.153519). \tag{19}$$

The ratios in Table I are computed from its conservatively rounded budgets. Thus 12.39 at $c = 5$ is a rounded budget ratio for these specific error targets. For specified windows, the full necessary rate comparison still includes T_e/T_{max}; both absolute budgets must also hold. This printed number is not a universal deployment constant. Direct evaluation at $\eta_m = 0.6$ gives $\mathcal{U}(c; 0.6) = (0.121901, 0.023115, 0.003358, 0.000394)$, so that $c_{sec}(10^{-3}) = 5$ for this illustrative exposure.

1) Effect of a common time window: Keeping T_{max} fixed while increasing T_e admits more unsafe exposure. Keeping T_e fixed while increasing an admissible T_{max} provides more honest opportunities. Changing both windows can alter the absolute budgets even when the effective rates and their ratio remain fixed.

Use the hypothetical effective rates $a = 0.2\,\mathrm{s}^{-1}$ and $b = 0.01\,\mathrm{s}^{-1}$ with the same error targets. For $T_e = T_{max} = 10\,\mathrm{s}$, the means are $\eta_m = 0.1$ and $\eta_h = 2$. Security requires at least $c = 3$, since $\mathcal{U}(2; 0.1) = 0.00467884 > 0.001$ and $\mathcal{U}(3; 0.1) = 0.000154653 < 0.001$. Even $c = 1$ fails the liveness certificate because $\mathcal{L}(1; 2) = e^{-2} = 0.135335 > 0.05$. Consequently, $(c_{sec}, c_{live}) = (3, 0)$.

Change only the common window to 50 s. Now $\eta_m = 0.5$ and $\eta_h = 10$. Direct substitution gives

$$\mathcal{U}(4; 0.5) = 0.00175162 > 0.001, \tag{20}$$

$$\mathcal{U}(5; 0.5) = 0.000172116 < 0.001, \tag{21}$$

$$\mathcal{L}(5; 10) = 0.0292527 < 0.05, \tag{22}$$

$$\mathcal{L}(6; 10) = 0.0670860 > 0.05. \tag{23}$$

The certified interval is therefore exactly $[5, 5]$. The rate ratio stayed at 20, but longer windows supplied enough honest opportunities while the unsafe exposure remained within the budget for $c = 5$. This example establishes that a change in the common time scale can repair an empty certificate interval. It is a direct evaluation of the probability bounds, not a simulated or hardware-measured result.

E. Reciprocal Validation Work

This subsection expands the stopping-case argument in main-manuscript Section IV-E.

1) Proof of Lemma 1: For a completed compliant invocation, the stopping condition gives $V_i \geq W_i$. Summing over a finite set $\mathcal{C}$ gives $\sum_{i\in\mathcal{C}} V_i \geq \sum_{i\in\mathcal{C}} W_i$. To bound the total work, consider which stopping condition is reached last.

If work coverage is reached on the final parent, the accumulated parent work before that draw was strictly less than W_i. Let $W_{last} \leq I_{max}$ be the final parent's work. Then

$$V_i < W_i + W_{last} \leq W_i + I_{max}, \tag{24}$$

and $W_i + V_i < 2W_i + I_{max} \leq 3I_{max}$.

If work coverage had already been reached but the minimum-parent condition had not, selection stops at exactly k_{min} accepted parents. Thus $V_i \leq k_{min} I_{max}$ and $W_i + V_i \leq (k_{min} + 1)I_{max}$. Taking the larger bound proves main-manuscript Eq. (8). For $k_{min} \leq 2$, the resulting bound is $3I_{max}$.

These bounds concern verified metered work for completed compliant attempts. Re-executing r_i rejected candidates adds at most $r_i I_{max}$ work units; fixed runtime and communication costs are additional. ■

F. Selection and Request Completion

This subsection expands main-manuscript Section IV-F, including the selection count, explicit timing requirements, full numerical parameter construction, and composition of completion stages.

1) Fixed-pool selection certificate: **Proposition 2 (A Sufficient Selection Certificate):** *Fix one compliant attempt with own work W_i. Suppose its initial selection pool consists of $N \in \mathbb{N}^+$ distinct retrievable candidates. There are no additions, evictions, or expiry during the attempt; only already drawn candidates are removed. A fixed subset of G candidates, where $G \in \{0, \ldots, N\}$, passes all parent checks and remains eligible at every possible final issuance time within the attempt deadline. Each of these guaranteed-valid parents has verified work at least $w_{min} > 0$. Sampling is uniform without replacement with maximum draw count d_{max}. Let $d = \min\{d_{max}, N\}$ be the available draw budget and*

$$K_i = \max\{k_{min}, \lceil W_i/w_{min} \rceil\}. \tag{25}$$

Assume all d draws and their checks, or an earlier successful stop, can finish within the attempt's deadline. Then the probability p_{sel} of successful parent selection satisfies

$$p_{sel} \geq S(N, G, d, K_i) = \sum_{j=K_i}^{\min\{d,G\}} \frac{\binom{G}{j}\binom{N-G}{d-j}}{\binom{N}{d}}. \tag{26}$$

With an all-valid pool and $d \geq K_i$, the guarantee is $S = 1$. Parent work greater than w_{min} can permit success with fewer than K_i parents. Equation (26) requires a fixed candidate set during the attempt. Bounds on a changing pool's size, number of valid candidates, and operation times alone do not preserve the uniform sampling-without-replacement argument. We therefore do not apply this certificate to the dynamic pools used in the evaluation.

2) Proof of Proposition 2: The ceiling $\lceil W_i/w_{min} \rceil$ in Eq. (25) rounds up the number of minimum-work parents sufficient to cover W_i. Taking its maximum with k_{min} enforces the work and parent-count requirements together.

Fix the pool and the guaranteed-valid subset described in the proposition. Its membership is fixed before the random draws; validity throughout the allowed attempt horizon must not be defined by a draw-dependent completion time. This requirement ensures that the success labels used in the following count are not selected after seeing the sampled order.

A uniform sequence of draws without replacement can be represented by a uniform random permutation of the N candidates. Let X_d be the number of the G guaranteed-valid candidates among its first d positions. There are $\binom{N}{d}$ equally likely unordered samples of size d. Here $\binom{n}{r}$ counts the size-r subsets of an n-element set, with $\binom{n}{r} = 0$ when $r < 0$ or $r > n$, and a sum with its lower endpoint above its upper endpoint is zero. Exactly $\binom{G}{j}\binom{N-G}{d-j}$ contain j guaranteed-valid candidates, so

$$\Pr\{X_d = j\} = \frac{\binom{G}{j}\binom{N-G}{d-j}}{\binom{N}{d}}. \tag{27}$$

Each of these candidates contributes at least w_{min} verified work. If $X_d \geq K_i$, there are at least k_{min} valid parents and their work is at least $K_i w_{min} \geq W_i$. The compliant stopping rule must therefore succeed by its dth draw or earlier. Summing over $j \geq K_i$ proves the selection lower bound. If the actual procedure stops early, the unused suffix of the permutation is only a mathematical extension and consumes no work. Other candidates or greater parent work may allow success even when $X_d < K_i$, which is why the bound is sufficient rather than an equality. The assumption that all required operations fit within the attempt deadline prevents timeout from invalidating this counting argument. ■

3) Explicit timing and resource conditions: Let T_{pre} be the budget from request creation to successful issuance, and let δ_{pre} bound the probability of failing to issue within that budget.

For a wall-clock certificate, let τ_{own} bound initial execution and payload preparation, d_{step} bound the duration of a candidate draw including retrieval, admission checks and bounded semantic re-execution, and τ_{submit} bound final body and admission-evidence preparation, timestamp checks, signing, and successful local submission. These quantities bound the allowed paths, including any waiting. Under the fixed-pool premises of Proposition 2, if

$$\tau_{own} + dd_{step} + \tau_{submit} \leq T_{pre}, \tag{28}$$

all guaranteed-valid candidates remain eligible until the final timestamp, and submission introduces no further failure, then $1 - S(N, G, d, K_i)$ bounds failure to issue within T_{pre}. The fixed-pool certificate does not require waiting for newly issued candidates. An issuance bound for a changing pool requires a separate selection-success argument together with bounds on waiting, eviction, expiry, retrieval failure, and restarts. No such certificate is asserted for the evaluated dynamic implementation.

For an explicit conditional fixed-pool design point, take $N = G = 21$, $d_{max} = 16$, $k_{min} = 4$, and $W_i \leq 16 w_{min}$. Then $d = 16$, $K_i \leq 16$, every sampled candidate is valid, and $S = 1$. Assume $\tau_{own} \leq 2\,\mathrm{ms}$, $d_{step} \leq 2\,\mathrm{ms}$, and $\tau_{submit} \leq 2\,\mathrm{ms}$, with any waiting included in the corresponding operation bound. Equation (28) then gives $2 + 16(2) + 2 = 36\,\mathrm{ms} \leq T_{pre} = 50\,\mathrm{ms}$. These values are assumed wall-clock bounds for the design point. Under the other selection and submission premises, they give $\delta_{pre} = 0$ for this conditional example.

Set $T_e = 0.30\,\mathrm{s}$ and $T_{max} = 0.15\,\mathrm{s}$, so $T_{pre} + T_{max} = 0.20\,\mathrm{s}$ is the request-to-semantic budget. A hypothetical honest opportunity budget with $N_{max} = 21$, $k_{min} = 4$, $v_{min} = 1$, and $\lambda_h^{loc} = 350\,\mathrm{s}^{-1}$ gives $q_h = 4/21$ and $a = 200/3\,\mathrm{s}^{-1}$. Take $q_m = 1$ and $\lambda_m^{max} = 2/3\,\mathrm{s}^{-1}$, so $b = 2/3\,\mathrm{s}^{-1}$. The two count means are

$$\eta_m = (2/3)(0.30) = 0.2, \qquad \eta_h = (200/3)(0.15) = 10. \tag{29}$$

For $\varepsilon = 10^{-3}$ and $\delta = 0.05$,

$$\mathcal{U}(3; 0.2) = 0.00114848 > 0.001, \tag{30}$$

$$\mathcal{U}(4; 0.2) = 0.0000568402 < 0.001, \tag{31}$$

$$\mathcal{L}(5; 10) = 0.0292527 < 0.05, \tag{32}$$

$$\mathcal{L}(6; 10) = 0.0670860 > 0.05. \tag{33}$$

Thus $[c_{sec}, c_{live}] = [4, 5]$. At $c = 4$, the tighter semantic failure bound is $\mathcal{L}(4; 10) = 0.0103361$. Substituting $\delta_{pre} = 0$ gives the same upper bound for failure to issue and reach the threshold within the $200\,\mathrm{ms}$ request budget. This substitution uses the uniform conditional opportunity assumptions after timely issuance; it does not infer them from the selection-success calculation.

Here N and N_{max} count candidate messages, not issuing nodes, even if a chosen node population has the same size. The example's cap and draw budget are 21 and 16, respectively; the evaluated dynamic-pool implementation uses 256 and 64. The example is a hypothetical fixed-pool configuration with explicit hypotheses. Sufficient approval supply across requests and ledger stabilization are separate conditions, and the $200\,\mathrm{ms}$ budget covers semantic completion.

4) Conditional composition of completion stages: For a valid request, let T_{pre} be its budget from request creation to successful issuance, T_{max} its subsequent semantic-confirmation budget, and T_{led} the additional budget for the consumer's ledger-stability rule. Let δ_{pre} bound the probability of failure to issue by T_{pre}. Conditional on successful timely issuance, require Theorem 2's assumptions uniformly over the resulting histories, giving failure probability at most δ. Conditional on both preceding stages succeeding on time, let δ_{led} bound failure of ledger stabilization within T_{led}. The stage failure bounds δ_{pre} and δ_{led} lie in $[0, 1]$; all time budgets are non-negative, with $0 < T_{max} \leq T_e$. If these bounds are justified, a conservative total budget and failure bound are

$$\begin{aligned} T_{end} &= T_{pre} + T_{max} + T_{led}, \\ \Pr\{\text{request incomplete by } T_{end}\} &\leq \delta_{pre} + \delta + \delta_{led}. \end{aligned} \tag{34}$$

No independence between stages is required. If ledger stabilization overlaps semantic confirmation, the sum remains conservative. With a justified ledger model, Eq. (34) yields a numerical end-to-end completion guarantee.

Let E_{pre} be the event that a valid request issues within T_{pre}. Let E_{sem} be the event that it reaches the semantic threshold within the next T_{max} after issuance, and E_{led} the event that the specified ledger-stability condition holds within a further T_{led}. Require

$$\Pr(E_{pre}^c) \leq \delta_{pre}, \tag{35}$$

$$\Pr(E_{sem}^c \mid E_{pre}) \leq \delta, \tag{36}$$

$$\Pr(E_{led}^c \mid E_{pre} \cap E_{sem}) \leq \delta_{led}. \tag{37}$$

Here a superscript c on an event means its complement, not the endorsement threshold. A conditional bound for a zero-probability preceding event is unneeded because that branch contributes zero probability. For the semantic bound to follow from Theorem 2, its assumptions must remain valid conditional on the timely-issuance histories.

The first stage at which completion fails partitions the failure event into three disjoint parts. Thus

$$\begin{aligned}\Pr\{(E_{pre} \cap E_{sem} \cap E_{led})^c\} &= \Pr(E_{pre}^c) \\ &\quad + \Pr(E_{pre} \cap E_{sem}^c) \\ &\quad + \Pr(E_{pre} \cap E_{sem} \cap E_{led}^c) \\ &\le \delta_{pre} + \delta \Pr(E_{pre}) \\ &\quad + \delta_{led} \Pr(E_{pre} \cap E_{sem}) \\ &\le \delta_{pre} + \delta + \delta_{led}. \end{aligned} \tag{38}$$

When all three events hold, completion occurs by the sum of their time budgets. Failure to complete by that sum is therefore contained in the union of the stage-failure events, proving Eq. (34). The construction allows dependence and concurrent ledger progress. It contributes no value for δ_{led} unless a separate, justified ledger model supplies one.

5) Proof of Corollary 3 and a conditional design example:

Proof of Corollary 3. Fix $\varepsilon \in (0,1)$ and suppose that at most M unsafe approvals can reach any incorrect target within its eligibility window. Honest issuers re-execute an incorrect target and reject it, so all its effective approvals are unsafe, and there are at most M of them. For every $c \ge M+1$ the target cannot reach c, and $P_{invalid} = 0 \le \varepsilon$. For $c \le M$, the count bound alone permits a count equal to M with probability one, and thus does not certify any $\varepsilon < 1$. This worst-case calculation uses all count distributions supported on $\{0,\dots,M\}$; it does not assert that a particular deployment's adversary can attain M. The thresholds certified from the count bound alone are therefore exactly $c \ge M+1$. The bound $\mathcal{L}(c;\eta_h(T_{max}))$ of main-manuscript Theorem 2 is nondecreasing in c, because it is the probability that a Poisson count falls below c. Consequently, its certified thresholds are $1 \le c \le c_{live}$. Intersecting the two sets gives the integers in $[M+1, c_{live}]$, which is nonempty exactly when $\mathcal{L}(M+1;\eta_h(T_{max})) \le \delta$. Failure of this test means that these bounds cannot certify a threshold; a smaller actual exposure or a sharper liveness bound could still permit one. When a separately justified issuance bound gives $\delta_{pre} = 0$, and Theorem 2 holds uniformly after timely issuance, Eq. (34) gives the same bound for request-to-semantic completion by $T_{pre} + T_{max}$. Ledger finality, if required, needs its own stage bound. ∎

Conditional design example. Use the configured values $T_{pre} = 50\,\text{ms}$, $T_{max} = 150\,\text{ms}$, $T_e = 300\,\text{ms}$, $c = k_{min} = 4$, $N_{max} = 256$, and $d_{max} = 64$ from Table II, with targets $\varepsilon = 10^{-3}$ and $\delta = 0.05$. The following additional assumptions define a numerical example; they are not inferred from the runs. Assume external admission enforces $m = 1$ approval per identity per target, with at most $f = 3$ unsafe identities, so $M = fm = 3$.

Let d_{vis} bound the time until the target is available to the relevant honest issuers, and let d_{obs} bound the time from their approval issuance to its availability at the observing node. Both bounds include serialization, retrieval, and queueing where applicable. Assume $d_{vis} = d_{obs} = 3\,\text{ms}$. On the remaining interval of duration $T_{eff} = T_{max} - d_{vis} - d_{obs} = 144\,\text{ms}$, assume Theorem 2's Poisson honest-opportunity and conditional selection conditions, uniformly over the history at the interval's start. Take $\lambda_h^{loc} = 4{,}384.6\,\text{s}^{-1}$ as an assumed honest countable opportunity rate. Its numerical equality to the lowest offered input rate does not derive it from that rate, particularly when some issuers are unsafe. Assume also $v_{min} = 1$ and the candidate-sampling, retention, verification, and transmission conditions of Eq. (8), so $q_h = 4/256$ is a lower bound until the threshold is reached. Candidate removal, eviction, and expiry must be accounted for after every history before that threshold. The assumed rate must apply to opportunities satisfying these conditions; neither the pool cap nor an average visible fraction establishes them.

Counting only the opportunities in this interval gives $a = \lambda_h^{loc} q_h = 68.509375\,\text{s}^{-1}$ and an auxiliary honest count mean $\eta_{eff} = aT_{eff} = 9.86535$. Theorem 2 applied on that interval gives $\mathcal{L}(4;\eta_{eff}) \approx 0.0114 < 0.05$; discarded earlier approvals can only help. Together with the external cap, this certifies $c = 4$ and $P_{invalid} = 0$ under the stated assumptions.

The issuance stage needs a separate bound. The measured work range, 1,376,262 to 20,971,526 fuel units, gives $K_i \le 16$. For a fixed all-valid pool with at least K_i candidates, $S = 1$ follows provided they remain eligible and available and the draws and checks finish within the deadline. The largest calibrated task service is $0.5294\,\text{ms}$, and control service is assumed to be $10\,\mu\text{s}$. Rounding task service up to $0.54\,\text{ms}$ gives $0.54 + 16(0.54) + 0.01 = 9.19\,\text{ms}$ for the modeled own, accepted-parent, and final control service. This sum excludes queueing, retrieval, and other waiting; it is not a wall-clock issuance guarantee. Likewise, the average CPU utilization of approximately 0.31 does not prove $\delta_{pre} = 0$. Only if the additional issuance conditions establish that value does the example also certify a $200\,\text{ms}$ request-to-semantic budget with failure probability at most approximately 0.0114.

Conditional budget sensitivity. The required count mean is $\eta_h^{min}(4;0.05) \approx 7.75366$. With the preceding assumptions and $T_{eff} = 144\,\text{ms}$, the required honest opportunity rate is approximately $3{,}446.1$ messages/s. If instead the total delay bounds are each $40\,\text{ms}$, then $T_{eff} = 70\,\text{ms}$ and $\eta_{eff} = 4.79565625$, giving $\mathcal{L}(4;\eta_{eff}) \approx 0.295$. The same assumed rate and selection probability require $T_{max} \ge d_{vis} + d_{obs} + \eta_h^{min}(4;0.05)/a \approx 193.1766\,\text{ms}$. A $193.2\,\text{ms}$ endorsement budget therefore passes this certificate, or a $243.2\,\text{ms}$ request budget if the preceding $50\,\text{ms}$ issuance stage is guaranteed. These are conditional calculations, not certified deadlines for the simulated links, whose configured propagation ranges do not bound serialization and queueing.

Simulation observations. At $0.2C_{ref}$ on normal links with $c = k = 4$, all 5,120 followed requests of the ten runs and all 2,560 of the separate five-seed check reach the threshold within $200\,\text{ms}$, with mean delay $33.53\,\text{ms}$ and P95 $62.24\,\text{ms}$. At $0.4C_{ref}$, the observed deadline-miss fraction is 1.02%. On restricted links at $0.2C_{ref}$, MBSC records 28.59% misses. With compliant parent checking, none of the 521 invalid cohort targets of the attack study reaches the threshold (Table X). The external cap, the honest-opportunity process, and the total-delay bounds remain the stated premises of the conditional example.

APPENDIX V
ADDITIONAL EVALUATION DETAILS

This appendix specifies the measurements, simulated resources, statistical units, and additional results for main-manuscript Section V. The WASM timing measurements use the execution platform specified in main-manuscript Section V-A. Network traffic, node queues, and application endorsement events are simulated. This appendix retains the earlier 620-run evaluation and its original seeds. The new 560-run, four-method comparison is reported separately in Appendix VII.

A. WASM Kernels and Measurement Procedure

This subsection expands the kernel qualification and timing procedures in main-manuscript Section V-A. Measurements use an Apple M4 Mac mini with ten CPU cores, 24 GB memory, and macOS 15.8, with Python 3.11.16 and Wasmtime 48.0.0. The heaviest warm execution, 0.5294 ms, is over two orders of magnitude below the 200 ms deadline, but excludes queueing and network delay. Cold setup dominates the smallest kernel, motivating cached code and warm service times in simulation. The same procedure can calibrate gateway-class devices. Slower execution lowers C_{ref} and each load factor's absolute rate; completion must then be evaluated with those costs, network delays, and deadline.

The three kernels are constructed integer programs with fixed public inputs. Let i be the loop index, n the number of iterations, and s the input seed. For the threshold kernel, $(n, s) = (65536, 7)$ and the result counts indices for which $((17i + s)\ \&\ 1023) > 511$. For the access-rule kernel, $(n, s) = (262144, 13)$ and the result counts indices satisfying $(((31i + s)\ \&\ 255)\ \&\ 7) = 3$. For the squared-sum kernel, $(n, s) = (1048576, 29)$ and the result sums $((i + s)\ \&\ 1023)^2$ modulo 2^{32}. Here $\&$ denotes bitwise AND. The output is represented as a signed 32-bit integer with the same bit pattern. These synthetic inputs isolate bounded integer computation; the access-rule kernel is not a complete authentication system and the squared-sum kernel is not an energy-consumption measurement.

The modules are compiled from WebAssembly text, use no imported host function, and expose one fixed 64-KiB linear memory. The program bodies use integer operations and no memory growth. Runtime limits additionally reject a request to grow memory. An independently implemented Python computation checks the expected result. Three qualification executions per module check that the output and fuel count repeat exactly; a separate execution with one unit of fuel must trap. Every timed invocation also checks its result and fuel.

Wasmtime 48.0.0 uses fuel consumption enabled and parallel compilation disabled. The fixed engine profile is part of the metering definition. The work range in this catalog is 1 376 262–20 971 526 fuel units; each invocation is given a 100 000 000-unit runtime allowance. A protocol configuration can use the catalog minimum as I_{min} and this allowance as I_{max}. Runtime is measured separately from the deterministic fuel counts. The simulator uses only the qualified fixed input of each kernel.

Ten independently launched Python processes each run the same measurement procedure on the platform specified in main-manuscript Section V-A. Task order is randomized within a process. Each task has one cold measurement, ten warm-up calls, and 100 retained warm calls. Within each process, mean warm time is computed before forming the across-process mean and sample standard deviation (SD). The cold measurement includes engine initialization, compilation, instantiation, and first execution, but not process startup or library import. The warm measurement includes resetting fuel, executing the reused instance, querying fuel, and checking agreement. It is not an isolated native-instruction timing.

For batch measurements, immutable compiled modules are shared but each task uses a new Store and Instance. Batches cycle through the three task classes in order and contain 1, 2, 4, or 8 tasks. Sequential execution is compared with a persistent two-worker thread pool. Each size has three warm-up pairs and twenty measured pairs per process; measured size/iteration order and the order of the two methods are randomized. The ten process-level means form the independent units for reporting. The mean sequential/parallel ratio is calculated from ten paired process-level ratios, rather than from pooled invocation times.

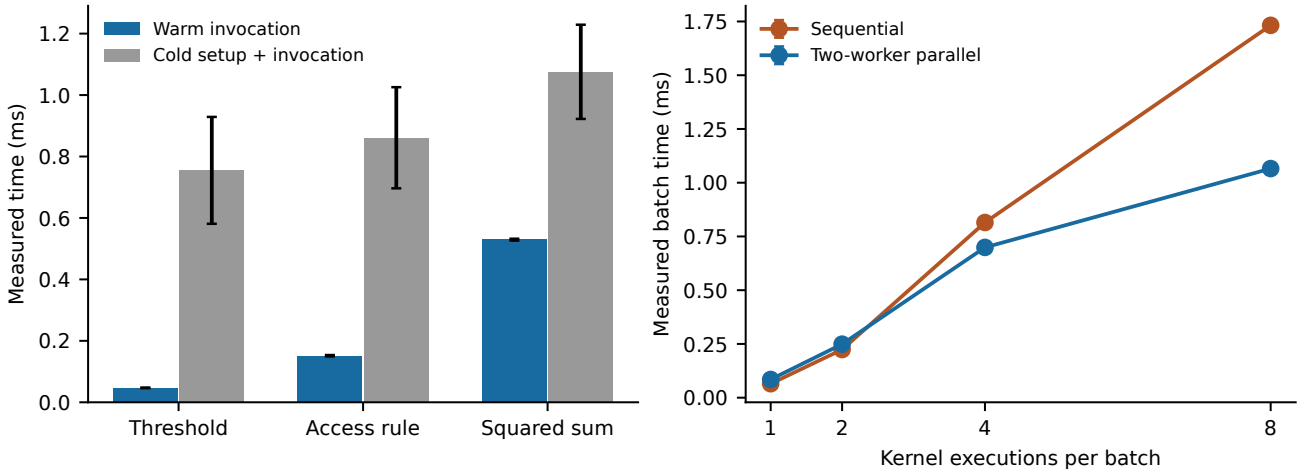


Fig. 1. WebAssembly (WASM) execution measurements. Left: warm invocation reuses an instance; cold execution includes engine creation, compilation, instantiation, and the first invocation. Right: sequential and two-worker execution of batches cycling through the three kernels; batch size counts kernel executions. Batch measurements include new Store/Instance creation and thread scheduling. Bars and points average measurements from ten independently launched processes; error bars are 95% Student-t intervals. Warm and batch observations are averaged within each process before computing across-process statistics.

Figure 1 compares warm and cold execution costs and sequential and parallel batch times. Cold setup has the largest relative impact on the smallest task; two-worker parallel execution becomes faster from batches of four kernel executions. The mean sequential/parallel time ratios are 0.77, 0.90, 1.17, and 1.63 for batches of 1, 2, 4, and 8 kernel executions, respectively.

The per-process high-water resident memory averages 32.57 MiB, with a range of 32.17–32.80 MiB. This includes Python, the execution engine, compiled code, and batch activity. It must not be equated with the 64-KiB per-instance linear memory limit or interpreted as an isolated kernel's peak memory. No energy measurement or hard real-time bound is inferred.

B. Node, Network, and Workload Model

This subsection expands the implementation parameters and workload design in main-manuscript Section V-A, including the attack configurations.

TABLE II
SIMULATION SETTINGS AND PARAMETER SOURCES

Quantity	Setting	Source or interpretation
Logical nodes and CPU	21 nodes; one serial service queue each	Simulated; not 21 physical devices
Task service	0.04696, 0.15156, 0.52938 ms	Means of ten measured warm process means; reused as fixed service demands
Task mixture	Equal probability of the three fixed-input tasks	Constructed workload
Local limits	256 candidate entries; 256 waiting CPU operations; 64 draws	Configured caps; known-ledger metadata is separately retained
Timing	50 ms pre-issuance; 200 ms request deadline; 300 ms approval window	Configured; no consensus-finality deadline
Application counts	$c = 4$; earlier matrices $k = 2, 4$; expanded controls in Appendix VII	Fixed before the corresponding runs
Normal links	1–3 ms uniform propagation; 1 Gbit/s	Assumed full-mesh network model
Restricted links	20–40 ms uniform propagation; 100 Mbit/s	Assumed full-mesh network model
Message size	256 bytes plus 48 bytes per parent reference	Assumed serialized length; 20 unicast copies per broadcast
Control service	10 μs per modeled control/admission operation	Assumed; not measured cryptographic verification
Bootstrap and follow-up	42 shared fixtures; 128 warm-up and 512 cohort arrivals	Background input continues through the last cohort deadline
Replications	Ten seeds per matrix cell; five per attack and initial completion cell	Paired inputs within each study and seed

c is the application endorsement threshold; k is the minimum parent count for MBSC and the fixed parent count for its control.

Table II distinguishes measured execution demands from configured network and resource parameters. CPU operations are queued and serviced serially at each logical node. An operation that cannot enter the finite queue is dropped; a local construction operation then defers its request. A pre-issuance timer independently terminates an unsuccessful attempt. Work already dispatched is charged even if the attempt's deadline prevents using its result. Own execution, parent execution, and received-message admission share the node's resource queue. Waiting for parents therefore cannot create unlimited CPU capacity.

A broadcast is represented by unicast copies to the other twenty nodes. Serialization uses the sender's shared outgoing link, and each copy adds a keyed uniform propagation delay. Receiver admission is a queued CPU operation. If a referenced parent has not arrived, the child waits for its dependency; parent metadata is consulted only after local availability. Missing dependencies do not provide free remote knowledge. Duplicate references, insufficient counts, out-of-window references, and, for MBSC, insufficient declared coverage are rejected. No hidden correctness label is consulted for structural admission.

Each catalog task has a fixed code/input pair and a prequalified work count, used equally by both methods. In an honest selection, the simulator charges the measured task service before exposing the modeled execution result to the selector. Re-execution failure adds to cost and removes that invalid parent locally. This abstracts deterministic re-execution; it does not actually launch WASM inside every event. The physical WASM measurements are the separate experiment in Appendix V-A. Both the skip-with-correct-output and blind-approval behaviors incur modeled control operations but omit own and parent WASM execution. In the former, the issuer copies the correct own output from the public fixed-task catalog; in the latter, it submits an incorrect own output and prioritizes known incorrect parent targets. A copied correct declaration can pass structural checks. The catalog's known weights do not turn that declaration into proof of private computation.

The issuing node observes its request's threshold using distinct received child messages. Eligibility is checked by the child's signed issuance time relative to its parent, including both the lower and upper age bounds. A message issued within the window may arrive later; its observation time includes that delay. A repeated arrival of the same message contributes no additional approval. Known-invalid targets are rejected independently of their approval count. Each draw permanently removes its candidate from that node's pool, including when the enclosing attempt is later deferred. Repeated delivery does not restore a known message. Candidate removal on confirmation, age expiry, or pool eviction also affects future selection.

Input requests follow a Poisson process and are independently assigned an issuer and task. The offered rates are based on

$$C_{ref} = \frac{21}{3\bar{s} + 23 s_{ctl}} = 21923.1 \text{ requests/s}, \tag{39}$$

where $\bar{s}$ is the equally weighted mean of the three measured warm service times and $s_{ctl} = 10\,\mu$s is the assumed control-service time. The denominator is a planning proxy for own plus two parent executions and control work; it is not a derived or measured capacity of either method. Both the main and additional $k = 4$ matrices use 0.20, 0.40, 0.55, 0.70, 0.85, 1.00, and 1.20 times C_{ref}. Queueing, additional parents, missing data, and failed attempts can make achieved completion differ substantially from these nominal load factors. Because C_{ref} is computed from the measured service times, it adapts to each platform: a device with larger service times, such as a gateway-class processor, yields a smaller C_{ref} and

hence lower absolute request rates at the same load factor. Propagation delays, message sizes, and the 200 ms deadline are specified in absolute time and do not scale with C_{ref}. Repeating the execution measurements and simulations with another platform's service times therefore yields its completion fractions.

For each seed, both methods use the same input trace and keyed network delays. Pairwise input hashes are checked. Their actual message sizes and selection histories can differ because work coverage changes the number of parents. Those differences are outcomes of the mechanism, rather than independently changed offered loads or link conditions. The fixed-k control uses $k = 2$ in the main and attack matrices and $k = 4$ in the additional matrix and initial completion check.

The attack matrix uses normal links, an offered rate of $0.70C_{ref}$, $c = 4$, and $k = 2$, with four unsafe nodes among the 21 nodes. Wrong-own-output nodes re-execute and check parents but publish an incorrect own result. Duplicate-parent nodes repeat a parent identifier. The skip and targeted blind-approval behaviors omit both own and parent execution, as specified above. The insufficient-coverage behavior submits a correct squared-sum result with two threshold-kernel parents. Their combined declared work is less than the child's work, so this configuration violates MBSC's coverage rule while satisfying the control's fixed parent count.

The trusted bootstrap consists of fourteen fixtures per task class, initially available to all nodes. Their single-copy execution reference cost is $14(s_1 + s_2 + s_3) = 10.19$ ms, where s_1, s_2, s_3 are the three warm service demands. This is reported separately from the request cost. Every later re-execution is charged. The followed cohort is fixed by arrival index, not by success. After its last arrival, background arrivals continue for 200 ms and scheduled events are drained. This provides later requests an opportunity to receive approvals.

TABLE III
FINITE-COHORT TIMING: MEAN ± SD ACROSS TEN INPUT TRACES

Load/C_{ref}	Start (ms)	Interval (ms)	Background
0.2	29.96 ± 2.13	115.82 ± 5.60	892.3 ± 25.3
0.4	14.98 ± 1.06	57.91 ± 2.80	1766.9 ± 45.3
0.55	10.89 ± 0.77	42.11 ± 2.04	2426.9 ± 46.5
0.7	8.56 ± 0.61	33.09 ± 1.60	3070.0 ± 45.2
0.85	7.05 ± 0.50	27.25 ± 1.32	3719.0 ± 53.9
1	5.99 ± 0.43	23.16 ± 1.12	4389.8 ± 73.7
1.2	4.99 ± 0.35	19.30 ± 0.93	5278.4 ± 84.3

$C_{ref} = 21923.1$ requests/s is the service-demand planning proxy; SD is the sample standard deviation. Start measures initialization to the first cohort request; Interval spans the first to last cohort arrival; Background counts subsequent arrivals. Traces are shared across methods and networks.

Table III reports the time from initialization to the first cohort request (Start), the time from its first to last arrival (Interval), and the number of later background arrivals (Background). Each row summarizes ten input traces, shared across methods and networks. The warm-up is a fixed number of arrivals, not a fixed duration or a test of stationarity. At higher load the followed cohort starts before even one restricted-link propagation interval has elapsed. Initialization, finite pools, and subsequent background traffic can therefore affect the results. The seven-point curve describes this finite-cohort experiment; it does not estimate a steady-state capacity or a steady-state load-response curve.

C. Replication, Metrics, and Cost Attribution

This subsection expands the replication and endpoint definitions in main-manuscript Section V-A and the cost accounting used in Section V-D.

The two parent-count matrices each contain 280 runs: seven loads, two networks, two methods, and ten seeds (2026092001–2026092010). Attack scenarios use five seeds (2026092101–2026092105) per method and behavior, giving 50 runs. The separate initial completion check uses five other seeds per method (2026092201–2026092205), giving ten runs and 620 runs overall. The load grid was expanded after inspecting initial results; each extension's settings were fixed before its additional runs. The simulator, calibration, task mixture, deadlines, and paired-input definitions remained unchanged. The initial five-seed check is not pooled with the ten-seed matrices. A separate pilot seed was used to check feasibility and resource accounting; its runs are excluded from the 620 reported runs.

Each run produces one metric per endpoint. Tables report means and sample standard deviations across these run metrics. For a metric with values $x_1, \ldots, x_n$ from n independent runs, the displayed mean is $\bar{x} = n^{-1} \sum_{j=1}^{n} x_j$, and its sample standard deviation is $s_x = [\sum_{j=1}^{n} (x_j - \bar{x})^2/(n-1)]^{1/2}$. A 95% interval is $\bar{x} \pm t_{0.975,n-1} s_x/\sqrt{n}$, where $t_{0.975,n-1}$ is the 97.5th percentile of a Student-t distribution with $n - 1$ degrees of freedom. Paired differences subtract fixed-k from MBSC within the same seed before forming this interval. These intervals summarize between-run variability.

A valid request is configured to produce a correct own output, even if its parent references violate an admission rule. The valid completion fraction is $F = n_{v,c}/n_v$, where n_v counts these requests in the followed cohort and $n_{v,c}$ counts those reaching c by their request deadline. For a timely request with birth time b_i and threshold-observation time f_i, end-to-end delay is $f_i - b_i$, including queueing, own execution, selection, and message delivery. A request can instead be deferred before issue, issued without a threshold, or reach the threshold after its deadline. These outcomes are retained. Delay means and nearest-rank 95th percentiles (P95) use only timely completions and are accompanied by the completion fraction. Cohort valid goodput divides timely completed valid

cohort requests by the time from the first to last cohort arrival. It describes a finite followed cohort with later background support, not stationary system capacity.

For attacks, the per-invalid-target empirical fraction divides invalid cohort requests reaching the threshold on time by all invalid cohort requests. The incorrect-share metric instead divides those incorrect completions by all timely threshold completions. Their denominators differ, and both describe the simulated scenarios, whereas Theorem 1 bounds the worst case over all admissible histories. A zero observed count is reported as a scenario outcome.

CPU cost attributed to a request includes the operations it initiates: its own execution, its selected-parent executions, and admission of its resulting message at receivers, including failed or redundant processing. Bytes include all its outgoing copies. Cohort-attributed totals sum these costs for every cohort request, including unsuccessful ones. Later background requests can validate cohort parents; their work belongs to those background requests and is not included in the cohort-attributed total. Consequently, dividing this total by cohort completions is an allocation statistic, not the entire system cost required to complete that cohort. Full-run CPU and byte totals additionally include warm-up and background traffic. Both accounting views are retained.

D. Additional Results

The two-method results below belong to the earlier evaluation; the expanded comparisons in Appendices VII and VIII provide the data for the current main-text Fig. 2.

This subsection expands the completion, cost, and failure results in main-manuscript Section V-D and the adversarial outcomes in Sections V-B and V-C.

Figure 2 provides a visual comparison of total CPU and communication cost; Tables V and VII retain the corresponding numerical means and standard deviations.

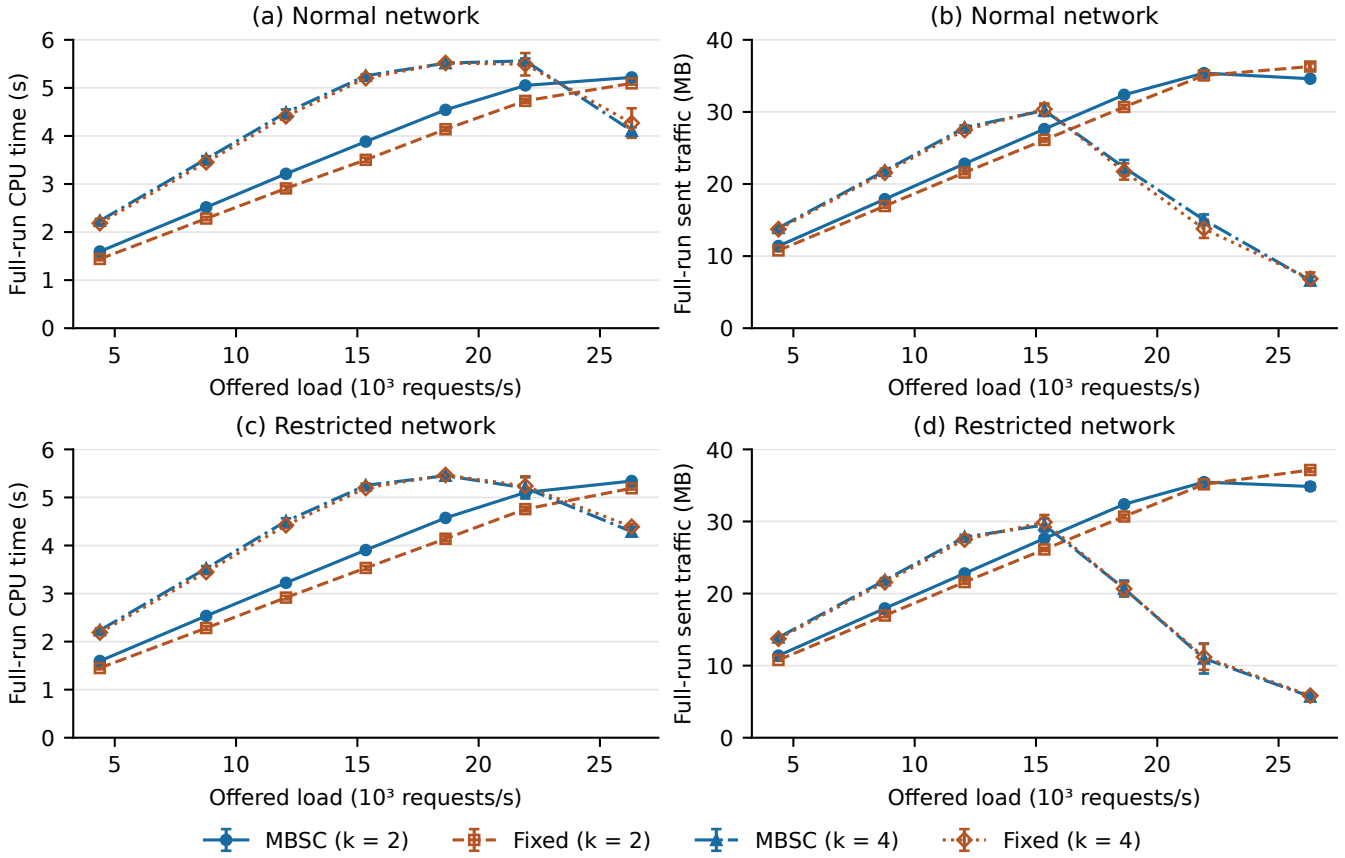


Fig. 2. Total simulated resource consumption across warm-up, cohort, and background requests. Points show ten-run means with 95% Student-t intervals; lines connect observed means. CPU time sums service across all 21 logical nodes and is not elapsed time; one MB is 10^6 bytes. Here k is MBSC's minimum parent count and the control's fixed count; both matrices use endorsement threshold $c = 4$ and seven offered loads. Normal links have uniform 1–3 ms propagation and 1 Gbit/s bandwidth; restricted links have 20–40 ms propagation and 100 Mbit/s bandwidth. Lower cost at high load accompanies deferred or unconfirmed requests and is read together with completion. The finite follow-up also makes the number of background requests load-dependent.

TABLE IV
$k = 2$ COMPARISON: MEAN $\pm$ SD ACROSS TEN RUNS PER SETTING

Network	Load/C_{ref}	Method	Completion (%)	Mean delay (ms)	P95 delay (ms)	CPU/request (ms)
Normal	0.2	MBSC	26.62 ± 2.36	52.20 ± 1.93	71.69 ± 2.56	1.043 ± 0.023
Normal	0.2	Fixed-2	15.78 ± 1.23	51.16 ± 1.91	68.78 ± 1.81	0.945 ± 0.020
Normal	0.4	MBSC	26.11 ± 1.97	29.64 ± 0.94	39.84 ± 1.68	1.039 ± 0.021
Normal	0.4	Fixed-2	15.25 ± 1.08	29.44 ± 0.88	38.37 ± 1.16	0.950 ± 0.021
Normal	0.55	MBSC	26.78 ± 1.87	24.73 ± 0.46	33.11 ± 1.11	1.047 ± 0.028
Normal	0.55	Fixed-2	14.84 ± 0.94	24.02 ± 0.56	31.42 ± 1.46	0.940 ± 0.021
Normal	0.7	MBSC	26.23 ± 2.45	23.84 ± 0.84	33.64 ± 4.17	1.035 ± 0.022
Normal	0.7	Fixed-2	15.51 ± 1.61	21.74 ± 0.81	29.49 ± 1.99	0.936 ± 0.029
Normal	0.85	MBSC	29.20 ± 2.41	27.27 ± 1.91	41.07 ± 4.28	1.029 ± 0.024
Normal	0.85	Fixed-2	16.21 ± 1.63	23.06 ± 1.08	32.78 ± 3.22	0.939 ± 0.018
Normal	1	MBSC	36.37 ± 3.22	35.59 ± 3.11	58.27 ± 7.38	1.028 ± 0.018
Normal	1	Fixed-2	19.02 ± 2.31	28.07 ± 2.25	42.77 ± 7.00	0.936 ± 0.026
Normal	1.2	MBSC	44.43 ± 3.96	49.20 ± 3.04	79.13 ± 3.69	1.017 ± 0.028
Normal	1.2	Fixed-2	28.20 ± 2.50	40.65 ± 3.45	63.56 ± 6.32	0.925 ± 0.026
Restricted	0.2	MBSC	22.99 ± 1.48	107.44 ± 2.14	130.78 ± 2.95	1.046 ± 0.023
Restricted	0.2	Fixed-2	14.12 ± 1.29	108.18 ± 2.62	129.61 ± 2.48	0.944 ± 0.025
Restricted	0.4	MBSC	25.76 ± 2.58	87.31 ± 1.18	102.40 ± 1.25	1.048 ± 0.026
Restricted	0.4	Fixed-2	15.66 ± 2.39	87.37 ± 1.27	102.07 ± 2.06	0.943 ± 0.028
Restricted	0.55	MBSC	28.75 ± 2.14	82.46 ± 0.63	97.62 ± 1.24	1.050 ± 0.021
Restricted	0.55	Fixed-2	17.99 ± 1.10	82.59 ± 1.19	96.86 ± 2.29	0.947 ± 0.026
Restricted	0.7	MBSC	35.53 ± 1.70	81.95 ± 1.19	96.76 ± 2.19	1.060 ± 0.020
Restricted	0.7	Fixed-2	21.23 ± 1.97	80.32 ± 0.83	93.54 ± 2.05	0.950 ± 0.016
Restricted	0.85	MBSC	41.39 ± 2.63	85.65 ± 1.45	105.09 ± 4.07	1.068 ± 0.015
Restricted	0.85	Fixed-2	26.11 ± 2.46	82.35 ± 0.77	97.73 ± 2.34	0.959 ± 0.015
Restricted	1	MBSC	45.86 ± 2.12	93.44 ± 1.99	120.83 ± 3.32	1.076 ± 0.017
Restricted	1	Fixed-2	32.30 ± 2.25	86.78 ± 1.60	105.47 ± 4.57	0.968 ± 0.019
Restricted	1.2	MBSC	47.81 ± 3.41	108.69 ± 1.86	141.02 ± 4.47	1.073 ± 0.021
Restricted	1.2	Fixed-2	38.28 ± 2.58	99.12 ± 2.02	125.08 ± 3.87	0.967 ± 0.015

k is the minimum parent count for MBSC and the fixed count for its control; the endorsement threshold is $c = 4$. $C_{ref} = 21923.1$ requests/s is a planning proxy. Normal links use uniform 1–3 ms propagation and 1 Gbit/s bandwidth; restricted links use 20–40 ms and 100 Mbit/s. Completion uses all valid cohort requests; delays use only those completing within 200 ms of creation. P95 is each run's 95th percentile. CPU/request divides cohort-attributed CPU time by all submitted cohort requests, including unsuccessful ones. SD is the sample standard deviation of the ten run metrics.

TABLE V
$k = 2$ FULL-RUN COST, INCLUDING WARM-UP AND BACKGROUND: MEAN $\pm$ SD

Network	Load/C_{ref}	Method	CPU time (s)	Sent traffic (MB)
Normal	0.2	MBSC	1.596 ± 0.034	11.399 ± 0.195
Normal	0.2	Fixed-2	1.440 ± 0.040	10.787 ± 0.178
Normal	0.4	MBSC	2.516 ± 0.049	17.893 ± 0.330
Normal	0.4	Fixed-2	2.282 ± 0.048	16.945 ± 0.319
Normal	0.55	MBSC	3.212 ± 0.062	22.816 ± 0.340
Normal	0.55	Fixed-2	2.909 ± 0.062	21.591 ± 0.328
Normal	0.7	MBSC	3.882 ± 0.063	27.625 ± 0.365
Normal	0.7	Fixed-2	3.504 ± 0.063	26.118 ± 0.318
Normal	0.85	MBSC	4.543 ± 0.067	32.373 ± 0.346
Normal	0.85	Fixed-2	4.139 ± 0.057	30.687 ± 0.380
Normal	1	MBSC	5.051 ± 0.040	35.386 ± 0.295
Normal	1	Fixed-2	4.730 ± 0.062	35.059 ± 0.341
Normal	1.2	MBSC	5.219 ± 0.035	34.605 ± 0.466
Normal	1.2	Fixed-2	5.095 ± 0.039	36.300 ± 0.577
Restricted	0.2	MBSC	1.596 ± 0.036	11.391 ± 0.224
Restricted	0.2	Fixed-2	1.450 ± 0.039	10.787 ± 0.178
Restricted	0.4	MBSC	2.534 ± 0.057	17.939 ± 0.338
Restricted	0.4	Fixed-2	2.282 ± 0.053	16.945 ± 0.319
Restricted	0.55	MBSC	3.223 ± 0.061	22.795 ± 0.358
Restricted	0.55	Fixed-2	2.915 ± 0.053	21.591 ± 0.328
Restricted	0.7	MBSC	3.905 ± 0.054	27.636 ± 0.320
Restricted	0.7	Fixed-2	3.533 ± 0.057	26.118 ± 0.318
Restricted	0.85	MBSC	4.573 ± 0.054	32.383 ± 0.355
Restricted	0.85	Fixed-2	4.141 ± 0.068	30.687 ± 0.380
Restricted	1	MBSC	5.103 ± 0.053	35.461 ± 0.358
Restricted	1	Fixed-2	4.757 ± 0.063	35.154 ± 0.375
Restricted	1.2	MBSC	5.342 ± 0.034	34.852 ± 0.653
Restricted	1.2	Fixed-2	5.191 ± 0.037	37.142 ± 0.343

k is the minimum parent count for MBSC and the fixed count for its control; the endorsement threshold is $c = 4$. $C_{ref} = 21923.1$ requests/s is a planning proxy. Normal links use uniform 1–3 ms propagation and 1 Gbit/s bandwidth; restricted links use 20–40 ms and 100 Mbit/s. Entries are means and sample standard deviations across ten runs. CPU time sums service over all 21 nodes, including warm-up and background requests; it is not elapsed time. Sent traffic includes the same requests; one MB is 10^6 bytes.

Delays in Table IV are conditional on timely completion. CPU/request is the cohort-attributed cost divided by all submitted cohort requests; its attribution follows Appendix V-C. The means of run-level P95 values are reported separately from means of run-level delays. No pooled cross-run request percentile is substituted for these quantities.

MBSC's completion exceeds the control's at every load on both networks. With two parents, completion also rises with offered load, from 22.99%–26.62% at $0.2C_{ref}$ to 44.43%–47.81% at $1.2C_{ref}$. Additional arrivals supply potential endorsers, although queueing and selection determine which opportunities become countable. Conditional mean delay is similar at the lowest load (52.20 versus 51.16 ms on normal links; 107.44 versus 108.18 ms on restricted links) and higher at the highest load (49.20 versus 40.65 ms; 108.69 versus 99.12 ms), consistent with MBSC completing additional requests later within the deadline.

In Table V, one MB is 10^6 bytes; CPU time sums service across all nodes and is not elapsed wall-clock time. At higher offered load, queue drops or deferred construction can reduce transmitted traffic while increasing incomplete requests. Traffic cost must therefore be interpreted alongside completion.

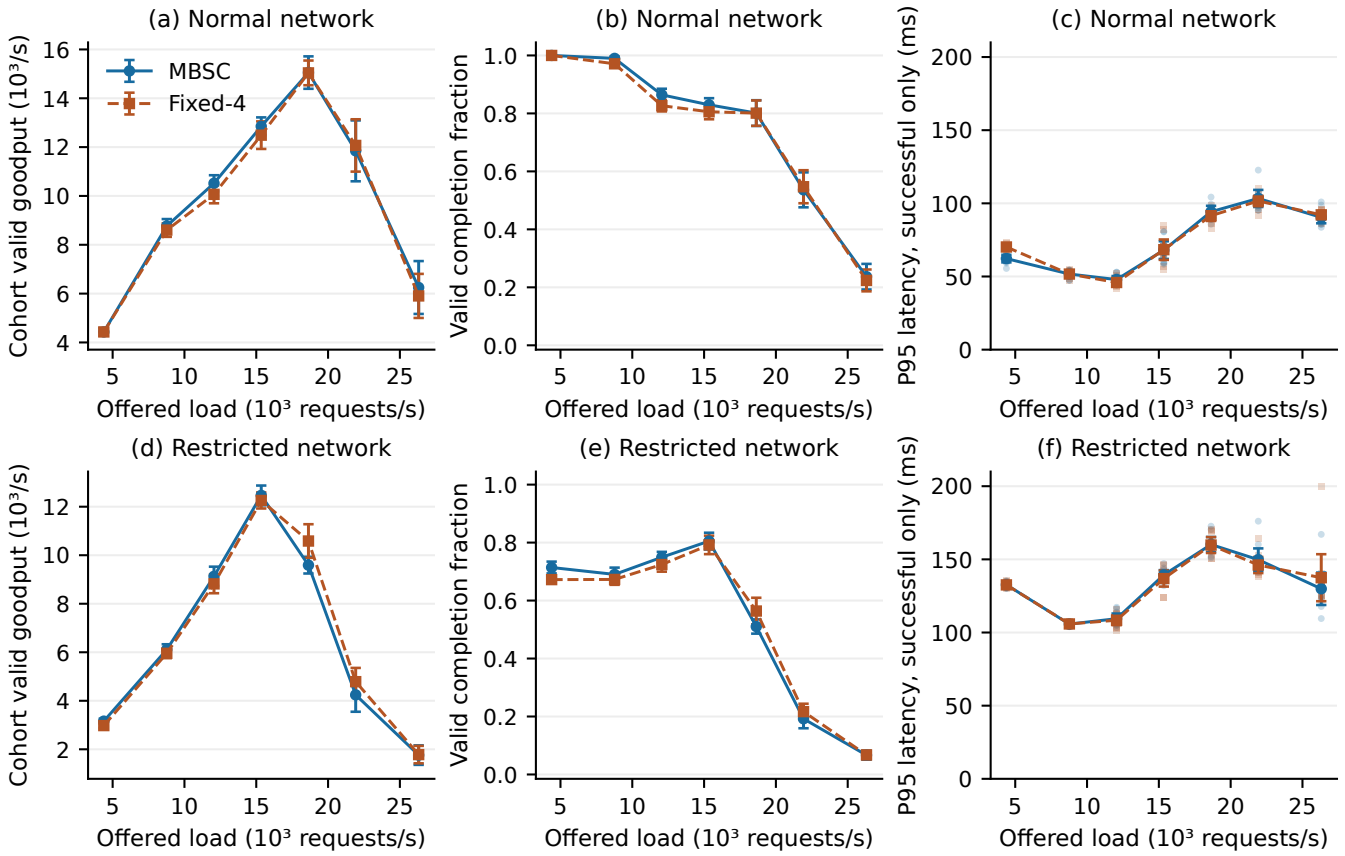


Fig. 3. Comparison with endorsement threshold $c = 4$ and parent count $k = 4$, the minimum for MBSC and the fixed count for Fixed-4. Normal links have uniform 1–3 ms propagation and 1 Gbit/s bandwidth; restricted links have 20–40 ms propagation and 100 Mbit/s bandwidth. The network and queues are simulated. Completion divides valid requests reaching the threshold within 200 ms of request creation by all valid cohort requests; goodput divides those completions by the cohort arrival interval. P95 is the within-run 95th percentile of latency among timely completions. Main markers average ten independent run metrics; bars are 95% Student-t intervals. Translucent markers in (c) and (f) show individual run-level P95 values on the same delay scale. Lines join observed means as visual guides. Both methods reach full completion at low load on normal links.

TABLE VI
$k = 4$ COMPARISON: MEAN ± SD ACROSS TEN RUNS PER SETTING

Network	Load/C_{ref}	Method	Completion (%)	Mean delay (ms)	P95 delay (ms)	CPU/request (ms)
Normal	0.2	MBSC	100.00 ± 0.00	33.53 ± 1.01	62.24 ± 3.64	1.465 ± 0.027
Normal	0.2	Fixed-4	100.00 ± 0.00	37.33 ± 0.45	70.17 ± 2.31	1.436 ± 0.031
Normal	0.4	MBSC	98.98 ± 0.89	29.54 ± 0.97	51.69 ± 2.56	1.459 ± 0.032
Normal	0.4	Fixed-4	97.11 ± 1.46	29.81 ± 0.94	51.66 ± 2.35	1.437 ± 0.035
Normal	0.55	MBSC	86.41 ± 2.94	30.30 ± 1.61	47.89 ± 3.40	1.465 ± 0.039
Normal	0.55	Fixed-4	82.66 ± 2.75	29.62 ± 1.01	45.89 ± 3.16	1.429 ± 0.041
Normal	0.7	MBSC	82.97 ± 3.20	40.68 ± 4.22	68.05 ± 8.45	1.441 ± 0.048
Normal	0.7	Fixed-4	80.57 ± 3.62	40.34 ± 4.95	68.33 ± 9.72	1.425 ± 0.047
Normal	0.85	MBSC	80.12 ± 6.20	57.12 ± 4.41	94.27 ± 5.64	1.426 ± 0.037
Normal	0.85	Fixed-4	80.10 ± 6.12	55.55 ± 3.33	91.55 ± 5.01	1.407 ± 0.038
Normal	1	MBSC	53.63 ± 8.43	63.85 ± 3.93	103.34 ± 8.17	1.403 ± 0.046
Normal	1	Fixed-4	54.71 ± 7.96	63.97 ± 4.12	101.61 ± 5.05	1.396 ± 0.039
Normal	1.2	MBSC	23.67 ± 6.17	60.84 ± 3.49	90.42 ± 5.67	1.346 ± 0.035
Normal	1.2	Fixed-4	22.38 ± 5.26	61.75 ± 3.21	92.15 ± 3.26	1.352 ± 0.037
Restricted	0.2	MBSC	71.41 ± 2.84	104.24 ± 1.44	132.64 ± 2.08	1.462 ± 0.036
Restricted	0.2	Fixed-4	67.23 ± 1.95	103.74 ± 1.05	132.55 ± 1.66	1.434 ± 0.036
Restricted	0.4	MBSC	69.02 ± 3.27	87.22 ± 0.63	105.75 ± 1.14	1.465 ± 0.044
Restricted	0.4	Fixed-4	67.29 ± 2.59	87.41 ± 0.68	105.85 ± 1.12	1.440 ± 0.040
Restricted	0.55	MBSC	74.94 ± 2.61	89.34 ± 2.33	109.45 ± 4.93	1.482 ± 0.040
Restricted	0.55	Fixed-4	72.42 ± 3.43	88.16 ± 1.68	108.40 ± 4.72	1.451 ± 0.048
Restricted	0.7	MBSC	80.51 ± 4.04	106.09 ± 3.10	139.41 ± 4.52	1.483 ± 0.036
Restricted	0.7	Fixed-4	79.20 ± 4.46	104.15 ± 3.05	136.87 ± 7.66	1.463 ± 0.043
Restricted	0.85	MBSC	51.04 ± 3.42	119.30 ± 3.41	160.14 ± 7.20	1.479 ± 0.034
Restricted	0.85	Fixed-4	56.39 ± 6.41	119.30 ± 4.41	159.78 ± 7.59	1.465 ± 0.039
Restricted	1	MBSC	19.22 ± 4.58	112.48 ± 5.26	149.79 ± 10.77	1.454 ± 0.030
Restricted	1	Fixed-4	21.68 ± 3.81	113.47 ± 3.02	146.13 ± 7.70	1.453 ± 0.034
Restricted	1.2	MBSC	6.66 ± 2.18	103.09 ± 3.09	129.89 ± 15.49	1.407 ± 0.030
Restricted	1.2	Fixed-4	6.74 ± 1.95	106.21 ± 7.54	137.46 ± 22.45	1.409 ± 0.031

k is the minimum parent count for MBSC and the fixed count for its control; the endorsement threshold is $c = 4$. $C_{ref} = 21923.1$ requests/s is a planning proxy. Normal links use uniform 1–3 ms propagation and 1 Gbit/s bandwidth; restricted links use 20–40 ms and 100 Mbit/s. Completion uses all valid cohort requests; delays use only those completing within 200 ms of creation. P95 is each run's 95th percentile. CPU/request divides cohort-attributed CPU time by all submitted cohort requests, including unsuccessful ones. SD is the sample standard deviation of the ten run metrics.

TABLE VII
$k = 4$ FULL-RUN COST, INCLUDING WARM-UP AND BACKGROUND: MEAN ± SD

Network	Load/C_{ref}	Method	CPU time (s)	Sent traffic (MB)
Normal	0.2	MBSC	2.234 ± 0.051	13.910 ± 0.228
Normal	0.2	Fixed-4	2.186 ± 0.056	13.729 ± 0.227
Normal	0.4	MBSC	3.522 ± 0.081	21.835 ± 0.416
Normal	0.4	Fixed-4	3.453 ± 0.076	21.566 ± 0.406
Normal	0.55	MBSC	4.486 ± 0.092	27.828 ± 0.428
Normal	0.55	Fixed-4	4.408 ± 0.085	27.479 ± 0.417
Normal	0.7	MBSC	5.253 ± 0.045	30.210 ± 1.078
Normal	0.7	Fixed-4	5.202 ± 0.052	30.380 ± 1.114
Normal	0.85	MBSC	5.515 ± 0.027	22.280 ± 1.474
Normal	0.85	Fixed-4	5.519 ± 0.026	21.736 ± 1.577
Normal	1	MBSC	5.564 ± 0.069	15.035 ± 1.029
Normal	1	Fixed-4	5.492 ± 0.326	13.786 ± 1.767
Normal	1.2	MBSC	4.108 ± 0.101	6.606 ± 0.663
Normal	1.2	Fixed-4	4.268 ± 0.430	6.854 ± 1.232
Restricted	0.2	MBSC	2.236 ± 0.053	13.904 ± 0.239
Restricted	0.2	Fixed-4	2.189 ± 0.057	13.729 ± 0.227
Restricted	0.4	MBSC	3.523 ± 0.070	21.839 ± 0.424
Restricted	0.4	Fixed-4	3.448 ± 0.066	21.566 ± 0.406
Restricted	0.55	MBSC	4.504 ± 0.087	27.832 ± 0.434
Restricted	0.55	Fixed-4	4.420 ± 0.078	27.479 ± 0.417
Restricted	0.7	MBSC	5.252 ± 0.052	29.509 ± 1.261
Restricted	0.7	Fixed-4	5.197 ± 0.060	29.899 ± 1.400
Restricted	0.85	MBSC	5.452 ± 0.029	20.692 ± 1.551
Restricted	0.85	Fixed-4	5.467 ± 0.029	20.656 ± 1.293
Restricted	1	MBSC	5.198 ± 0.309	10.990 ± 2.914
Restricted	1	Fixed-4	5.237 ± 0.284	11.208 ± 2.511
Restricted	1.2	MBSC	4.288 ± 0.141	5.709 ± 0.452
Restricted	1.2	Fixed-4	4.390 ± 0.079	5.832 ± 0.332

k is the minimum parent count for MBSC and the fixed count for its control; the endorsement threshold is $c = 4$. $C_{ref} = 21923.1$ requests/s is a planning proxy. Normal links use uniform 1–3 ms propagation and 1 Gbit/s bandwidth; restricted links use 20–40 ms and 100 Mbit/s. Entries are means and sample standard deviations across ten runs. CPU time sums service over all 21 nodes, including warm-up and background requests; it is not elapsed time. Sent traffic includes the same requests; one MB is 10^6 bytes.

Figure 3 and Table VI show the additional $k = 4$ matrix. Delays and CPU attribution use the same definitions as Table IV; Table VII retains all warm-up and background cost. At three representative loads on normal links, mean completion falls from 100% at $0.2C_{ref}$ to 82.97% for MBSC and 80.57% for fixed-4 at $0.7C_{ref}$, and then to 23.67% and 22.38% at $1.2C_{ref}$. On restricted links the corresponding pairs are 71.41%/67.23%, 80.51%/79.20%, and 6.66%/6.74%. With four parents, the coverage rule seldom adds a reference, so MBSC tracks fixed-4 closely and matches or exceeds its mean completion in ten of the fourteen settings. Comparing the two matrices shows the engineering balance set by the parent count. Four parents give higher completion in the tested low-load settings, whereas the added verification work saturates queues under overload. At $1.2C_{ref}$ on normal links and at $1.0C_{ref}$ and $1.2C_{ref}$ on restricted links, MBSC with two parents completes 44.43%, 45.86%, and 47.81% of valid requests at 1.017–1.076 ms per request, whereas MBSC with four parents completes 23.67%, 19.22%, and 6.66% and fixed-4 completes 22.38%, 21.68%, and 6.74% at 1.346–1.454 ms per request.

The wider P95 intervals reflect between-run variation among timely completed requests. For example, at $1.2C_{ref}$ on restricted links, the mean run-level P95 is 129.89 ± 11.08 ms for MBSC and 137.46 ± 16.06 ms for fixed-4, where $\pm$ denotes the 95% interval half-width. Individual runs contain 11–52 and 14–48 timely completions, respectively. With 14 completions, the nearest-rank P95 selects observation $\lceil 0.95 \times 14 \rceil = 14$, the largest delay; this is 199.89 ms in one fixed-4 run. Sparse completion samples can therefore make the tail estimate sensitive to a few requests. The intervals retain this uncertainty.

The separate initial low-load, normal-network check used five different seeds per method. All 512 followed requests completed in each run. Mean delay was 33.99 ± 2.52 ms for MBSC and 37.97 ± 2.70 ms for fixed-4; MBSC used 4.113 ± 0.029 parents per issued request, versus exactly four. Cohort-attributed cost was 1.487 ± 0.038 versus 1.468 ± 0.037 ms per submission. Here $\pm$ denotes the sample standard deviation across five runs. These five-seed results are reported separately from the ten-seed matrix.

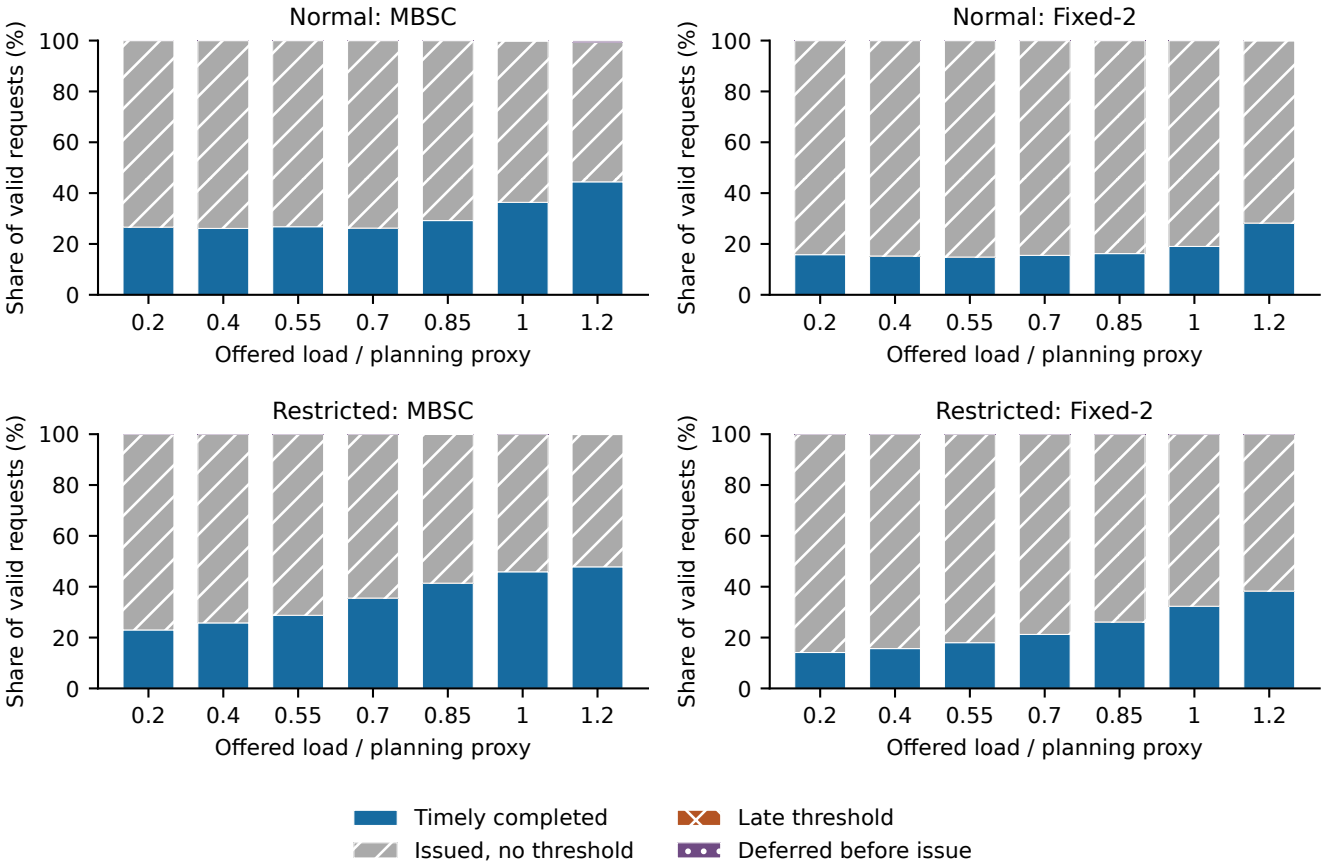


Fig. 4. Outcome composition for MBSC (minimum $k = 2$ parents) and Fixed-2 (exactly two parents), with endorsement threshold $c = 4$. Offered load is divided by the planning proxy $C_{ref} = 21923.1$ requests/s. Normal links use uniform 1–3 ms propagation and 1 Gbit/s bandwidth; restricted links use 20–40 ms and 100 Mbit/s. Each stack averages within-run percentages of all valid cohort requests across ten independent runs. The four mutually exclusive outcomes are: threshold completion within 200 ms of request creation; issuance without a threshold during the recorded run; a threshold after the deadline; and deferral before issuance. Stacks summarize composition without error bars; completion variability is reported in the main comparison figure and the numerical comparison table. Small or zero fractions can be visually indistinguishable and remain available in Table VIII.

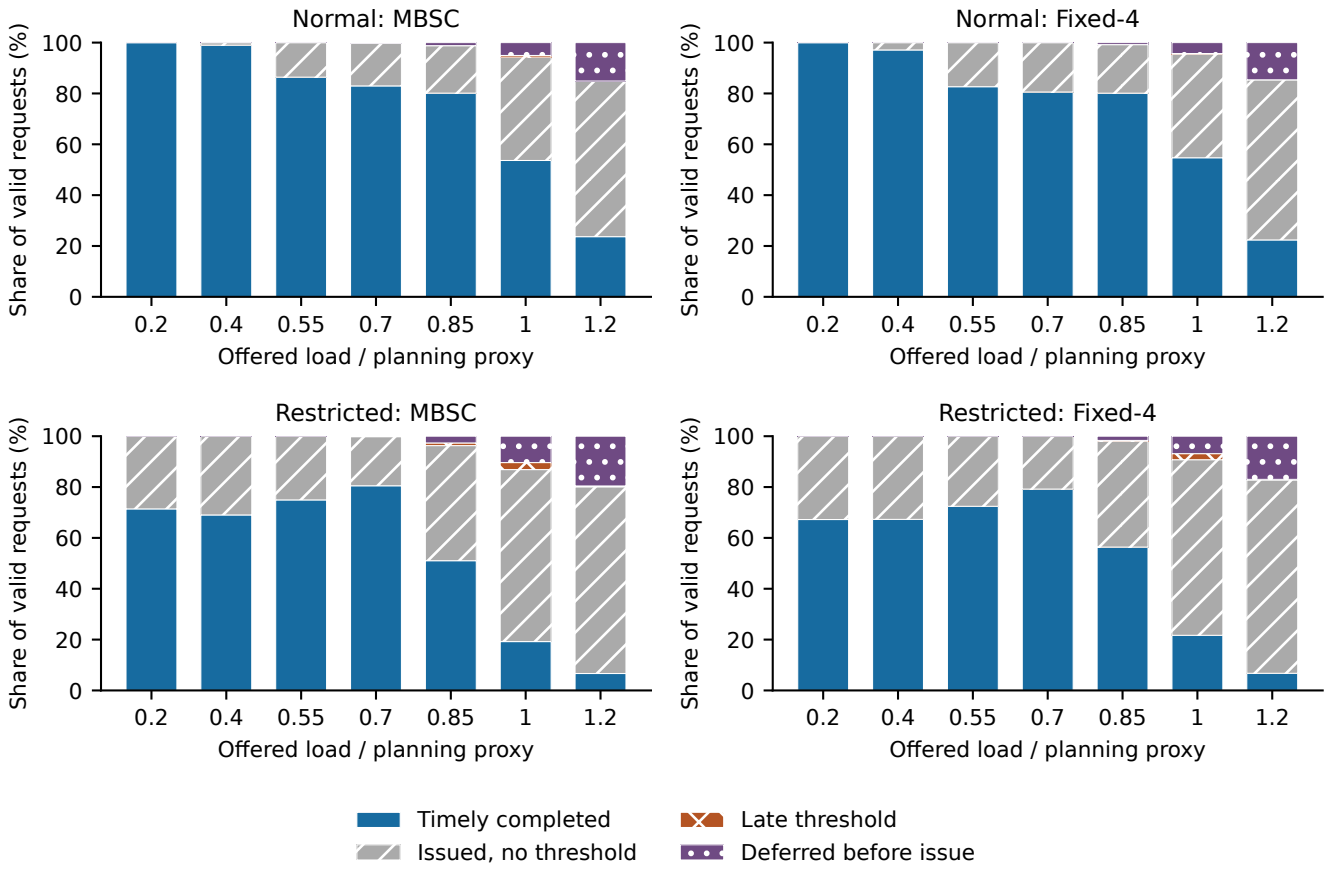


Fig. 5. Outcome composition for MBSC (minimum $k = 4$ parents) and Fixed-4 (exactly four parents), with endorsement threshold $c = 4$. Offered load is divided by the planning proxy $C_{ref} = 21923.1$ requests/s. Normal links use uniform 1–3 ms propagation and 1 Gbit/s bandwidth; restricted links use 20–40 ms and 100 Mbit/s. Stacks average within-run percentages of all valid cohort requests across ten independent runs. The four mutually exclusive outcomes are: threshold completion within 200 ms of request creation; issuance without a threshold during the recorded run; a threshold after the deadline; and pre-issuance deferral. Stacks show composition without error bars. The high-load restricted-link setting has a large issued-but-unconfirmed fraction and more deferrals. These categories describe observed outcomes, not uniquely identified causes. Completion uncertainty is reported in Fig. 3; Table IX retains precise fractions, including small late-threshold components.

Figures 4 and 5 show how successful and unsuccessful outcomes share the full cohort denominator. Tables VIII and IX partition all valid cohort requests into four mutually exclusive outcomes. Timely requests reach the threshold by their deadline. Unconfirmed requests are issued but never reach it during the recorded run. Late requests reach it after the deadline. Deferred requests are not issued. Each row reports the mean of ten within-run percentages and sums to 100%, up to rounding. These are outcome categories, not an attribution of each failure to a unique cause. For example, queue rejection and parent unavailability can both prevent issuance before the construction deadline; the final deferred category does not distinguish their causal effects.

In the high-load restricted $k = 4$ setting, MBSC has 19.57% deferred, 73.44% unconfirmed, and 0.33% late requests; fixed-4 has 16.97%, 76.05%, and 0.23%, respectively. This breakdown prevents the short conditional latency of the few timely completions from being mistaken for a general deadline guarantee. Four parents increase the available reference supply but also add re-execution and delivery demand. Dynamic visibility, resource queues, and deadlines remain relevant to completion.

TABLE VIII
$k = 2$ FAILURE COMPOSITION: MEAN PERCENTAGES ACROSS TEN RUNS

Network	Load/C_{ref}	Method	Timely	Unconfirmed	Late	Deferred
Normal	0.2	MBSC	26.62	73.38	0.00	0.00
Normal	0.2	Fixed-2	15.78	84.22	0.00	0.00
Normal	0.4	MBSC	26.11	73.89	0.00	0.00
Normal	0.4	Fixed-2	15.25	84.75	0.00	0.00
Normal	0.55	MBSC	26.78	73.22	0.00	0.00
Normal	0.55	Fixed-2	14.84	85.16	0.00	0.00
Normal	0.7	MBSC	26.23	73.77	0.00	0.00
Normal	0.7	Fixed-2	15.51	84.49	0.00	0.00
Normal	0.85	MBSC	29.20	70.80	0.00	0.00
Normal	0.85	Fixed-2	16.21	83.79	0.00	0.00
Normal	1	MBSC	36.37	63.50	0.00	0.14
Normal	1	Fixed-2	19.02	80.98	0.00	0.00
Normal	1.2	MBSC	44.43	55.00	0.00	0.57
Normal	1.2	Fixed-2	28.20	71.74	0.00	0.06
Restricted	0.2	MBSC	22.99	77.01	0.00	0.00
Restricted	0.2	Fixed-2	14.12	85.88	0.00	0.00
Restricted	0.4	MBSC	25.76	74.24	0.00	0.00
Restricted	0.4	Fixed-2	15.66	84.34	0.00	0.00
Restricted	0.55	MBSC	28.75	71.25	0.00	0.00
Restricted	0.55	Fixed-2	17.99	82.01	0.00	0.00
Restricted	0.7	MBSC	35.53	64.47	0.00	0.00
Restricted	0.7	Fixed-2	21.23	78.77	0.00	0.00
Restricted	0.85	MBSC	41.39	58.59	0.00	0.02
Restricted	0.85	Fixed-2	26.11	73.89	0.00	0.00
Restricted	1	MBSC	45.86	54.14	0.00	0.00
Restricted	1	Fixed-2	32.30	67.70	0.00	0.00
Restricted	1.2	MBSC	47.81	52.05	0.00	0.14
Restricted	1.2	Fixed-2	38.28	61.72	0.00	0.00

k is the minimum parent count for MBSC and the fixed count for its control; the endorsement threshold is $c = 4$. $C_{ref} = 21923.1$ requests/s is a planning proxy. Normal links use uniform 1–3 ms propagation and 1 Gbit/s bandwidth; restricted links use 20–40 ms and 100 Mbit/s. All percentages use valid cohort requests. Timely means threshold completion within 200 ms of creation; Unconfirmed means issued without a threshold by run end; Late means a threshold after the deadline; Deferred means not issued. These mutually exclusive outcomes sum to 100%, up to rounding; they do not identify unique failure causes.

TABLE IX
$k = 4$ FAILURE COMPOSITION: MEAN PERCENTAGES ACROSS TEN RUNS

Network	Load/C_{ref}	Method	Timely	Unconfirmed	Late	Deferred
Normal	0.2	MBSC	100.00	0.00	0.00	0.00
Normal	0.2	Fixed-4	100.00	0.00	0.00	0.00
Normal	0.4	MBSC	98.98	1.02	0.00	0.00
Normal	0.4	Fixed-4	97.11	2.89	0.00	0.00
Normal	0.55	MBSC	86.41	13.59	0.00	0.00
Normal	0.55	Fixed-4	82.66	17.34	0.00	0.00
Normal	0.7	MBSC	82.97	16.78	0.00	0.25
Normal	0.7	Fixed-4	80.57	19.43	0.00	0.00
Normal	0.85	MBSC	80.12	18.69	0.00	1.19
Normal	0.85	Fixed-4	80.10	19.12	0.00	0.78
Normal	1	MBSC	53.63	40.64	0.66	5.06
Normal	1	Fixed-4	54.71	40.72	0.27	4.30
Normal	1.2	MBSC	23.67	61.21	0.00	15.12
Normal	1.2	Fixed-4	22.38	62.99	0.00	14.63
Restricted	0.2	MBSC	71.41	28.59	0.00	0.00
Restricted	0.2	Fixed-4	67.23	32.77	0.00	0.00
Restricted	0.4	MBSC	69.02	30.98	0.00	0.00
Restricted	0.4	Fixed-4	67.29	32.71	0.00	0.00
Restricted	0.55	MBSC	74.94	25.06	0.00	0.00
Restricted	0.55	Fixed-4	72.42	27.58	0.00	0.00
Restricted	0.7	MBSC	80.51	19.34	0.00	0.16
Restricted	0.7	Fixed-4	79.20	20.78	0.02	0.00
Restricted	0.85	MBSC	51.04	45.27	1.04	2.66
Restricted	0.85	Fixed-4	56.39	41.60	0.35	1.66
Restricted	1	MBSC	19.22	67.66	2.75	10.37
Restricted	1	Fixed-4	21.68	69.08	2.38	6.86
Restricted	1.2	MBSC	6.66	73.44	0.33	19.57
Restricted	1.2	Fixed-4	6.74	76.05	0.23	16.97

k is the minimum parent count for MBSC and the fixed count for its control; the endorsement threshold is $c = 4$. $C_{ref} = 21923.1$ requests/s is a planning proxy. Normal links use uniform 1–3 ms propagation and 1 Gbit/s bandwidth; restricted links use 20–40 ms and 100 Mbit/s. All percentages use valid cohort requests. Timely means threshold completion within 200 ms of creation; Unconfirmed means issued without a threshold by run end; Late means a threshold after the deadline; Deferred means not issued. These mutually exclusive outcomes sum to 100%, up to rounding; they do not identify unique failure causes.

TABLE X
ATTACK SCENARIOS: MEAN ± SD ACROSS FIVE RUNS PER SETTING

Behavior	Method	Valid completion (%)	Invalid-target endorsement (%)	CPU/request (ms)
Skip, correct output	MBSC	26.45 ± 1.77	—	0.887 ± 0.009
Skip, correct output	Fixed-2	15.43 ± 1.02	—	0.814 ± 0.022
Wrong own output	MBSC	46.71 ± 2.16	0.00 ± 0.00	1.196 ± 0.036
Wrong own output	Fixed-2	28.29 ± 3.65	0.00 ± 0.00	1.081 ± 0.025
Targeted blind approval	MBSC	28.26 ± 1.43	18.46 ± 3.10	1.010 ± 0.012
Targeted blind approval	Fixed-2	17.06 ± 2.00	8.90 ± 2.50	0.923 ± 0.027
Duplicate parent	MBSC	21.88 ± 2.37	—	1.042 ± 0.012
Duplicate parent	Fixed-2	12.54 ± 0.42	—	0.948 ± 0.021
Insufficient coverage	MBSC	23.20 ± 2.99	—	0.993 ± 0.008
Insufficient coverage	Fixed-2	18.09 ± 1.24	—	1.038 ± 0.017

SD denotes the sample standard deviation of five run metrics. Four of 21 nodes use the listed behavior, with 1–3 ms uniform propagation, 1 Gbit/s links, and offered load $0.70C_{ref}$, where $C_{ref} = 21923.1$ requests/s is the planning proxy. The parent count is two, the minimum for MBSC and the fixed count for its control. Valid completion divides timely completions of valid cohort requests by all valid cohort requests; invalid-target endorsement divides timely invalid completions by all invalid cohort requests. The request deadline is 200 ms and the endorsement threshold is $c = 4$. A dash indicates that no incorrect outputs were submitted, leaving no invalid-target denominator. CPU/request is cohort-attributed CPU time per submitted request, including failures.

Under all five behaviors in Table X, MBSC completes more valid requests than fixed-2, 21.88%–46.71% versus 12.54%–28.29%. As in the benign comparison, this same-k contrast includes the additional references that coverage induces. The dash indicates that no incorrect own outputs were submitted, so the invalid-target denominator is zero. Both wrong-output scenarios have 521 invalid cohort targets across their five paired runs. Targeted blind approval yields 96 invalid timely endorsements for MBSC and 47 for fixed-2. The table reports means of five within-run fractions, not a pooled fraction. The simulated unsafe issuers still remove each drawn candidate from their local pool. The receiving rules impose no per-identity target quota in these runs. Because the extra coverage references also come from unsafe issuers, protection against this attack comes from the admission cap of main-manuscript Corollary 3, under which $c = M + 1$ excludes invalid endorsement. Duplicate-parent and insufficient-coverage behaviors can submit a correct own output with a malformed or noncompliant endorsement set; the output-valid completion denominator retains those requests.

The observable-rule attacks can also be checked directly against actor-specific outcomes. Each attack/method setting has 104.2 ± 9.36 unsafe cohort requests per run, all issued, where $\pm$ denotes sample SD across five runs. Duplicate-parent requests have zero timely completions in both methods; insufficient-coverage requests have zero in MBSC and a mean completion fraction of $8.49 \pm 2.18\%$ in fixed-2. Whole-run duplicate-reference rejection counts are 14952.0 ± 488.2 per run in both methods; coverage-rejection counts are 14952.0 ± 488.2 for MBSC and zero for fixed-2. These event counts include warm-up and background traffic and repeated observations at multiple nodes, including the sender. They are not counts of distinct rejected requests or per-request admission rates. Issuance and receiver admission are distinct events.

In the skip-with-correct-output scenario, the unsafe nodes incur zero modeled WASM fuel, yet the mean fractions of their requests reaching the threshold are $29.94 \pm 5.06\%$ for MBSC and $18.10 \pm 3.02\%$ for fixed-2 across five runs. These fractions use unsafe requests as the denominator and differ from the all-valid-request completion column in Table X. They provide a direct modeled example of successful admission without actual re-execution; they come from the calibrated discrete-event model of Appendix V-B.

E. Scope of Evaluation

This subsection expands the empirical scope and conditional-certificate interpretation in main-manuscript Section V-C.

The experiments test computation, observable checks, and finite-cohort behavior under the compared validation rules. Their topology, control cost, serialization length, cached-code assumption, and workload distribution are explicit model choices. The evaluation isolates the validation rules from identity and rate control, cryptographic signature costs, and ledger consensus.

The finite Poisson sums and selection certificates in Appendix IV are analytical calculations under stated assumptions. Poisson request arrivals do not establish a Poisson process of successful, locally available honest messages. Configured propagation ranges and average CPU utilization do not establish hard bounds on total delivery or issuance delay. Similarly, local pool removal by the simulated attackers does not establish an externally enforced per-identity target quota. The conditional design example therefore illustrates the bounds without certifying the simulated deployment, and zero observed failures in a finite cohort do not establish zero failure probability.

APPENDIX VI
PROTOCOL LISTINGS AND ARCHITECTURE COMPARISON

This appendix supplies the detailed message fields and procedure listings for main-manuscript Sections III-A–III-B. The parent-selection loop appears in Appendix VI-E as Algorithm S3. The contracts in Appendix II apply to every procedure below.

A. Deployment and Invocation Fields

Tables XI and XII specify the payloads for main-manuscript Section III-A. SC denotes smart contract, and `byte[]` denotes a Protobuf-serialized byte array. Deterministic encoding and profile semantics follow Appendix II-A. A Deployment is immutable and available before an Invocation attempts execution.

TABLE XI
SC DEPLOYMENT TRANSACTION PAYLOAD

Field	Type	Description
Payload Type	`uint32`	Protocol-local identifier (e.g., 70) used by MBSC-capable peers; illustrative, not a public-IOTA registry assignment.
SC Data	`byte[]`	Immutable WASM bytecode for the versioned execution profile.
Version	`uint8`	MBSC protocol and execution-profile version.

The Invocation transaction (Table XII) represents a verifiable execution request. It records the execution intent and a declaration of computational work. Key fields include *Invoked SC*, which points to the immutable Deployment transaction. The *Metered Work* field (`insCount`) declares the invocation's work under the versioned execution profile. A verifier treats it as a trusted weight only after reproducing the count by local re-execution. The *Output* field stores the execution result and a result digest for later re-execution checks. The *Timestamp* field stores a protocol issuance value returned by `protocolTime()`, covered by the issuer signature, and accepted only after network protocol-time validation.

TABLE XII
SC INVOCATION TRANSACTION PAYLOAD

Field	Type	Description
Payload Type	`uint32`	Protocol-local identifier (e.g., 71) used by MBSC-capable peers; illustrative, not a public-IOTA registry assignment.
Invoked SC	`Hash`	Reference to the immutable Deployment transaction.
Function	`string`	Exported WASM function name.
Metered Work	`uint32`	Declared versioned work units; locally trusted after re-execution.
Input	`byte[]`	Serialized function arguments.
Output	`byte[]`	Serialized execution result and digest.
Timestamp	`uint64`	Network-validated protocol issuance value used for two-sided parent-age eligibility.

B. Complete Invocation Attempt

Algorithm S1 expands the execute–select–issue workflow in main-manuscript Section III-B. The inputs $scID$, $func$, and $input$ identify the deployment, exported function, and arguments; pk and sk are the public and secret signing keys. The ledger's admission-work parameter is $Target$, the specified digest function is H, and $\perp$ means no result. The work bounds I_{min} and I_{max}, attempt duration $t_{attempt}$, and absolute local deadline D have the meanings given in main-manuscript Sections III-A–III-B.

The helper `execute` retrieves and runs the code within the work and time limits; `serialize` applies the profile's deterministic encoding. The call `tipSelection` invokes Algorithm S3. The constructor SCINVTX forms the payload in Table XII. The helper `canonicalEncode` fixes the unsigned body containing the ordered parents, public key, and complete payload, including its timestamp. The constructor TANGLEMSG preserves those bytes and adds the nonce. The helper $rateControlNonce(body, Target, D)$ prepares evidence valid for that exact body and $Target$, or fails by D. The subsequent signature covers the body and nonce. This is a ledger-adapter contract, not a measured implementation of a particular admission mechanism.

All retrieval, execution, waiting, preparation, signing, and submission calls respect D. An expired selected parent or failed own execution defers the attempt. The helper $submit(m, D)$ succeeds only if the local ledger interface accepts the signed message for dissemination before D, including timestamp and admission-evidence checks. If preparation makes the fixed timestamp inadmissible, submission fails without retiming the signed body. A later attempt that changes a bound field must prepare its nonce and signature again; its work and delay are counted. Local issuance establishes neither network delivery, semantic confirmation, nor ledger finality. Appendix II-B gives the full state and failure semantics.

Algorithm S1 A bounded invocation attempt

Input: $scID$, $func$, $input$, $Target$, pk, sk
Output: $status$, out

$D \leftarrow monotonicTime() + t_{attempt}$
$(ok, out, W_i) \leftarrow execute(scID, func, input, I_{max}, D)$
if $\neg ok$ or $W_i < I_{min}$ **then**
    return (DEFERRED, $\perp$)
end if
$(status, parents) \leftarrow tipSelection(W_i, D)$
if $status \neq$ SUCCESS **then**
    return (DEFERRED, $\perp$)
end if
$ts \leftarrow protocolTime()$
if any parent is ineligible at ts **then**
    return (DEFERRED, $\perp$)
end if
$scOut \leftarrow (serialize(out), H(serialize(out)))$
$tx \leftarrow$ SCINVTX$(scID, func, W_i, input, scOut, ts)$
$body \leftarrow canonicalEncode(parents, pk, tx)$
$(ok, nonce) \leftarrow rateControlNonce(body, Target, D)$
if $\neg ok$ **then**
    return (DEFERRED, $\perp$)
end if
$m \leftarrow$ TANGLEMSG$(body, nonce)$
sign m with sk within deadline D
if $monotonicTime() \geq D$ or signing failed **then**
    return (DEFERRED, $\perp$)
end if
$ok \leftarrow submit(m, D)$
if $\neg ok$ **then**
    return (DEFERRED, $\perp$)
end if
return (SUCCESS, out)

C. Candidate Re-execution

Algorithm S2 is called within Algorithm S3. Its input m is the candidate message and D is the current attempt deadline. The returned ok flag indicates successful verification, W is the reproduced work, and $reject$ indicates objective invalidity. A timeout is not sufficient to set $reject$. Admission checks precede re-execution of the candidate's own invocation; its parents are not recursively re-executed. The returned trusted weight is used only after output bytes, digest, and work count match their declarations. Appendix II-C explains why this does not prove the candidate issuer's private parent-verification history.

Algorithm S2 Re-execution of a candidate invocation

Input: candidate message m, deadline D
Output: ok, trusted work W, proven-invalid flag $reject$

if required data are unavailable before D **then**
  return $(\text{False}, 0, \text{False})$
end if
if structural admission checks on m fail **then**
  return $(\text{False}, 0, \text{True})$
end if
$tx \leftarrow m.payload$
$sc \leftarrow tx.invokedSC$; $f \leftarrow tx.invokedFunc$
$args \leftarrow tx.input$
$(ok, out, W) \leftarrow execute(sc, f, args, I_{max}, D)$
if $\neg ok$ **then**
  return $(\text{False}, 0, \text{False})$
end if
if $W < I_{min}$ **then**
  return $(\text{False}, 0, \text{True})$
end if
if $tx.output.result \neq serialize(out)$ or $tx.output.result_digest \neq H(serialize(out))$ or $tx.insCount \neq W$ **then**
  return $(\text{False}, 0, \text{True})$
end if
return $(\text{True}, W, \text{False})$

D. Execution and Acceptance Roles

Table XIII extends the architectural comparison in main-manuscript Sections II and V-A. It distinguishes the computation model, checking responsibility, acceptance event, and fee policy. Security and performance are established separately, through main-manuscript Theorem 1 and the matched experiments of Section V.

TABLE XIII
EXECUTION AND ACCEPTANCE ROLES ACROSS LEDGER ARCHITECTURES

Dimension	MBSC	ISC [20]	IOTA Rebased [21], [22]
Computation	Bounded, explicit-input stateless WASM tasks	Stateful Layer-2 Wasp chains	Object-based, stateful Layer-1 MoveVM execution
Checking responsibility	Compliant issuers re-execute selected earlier invocations	A designated, potentially rotating execution committee	A delegated-proof-of-stake validator set
Acceptance meaning	Application endorsement count; ledger stability is additional	Committee agreement on chain state, anchored to Layer 1	Validator execution and consensus for ledger state
Fee scope	No monetary payment rule added by reciprocal work; underlying ledger policy remains separate	Fee policy is configurable; private or corporate chains can be fee-free	Ledger transaction and execution policies are separate from the MBSC work rule

E. Bounded Reciprocal Parent Selection

Algorithm S3 specifies the work-coverage selection loop, storing the accumulated work in V. The helper $verifyResult(m, D)$ checks candidate message m and returns a success flag ok, reproduced work W, and an objective-invalidity flag $reject$. It re-executes the candidate's own invocation, without recursively executing ancestors. Only objectively invalid records are cached as rejected; a retrieval timeout alone does not establish invalidity. Appendices VI-B and VI-C give the complete invocation and verification algorithms, respectively; Appendix II-B specifies their helper contracts and state transitions.

Algorithm S3 Bounded reciprocal parent selection

Input: own work W_i, attempt deadline D
Output: $status$, $parents$

$parents \leftarrow \emptyset$; $V \leftarrow 0$; $draws \leftarrow 0$
$D_{pool} \leftarrow \min\{D, monotonicTime() + t_{pool}\}$
while $V < W_i$ or $|parents| < k_{min}$ **do**
    if $monotonicTime() \geq D$ or $draws \geq d_{max}$ **then**
        return $(\text{DEFERRED}, \emptyset)$
    end if
    if local eligible pool is empty **then**
        if $monotonicTime() \geq D_{pool}$ **then**
            return $(\text{DEFERRED}, \emptyset)$
        end if
        wait for a candidate, but no later than D_{pool}
        continue
    end if
    draw m uniformly; permanently remove it from the local pool
    $draws \leftarrow draws + 1$; $t \leftarrow protocolTime()$
    if $m.ID$ is selected, cached rejected, or ineligible at t **then**
        continue
    end if
    $(ok, W, reject) \leftarrow verifyResult(m, D)$
    if ok **then**
        append m to $parents$; $V \leftarrow V + W$
    else
        if $reject$ **then**
            cache $m.ID$ as objectively invalid
        end if
    end if
end while
return $(\text{SUCCESS}, parents)$

APPENDIX VII
EXPANDED PARENT-COUNT COMPARISON

A. Design and Replication

This comparison addresses the distinction between enforcing work coverage and supplying additional parent references. It uses MBSC with $k_{min} = 2$, Fixed-2, Fixed-3, and Rand-2/3. The fixed-count controls re-execute candidates using the same checks and selection rule, but stop at their specified number of valid parents. Rand-2/3 assigns each arriving request a target $K = 2 + B$, where B is a Bernoulli random variable with $\Pr(B = 1) = \Pr(B = 0) = 1/2$. The assignment precedes execution and does not use the request's task, own work, candidate pool, or eventual outcome. The receiver uses that request's same target count. Its expectation, $\mathrm{E}[K] = 2.5$, is a parent-count control, not an equality constraint on realized resources.

The seven offered loads, two networks, measured service demands, task mixture, node and queue limits, bootstrap procedure, and deadlines are those of Table II; the application threshold remains $c = 4$. Within each seed, all four methods share exogenous arrivals, issuers, tasks, and keyed link-delay draws. The random target uses a separate keyed random namespace, so changing the stopping rule does not advance the exogenous random stream. The methods can nevertheless reach different queue, pool, and delivery states. The finite warm-up and cohort interpretation of Appendix V-B still applies; these are not steady-state capacity estimates.

The design, complete load grid, and ten new seeds (2026101801–2026101810) were fixed before the campaign. All 560 planned runs completed; none was excluded or replaced. They are not pooled with the earlier 620 runs. Each run supplies one observation per endpoint, and all tables report means and sample SDs of the ten observations unless stated otherwise. For paired comparisons, the control's metric is subtracted from MBSC's within each seed before computing the mean and the pointwise 95% Student-t interval with nine degrees of freedom, using the formula in Appendix V-C. These intervals describe this finite grid and do not constitute multiplicity-adjusted superiority tests. An interval containing zero does not establish equivalence.

B. Accounting and Coverage Definitions

Every run follows 512 cohort submissions after 128 warm-up arrivals. Completion counts all valid submissions, including requests that never issue. Cohort CPU service and bytes are charged to the request performing the operation, as in Appendix V-C, and divided by all 512 submissions. This includes attempted work and traffic of unsuccessful requests. The primary tables' CPU therefore includes own execution, candidate re-execution, and modeled control service; it is not just the work of completed requests.

Let V be the fuel of accepted parents and W the catalog fuel of a request's own kernel. The ratio V/W and its quantiles are computed among issued cohort requests within each run, then summarized across runs. Nonissued requests are not classified as satisfying coverage; the issued fraction is reported alongside these conditional statistics. The quantity $\overline{k}$ is the within-run mean number of references per issued cohort request. References per submitted request instead divide the total emitted references by 512. The distinction preserves the cost and outcome of nonissued requests.

Candidate CPU and accepted-parent fuel per submission are reported separately. Across all 560 runs, dispatched candidate fuel not subsequently accepted was zero in this benign comparison. This observed zero does not remove failed-work accounting: accepted work performed by a request that never issues is still charged. It also does not describe the separate attack runs. Full-run CPU and bytes include warm-up, cohort, background, and drain activity. Their per-request columns divide by all generated requests in that run, before averaging across runs. Provisioning the shared bootstrap fixtures is separate; later re-execution of these fixtures is charged.

C. Results and Interpretation

Tables XIV–XV report all method/load/network cells. MBSC's completion difference from Fixed-2 is 8.89–15.98 percentage points. Against Fixed-3, the difference ranges from -26.50 to 17.50 points, with higher MBSC mean completion at the highest restricted-link load (47.40% versus 29.90%). Against Rand-2/3 it ranges from -5.59 to 0.00 points. The paired results in Tables XVI–XVII therefore attribute the completion differences to reference supply; the coverage results below isolate the property specific to MBSC. MBSC's cohort-attributed CPU per submission is higher than Fixed-2's and lower than Fixed-3's or Rand-2/3's in every cell. Its cohort traffic has the same ordering, whereas the full-run traffic per generated request can reverse the latter comparison. Both scopes are reported rather than treating parent count as an equal-resource budget.

MBSC has no observed $V < W$ event among its 71,611 issued cohort requests. The corresponding mean fractions for Fixed-2, Fixed-3, and Rand-2/3 are 17.32–19.32%, 10.55–12.42%, and 14.36–16.11%, respectively. Even a high mean ratio V/W can coexist with individual shortfalls; Tables XX–XXI therefore report within-run quantiles rather than only the mean. Tables XXIV–XXV separate the three own-kernel classes and the realized parent-count distribution. Beyond the minimum count, MBSC adds a parent only while accepted parent work is below the request's own work, so 78.5–81.8% of its issued requests use two parents and 10.6–13.0% use four or more. At almost the same mean count, Rand-2/3 leaves 35.2–40.2% of squared-sum requests below their own work. For the heaviest squared-sum kernel, Fixed-3 still leaves 27.18–33.32% of issued requests below their own work. Coverage and completion are thus distinct design criteria, and MBSC is the only tested policy that meets the coverage criterion for every task class.

TABLE XIV
COMPLETION, COHORT CPU, AND COVERAGE (NORMAL LINKS)

λ/C_{ref}	Method	Completion (%)	CPU (ms)	$\overline{k}$	$V < W$ (%)
0.20	MBSC	26.95 ± 1.40	1.056 ± 0.024	2.412 ± 0.046	0.00 ± 0.00
0.20	Fixed-2	15.02 ± 1.28	0.958 ± 0.022	2.000 ± 0.000	18.28 ± 1.60
0.20	Fixed-3	53.46 ± 1.68	1.211 ± 0.036	3.000 ± 0.000	10.96 ± 1.08
0.20	Rand-2/3	29.98 ± 1.67	1.073 ± 0.030	2.499 ± 0.028	15.45 ± 1.64
0.40	MBSC	26.11 ± 1.51	1.059 ± 0.027	2.421 ± 0.043	0.00 ± 0.00
0.40	Fixed-2	15.08 ± 1.35	0.964 ± 0.026	2.000 ± 0.000	17.32 ± 1.44
0.40	Fixed-3	48.91 ± 1.51	1.202 ± 0.042	3.000 ± 0.000	11.11 ± 1.21
0.40	Rand-2/3	29.55 ± 1.93	1.070 ± 0.029	2.499 ± 0.028	14.41 ± 1.06
0.55	MBSC	26.56 ± 2.49	1.052 ± 0.027	2.418 ± 0.041	0.00 ± 0.00
0.55	Fixed-2	15.31 ± 1.42	0.953 ± 0.027	2.000 ± 0.000	18.63 ± 1.35
0.55	Fixed-3	46.76 ± 2.22	1.198 ± 0.045	3.000 ± 0.000	11.05 ± 0.97
0.55	Rand-2/3	28.34 ± 2.13	1.078 ± 0.038	2.499 ± 0.028	14.37 ± 1.82
0.70	MBSC	25.64 ± 1.98	1.048 ± 0.027	2.418 ± 0.046	0.00 ± 0.00
0.70	Fixed-2	15.41 ± 2.05	0.948 ± 0.031	2.000 ± 0.000	18.96 ± 1.19
0.70	Fixed-3	47.29 ± 2.52	1.193 ± 0.028	3.000 ± 0.000	10.82 ± 1.52
0.70	Rand-2/3	28.73 ± 2.08	1.076 ± 0.036	2.499 ± 0.028	14.86 ± 1.70
0.85	MBSC	30.16 ± 3.65	1.043 ± 0.026	2.465 ± 0.059	0.00 ± 0.00
0.85	Fixed-2	16.70 ± 2.05	0.953 ± 0.025	2.000 ± 0.000	17.60 ± 0.76
0.85	Fixed-3	53.96 ± 4.24	1.192 ± 0.038	3.000 ± 0.000	10.96 ± 0.93
0.85	Rand-2/3	32.48 ± 3.13	1.070 ± 0.032	2.499 ± 0.028	15.10 ± 1.31
1.00	MBSC	35.64 ± 3.35	1.038 ± 0.028	2.457 ± 0.048	0.00 ± 0.00
1.00	Fixed-2	20.72 ± 2.75	0.945 ± 0.028	2.000 ± 0.000	18.11 ± 1.55
1.00	Fixed-3	61.72 ± 4.24	1.194 ± 0.041	3.000 ± 0.000	11.37 ± 1.26
1.00	Rand-2/3	40.16 ± 4.14	1.073 ± 0.025	2.499 ± 0.028	14.39 ± 1.61
1.20	MBSC	43.57 ± 3.34	1.021 ± 0.024	2.493 ± 0.062	0.00 ± 0.00
1.20	Fixed-2	28.59 ± 3.95	0.946 ± 0.030	2.000 ± 0.000	18.77 ± 1.44
1.20	Fixed-3	49.49 ± 6.87	1.169 ± 0.037	3.000 ± 0.000	12.11 ± 1.83
1.20	Rand-2/3	49.16 ± 3.91	1.059 ± 0.025	2.499 ± 0.027	15.54 ± 1.30

All entries are mean $\pm$ sample SD over ten runs. CPU is service per submitted cohort request; $\overline{k}$ is parents per issued request; $V < W$ is the percentage of issued requests below their own work.

TABLE XV
COMPLETION, COHORT CPU, AND COVERAGE (RESTRICTED LINKS)

λ/C_{ref}	Method	Completion (%)	CPU (ms)	$\overline{k}$	$V < W$ (%)
0.20	MBSC	23.67 ± 2.51	1.057 ± 0.030	2.412 ± 0.048	0.00 ± 0.00
0.20	Fixed-2	14.79 ± 1.60	0.969 ± 0.026	2.000 ± 0.000	17.85 ± 1.55
0.20	Fixed-3	40.21 ± 2.52	1.206 ± 0.038	3.000 ± 0.000	10.61 ± 1.53
0.20	Rand-2/3	26.27 ± 2.34	1.084 ± 0.028	2.499 ± 0.028	14.36 ± 1.49
0.40	MBSC	25.66 ± 3.47	1.055 ± 0.021	2.412 ± 0.046	0.00 ± 0.00
0.40	Fixed-2	16.29 ± 1.20	0.960 ± 0.025	2.000 ± 0.000	18.18 ± 1.59
0.40	Fixed-3	40.59 ± 2.12	1.206 ± 0.033	3.000 ± 0.000	10.55 ± 1.87
0.40	Rand-2/3	28.30 ± 3.37	1.083 ± 0.026	2.499 ± 0.028	14.51 ± 1.38
0.55	MBSC	28.89 ± 2.47	1.060 ± 0.020	2.424 ± 0.040	0.00 ± 0.00
0.55	Fixed-2	18.69 ± 2.82	0.960 ± 0.016	2.000 ± 0.000	19.06 ± 1.73
0.55	Fixed-3	46.93 ± 1.83	1.210 ± 0.032	3.000 ± 0.000	11.54 ± 1.89
0.55	Rand-2/3	31.97 ± 2.56	1.083 ± 0.023	2.499 ± 0.028	14.77 ± 1.15
0.70	MBSC	34.12 ± 1.94	1.068 ± 0.023	2.447 ± 0.034	0.00 ± 0.00
0.70	Fixed-2	21.15 ± 1.41	0.962 ± 0.020	2.000 ± 0.000	19.08 ± 1.42
0.70	Fixed-3	53.98 ± 2.15	1.215 ± 0.030	3.000 ± 0.000	11.19 ± 1.23
0.70	Rand-2/3	37.36 ± 3.02	1.090 ± 0.020	2.499 ± 0.028	16.11 ± 1.29
0.85	MBSC	40.04 ± 3.15	1.076 ± 0.025	2.477 ± 0.047	0.00 ± 0.00
0.85	Fixed-2	25.51 ± 2.36	0.967 ± 0.022	2.000 ± 0.000	19.10 ± 1.02
0.85	Fixed-3	62.23 ± 2.67	1.231 ± 0.026	3.000 ± 0.000	12.42 ± 1.25
0.85	Rand-2/3	43.91 ± 3.34	1.098 ± 0.024	2.499 ± 0.028	15.53 ± 1.36
1.00	MBSC	47.07 ± 3.55	1.077 ± 0.018	2.454 ± 0.067	0.00 ± 0.00
1.00	Fixed-2	31.09 ± 2.22	0.972 ± 0.019	2.000 ± 0.000	18.52 ± 1.13
1.00	Fixed-3	60.92 ± 4.85	1.244 ± 0.030	3.000 ± 0.000	11.94 ± 1.99
1.00	Rand-2/3	50.25 ± 2.97	1.107 ± 0.025	2.499 ± 0.028	15.70 ± 1.64
1.20	MBSC	47.40 ± 2.96	1.077 ± 0.021	2.441 ± 0.041	0.00 ± 0.00
1.20	Fixed-2	36.88 ± 1.80	0.972 ± 0.019	2.000 ± 0.000	19.32 ± 1.12
1.20	Fixed-3	29.90 ± 4.69	1.237 ± 0.031	3.000 ± 0.000	12.06 ± 1.10
1.20	Rand-2/3	47.40 ± 3.47	1.104 ± 0.027	2.499 ± 0.028	15.81 ± 1.56

All entries are mean $\pm$ sample SD over ten runs. CPU is service per submitted cohort request; $\overline{k}$ is parents per issued request; $V < W$ is the percentage of issued requests below their own work.

TABLE XVI
PAIRED DIFFERENCES: MBSC MINUS EACH CONTROL (NORMAL LINKS)

λ/C_{ref}	Control	Δ completion (pp)	Δ CPU (ms)
0.20	Fixed-2	$11.93\ [10.76, 13.11]$	$0.098\ [0.089, 0.107]$
0.20	Fixed-3	$-26.50\ [-27.28, -25.73]$	$-0.154\ [-0.167, -0.142]$
0.20	Rand-2/3	$-3.03\ [-4.54, -1.52]$	$-0.017\ [-0.031, -0.003]$
0.40	Fixed-2	$11.04\ [9.35, 12.72]$	$0.096\ [0.084, 0.107]$
0.40	Fixed-3	$-22.79\ [-24.39, -21.20]$	$-0.142\ [-0.157, -0.128]$
0.40	Rand-2/3	$-3.44\ [-5.24, -1.63]$	$-0.011\ [-0.018, -0.005]$
0.55	Fixed-2	$11.25\ [9.93, 12.57]$	$0.099\ [0.085, 0.113]$
0.55	Fixed-3	$-20.20\ [-21.98, -18.41]$	$-0.146\ [-0.167, -0.125]$
0.55	Rand-2/3	$-1.78\ [-3.67, 0.12]$	$-0.025\ [-0.044, -0.007]$
0.70	Fixed-2	$10.23\ [8.76, 11.70]$	$0.100\ [0.091, 0.108]$
0.70	Fixed-3	$-21.64\ [-23.68, -19.60]$	$-0.145\ [-0.162, -0.129]$
0.70	Rand-2/3	$-3.09\ [-4.42, -1.75]$	$-0.029\ [-0.041, -0.016]$
0.85	Fixed-2	$13.46\ [11.77, 15.15]$	$0.090\ [0.084, 0.097]$
0.85	Fixed-3	$-23.81\ [-25.42, -22.20]$	$-0.149\ [-0.162, -0.135]$
0.85	Rand-2/3	$-2.32\ [-3.48, -1.17]$	$-0.027\ [-0.037, -0.018]$
1.00	Fixed-2	$14.92\ [13.35, 16.49]$	$0.093\ [0.075, 0.110]$
1.00	Fixed-3	$-26.07\ [-29.78, -22.37]$	$-0.156\ [-0.176, -0.136]$
1.00	Rand-2/3	$-4.51\ [-6.65, -2.37]$	$-0.035\ [-0.050, -0.021]$
1.20	Fixed-2	$14.98\ [13.52, 16.44]$	$0.074\ [0.063, 0.086]$
1.20	Fixed-3	$-5.92\ [-12.06, 0.22]$	$-0.148\ [-0.164, -0.133]$
1.20	Rand-2/3	$-5.59\ [-7.42, -3.76]$	$-0.038\ [-0.049, -0.028]$

Entries are mean paired difference [pointwise 95% Student-t interval], from ten paired runs. The unit pp denotes percentage points; CPU is service per submitted cohort request.

TABLE XVII
PAIRED DIFFERENCES: MBSC MINUS EACH CONTROL (RESTRICTED LINKS)

λ/C_{ref}	Control	Δ completion (pp)	Δ CPU (ms)
0.20	Fixed-2	$8.89\ [6.81, 10.96]$	$0.088\ [0.075, 0.101]$
0.20	Fixed-3	$-16.54\ [-18.10, -14.98]$	$-0.149\ [-0.165, -0.133]$
0.20	Rand-2/3	$-2.60\ [-4.46, -0.74]$	$-0.027\ [-0.035, -0.018]$
0.40	Fixed-2	$9.38\ [6.97, 11.78]$	$0.096\ [0.087, 0.104]$
0.40	Fixed-3	$-14.92\ [-16.32, -13.52]$	$-0.151\ [-0.163, -0.139]$
0.40	Rand-2/3	$-2.64\ [-3.51, -1.76]$	$-0.027\ [-0.034, -0.021]$
0.55	Fixed-2	$10.20\ [8.71, 11.68]$	$0.100\ [0.094, 0.106]$
0.55	Fixed-3	$-18.05\ [-18.96, -17.13]$	$-0.150\ [-0.163, -0.137]$
0.55	Rand-2/3	$-3.09\ [-4.98, -1.19]$	$-0.023\ [-0.032, -0.013]$
0.70	Fixed-2	$12.97\ [11.57, 14.36]$	$0.106\ [0.093, 0.119]$
0.70	Fixed-3	$-19.86\ [-21.19, -18.54]$	$-0.147\ [-0.158, -0.135]$
0.70	Rand-2/3	$-3.24\ [-5.44, -1.05]$	$-0.021\ [-0.029, -0.013]$
0.85	Fixed-2	$14.53\ [12.99, 16.07]$	$0.109\ [0.099, 0.119]$
0.85	Fixed-3	$-22.19\ [-24.99, -19.38]$	$-0.155\ [-0.164, -0.146]$
0.85	Rand-2/3	$-3.87\ [-5.67, -2.06]$	$-0.022\ [-0.033, -0.011]$
1.00	Fixed-2	$15.98\ [14.23, 17.72]$	$0.105\ [0.099, 0.112]$
1.00	Fixed-3	$-13.85\ [-17.62, -10.07]$	$-0.167\ [-0.179, -0.154]$
1.00	Rand-2/3	$-3.18\ [-5.32, -1.05]$	$-0.030\ [-0.042, -0.018]$
1.20	Fixed-2	$10.53\ [8.02, 13.03]$	$0.105\ [0.098, 0.113]$
1.20	Fixed-3	$17.50\ [13.53, 21.47]$	$-0.160\ [-0.171, -0.150]$
1.20	Rand-2/3	$0.00\ [-2.58, 2.58]$	$-0.027\ [-0.036, -0.017]$

Entries are mean paired difference [pointwise 95% Student-t interval], from ten paired runs. The unit pp denotes percentage points; CPU is service per submitted cohort request.

TABLE XVIII
FULL-RUN COSTS INCLUDING WARM-UP AND BACKGROUND TRAFFIC (NORMAL LINKS)

λ/C_{ref}	Method	CPU (s)	Bytes (MiB)	CPU/req. (ms)	Bytes/req.
0.20	MBSC	1.606 ± 0.028	10.801 ± 0.237	1.056 ± 0.021	7446.2 ± 31.7
0.20	Fixed-2	1.451 ± 0.029	10.212 ± 0.200	0.954 ± 0.020	7040.0 ± 0.0
0.20	Fixed-3	1.825 ± 0.040	11.604 ± 0.228	1.200 ± 0.030	8000.0 ± 0.0
0.20	Rand-2/3	1.627 ± 0.039	10.905 ± 0.217	1.070 ± 0.026	7517.8 ± 8.5
0.40	MBSC	2.521 ± 0.053	16.955 ± 0.343	1.055 ± 0.018	7440.9 ± 29.7
0.40	Fixed-2	2.282 ± 0.060	16.041 ± 0.299	0.955 ± 0.024	7040.0 ± 0.0
0.40	Fixed-3	2.866 ± 0.069	18.229 ± 0.340	1.200 ± 0.029	8000.0 ± 0.0
0.40	Rand-2/3	2.564 ± 0.064	17.127 ± 0.324	1.073 ± 0.025	7516.2 ± 9.6
0.55	MBSC	3.205 ± 0.067	21.589 ± 0.416	1.052 ± 0.017	7432.3 ± 23.2
0.55	Fixed-2	2.899 ± 0.055	20.450 ± 0.381	0.952 ± 0.016	7040.0 ± 0.0
0.55	Fixed-3	3.643 ± 0.081	23.238 ± 0.433	1.196 ± 0.025	8000.0 ± 0.0
0.55	Rand-2/3	3.270 ± 0.066	21.831 ± 0.413	1.074 ± 0.019	7515.4 ± 6.9
0.70	MBSC	3.888 ± 0.072	26.272 ± 0.454	1.050 ± 0.017	7440.7 ± 8.7
0.70	Fixed-2	3.521 ± 0.068	24.857 ± 0.423	0.951 ± 0.015	7040.0 ± 0.0
0.70	Fixed-3	4.422 ± 0.104	28.247 ± 0.481	1.194 ± 0.026	8000.0 ± 0.0
0.70	Rand-2/3	3.973 ± 0.074	26.540 ± 0.459	1.073 ± 0.018	7516.5 ± 7.8
0.85	MBSC	4.577 ± 0.054	30.910 ± 0.315	1.048 ± 0.014	7422.5 ± 31.2
0.85	Fixed-2	4.157 ± 0.061	29.317 ± 0.326	0.952 ± 0.015	7040.0 ± 0.0
0.85	Fixed-3	5.106 ± 0.043	30.565 ± 1.016	1.169 ± 0.016	7340.1 ± 248.3
0.85	Rand-2/3	4.682 ± 0.060	31.223 ± 0.363	1.072 ± 0.016	7497.7 ± 42.8
1.00	MBSC	5.072 ± 0.033	33.481 ± 0.451	1.007 ± 0.012	6970.3 ± 107.9
1.00	Fixed-2	4.758 ± 0.059	33.465 ± 0.288	0.945 ± 0.011	6966.6 ± 42.9
1.00	Fixed-3	5.408 ± 0.037	24.055 ± 1.865	1.074 ± 0.012	5008.8 ± 399.5
1.00	Rand-2/3	5.169 ± 0.038	32.404 ± 0.860	1.026 ± 0.012	6746.4 ± 205.4
1.20	MBSC	5.218 ± 0.041	32.695 ± 0.919	0.882 ± 0.011	5798.1 ± 182.7
1.20	Fixed-2	5.104 ± 0.029	34.333 ± 0.535	0.863 ± 0.010	6088.6 ± 122.2
1.20	Fixed-3	5.480 ± 0.040	14.760 ± 1.349	0.927 ± 0.012	2618.5 ± 255.1
1.20	Rand-2/3	5.381 ± 0.038	23.449 ± 1.892	0.910 ± 0.012	4159.2 ± 352.2

All entries are mean $\pm$ sample SD over ten runs. CPU and bytes include all simulated traffic; per-request costs divide by all generated requests. One MiB is 2^{20} bytes.

TABLE XIX
FULL-RUN COSTS INCLUDING WARM-UP AND BACKGROUND TRAFFIC (RESTRICTED LINKS)

λ/C_{ref}	Method	CPU (s)	Bytes (MiB)	CPU/req. (ms)	Bytes/req.
0.20	MBSC	1.604 ± 0.035	10.793 ± 0.216	1.055 ± 0.026	7440.8 ± 26.3
0.20	Fixed-2	1.459 ± 0.031	10.212 ± 0.200	0.960 ± 0.020	7040.0 ± 0.0
0.20	Fixed-3	1.832 ± 0.038	11.604 ± 0.228	1.205 ± 0.031	8000.0 ± 0.0
0.20	Rand-2/3	1.640 ± 0.035	10.905 ± 0.217	1.078 ± 0.025	7517.8 ± 8.5
0.40	MBSC	2.522 ± 0.053	16.934 ± 0.356	1.056 ± 0.017	7431.3 ± 24.6
0.40	Fixed-2	2.287 ± 0.052	16.041 ± 0.299	0.957 ± 0.019	7040.0 ± 0.0
0.40	Fixed-3	2.863 ± 0.067	18.229 ± 0.340	1.199 ± 0.026	8000.0 ± 0.0
0.40	Rand-2/3	2.578 ± 0.068	17.127 ± 0.324	1.079 ± 0.026	7516.2 ± 9.6
0.55	MBSC	3.223 ± 0.060	21.602 ± 0.414	1.058 ± 0.014	7436.6 ± 17.2
0.55	Fixed-2	2.916 ± 0.054	20.450 ± 0.381	0.957 ± 0.014	7040.0 ± 0.0
0.55	Fixed-3	3.656 ± 0.073	23.238 ± 0.433	1.201 ± 0.023	8000.0 ± 0.0
0.55	Rand-2/3	3.280 ± 0.073	21.831 ± 0.413	1.077 ± 0.019	7515.4 ± 6.9
0.70	MBSC	3.920 ± 0.070	26.299 ± 0.459	1.059 ± 0.017	7448.2 ± 16.6
0.70	Fixed-2	3.533 ± 0.075	24.857 ± 0.423	0.954 ± 0.016	7040.0 ± 0.0
0.70	Fixed-3	4.442 ± 0.090	28.247 ± 0.481	1.200 ± 0.020	8000.0 ± 0.0
0.70	Rand-2/3	3.990 ± 0.070	26.540 ± 0.459	1.078 ± 0.017	7516.5 ± 7.8
0.85	MBSC	4.603 ± 0.058	30.928 ± 0.351	1.054 ± 0.013	7426.7 ± 22.6
0.85	Fixed-2	4.167 ± 0.058	29.317 ± 0.326	0.954 ± 0.015	7040.0 ± 0.0
0.85	Fixed-3	5.147 ± 0.038	30.760 ± 0.912	1.179 ± 0.016	7387.5 ± 241.3
0.85	Rand-2/3	4.700 ± 0.062	31.215 ± 0.329	1.076 ± 0.014	7495.9 ± 36.5
1.00	MBSC	5.126 ± 0.038	33.687 ± 0.494	1.018 ± 0.013	7013.1 ± 116.2
1.00	Fixed-2	4.788 ± 0.056	33.496 ± 0.299	0.951 ± 0.014	6973.0 ± 38.9
1.00	Fixed-3	5.412 ± 0.032	23.888 ± 1.520	1.075 ± 0.012	4973.6 ± 328.2
1.00	Rand-2/3	5.204 ± 0.041	32.721 ± 0.783	1.033 ± 0.012	6812.6 ± 194.5
1.20	MBSC	5.347 ± 0.032	32.552 ± 1.072	0.904 ± 0.012	5773.2 ± 218.2
1.20	Fixed-2	5.181 ± 0.034	35.051 ± 0.661	0.876 ± 0.011	6215.9 ± 141.6
1.20	Fixed-3	5.417 ± 0.107	14.668 ± 1.694	0.916 ± 0.021	2601.5 ± 306.3
1.20	Rand-2/3	5.413 ± 0.027	24.448 ± 1.777	0.916 ± 0.011	4336.7 ± 336.7

All entries are mean $\pm$ sample SD over ten runs. CPU and bytes include all simulated traffic; per-request costs divide by all generated requests. One MiB is 2^{20} bytes.

TABLE XX
DISTRIBUTION OF THE WORK-COVERAGE RATIO AMONG ISSUED REQUESTS (NORMAL LINKS)

λ/C_{ref}	Method	Mean V/W	$q_{.05}$	$q_{.50}$	$q_{.95}$	Issued (%)
0.20	MBSC	6.255 ± 0.200	1.066 ± 0.000	3.024 ± 0.881	19.619 ± 0.000	100.00 ± 0.00
0.20	Fixed-2	6.045 ± 0.243	0.353 ± 0.000	2.341 ± 0.720	21.790 ± 4.578	100.00 ± 0.00
0.20	Fixed-3	9.205 ± 0.279	0.530 ± 0.117	4.629 ± 0.244	32.828 ± 1.746	100.00 ± 0.00
0.20	Rand-2/3	7.414 ± 0.286	0.363 ± 0.033	3.659 ± 0.242	29.181 ± 2.731	100.00 ± 0.00
0.40	MBSC	6.325 ± 0.208	1.066 ± 0.000	2.853 ± 0.899	20.705 ± 3.433	100.00 ± 0.00
0.40	Fixed-2	6.118 ± 0.257	0.331 ± 0.070	2.341 ± 0.720	21.790 ± 4.578	100.00 ± 0.00
0.40	Fixed-3	9.157 ± 0.358	0.508 ± 0.115	4.629 ± 0.244	34.181 ± 1.426	100.00 ± 0.00
0.40	Rand-2/3	7.397 ± 0.255	0.353 ± 0.000	3.636 ± 0.223	29.281 ± 2.801	100.00 ± 0.00
0.55	MBSC	6.184 ± 0.369	1.066 ± 0.000	2.512 ± 0.824	19.619 ± 0.000	100.00 ± 0.00
0.55	Fixed-2	6.004 ± 0.265	0.331 ± 0.070	2.171 ± 0.540	19.619 ± 0.000	100.00 ± 0.00
0.55	Fixed-3	9.068 ± 0.256	0.552 ± 0.115	4.552 ± 0.325	33.505 ± 1.746	100.00 ± 0.00
0.55	Rand-2/3	7.547 ± 0.399	0.386 ± 0.035	3.469 ± 0.532	30.028 ± 2.159	100.00 ± 0.00
0.70	MBSC	6.115 ± 0.324	1.066 ± 0.000	2.853 ± 0.899	19.619 ± 0.000	100.00 ± 0.00
0.70	Fixed-2	5.923 ± 0.329	0.353 ± 0.000	2.171 ± 0.540	19.619 ± 0.000	100.00 ± 0.00
0.70	Fixed-3	8.939 ± 0.257	0.574 ± 0.107	4.398 ± 0.399	33.505 ± 1.746	100.00 ± 0.00
0.70	Rand-2/3	7.500 ± 0.330	0.366 ± 0.028	3.713 ± 0.349	29.381 ± 2.865	100.00 ± 0.00
0.85	MBSC	6.062 ± 0.167	1.066 ± 0.000	2.512 ± 0.824	19.619 ± 0.000	100.00 ± 0.00
0.85	Fixed-2	6.046 ± 0.199	0.331 ± 0.070	2.000 ± 0.000	19.619 ± 0.000	100.00 ± 0.00
0.85	Fixed-3	8.879 ± 0.351	0.530 ± 0.117	4.629 ± 0.244	32.828 ± 1.746	100.00 ± 0.00
0.85	Rand-2/3	7.454 ± 0.317	0.353 ± 0.000	3.659 ± 0.242	30.228 ± 2.244	100.00 ± 0.00
1.00	MBSC	5.967 ± 0.137	1.066 ± 0.000	2.512 ± 0.824	20.705 ± 3.433	99.86 ± 0.19
1.00	Fixed-2	5.846 ± 0.424	0.309 ± 0.094	2.171 ± 0.540	21.790 ± 4.578	100.00 ± 0.00
1.00	Fixed-3	8.919 ± 0.394	0.552 ± 0.115	4.552 ± 0.325	32.828 ± 1.746	100.00 ± 0.00
1.00	Rand-2/3	7.423 ± 0.459	0.344 ± 0.056	3.517 ± 0.364	28.833 ± 3.364	100.00 ± 0.00
1.20	MBSC	5.781 ± 0.256	1.066 ± 0.000	2.000 ± 0.000	19.619 ± 0.000	99.04 ± 0.42
1.20	Fixed-2	5.847 ± 0.204	0.331 ± 0.070	2.341 ± 0.720	19.619 ± 0.000	100.00 ± 0.00
1.20	Fixed-3	8.631 ± 0.366	0.552 ± 0.115	4.398 ± 0.399	31.814 ± 1.069	98.28 ± 1.61
1.20	Rand-2/3	7.369 ± 0.292	0.360 ± 0.021	3.375 ± 0.524	29.281 ± 2.801	99.94 ± 0.19

Entries summarize ten run metrics as mean $\pm$ sample SD. The symbols $q_{.05}$, $q_{.50}$, and $q_{.95}$ denote the within-run 5th, 50th, and 95th percentiles of V/W among issued requests. Issued is the percentage of all cohort submissions that issue.

TABLE XXI
DISTRIBUTION OF THE WORK-COVERAGE RATIO AMONG ISSUED REQUESTS (RESTRICTED LINKS)

λ/C_{ref}	Method	Mean V/W	$q_{.05}$	$q_{.50}$	$q_{.95}$	Issued (%)
0.20	MBSC	6.290 ± 0.261	1.066 ± 0.000	3.024 ± 0.881	20.705 ± 3.433	100.00 ± 0.00
0.20	Fixed-2	6.250 ± 0.251	0.353 ± 0.000	2.683 ± 0.881	22.876 ± 5.244	100.00 ± 0.00
0.20	Fixed-3	9.186 ± 0.367	0.552 ± 0.115	4.552 ± 0.325	34.519 ± 1.069	100.00 ± 0.00
0.20	Rand-2/3	7.658 ± 0.271	0.360 ± 0.021	3.659 ± 0.242	30.976 ± 0.527	100.00 ± 0.00
0.40	MBSC	6.077 ± 0.293	1.066 ± 0.000	2.341 ± 0.720	19.619 ± 0.000	100.00 ± 0.00
0.40	Fixed-2	6.062 ± 0.234	0.353 ± 0.000	2.341 ± 0.720	20.705 ± 3.433	100.00 ± 0.00
0.40	Fixed-3	8.952 ± 0.381	0.552 ± 0.115	4.629 ± 0.244	33.167 ± 1.782	100.00 ± 0.00
0.40	Rand-2/3	7.396 ± 0.385	0.366 ± 0.028	3.517 ± 0.495	30.128 ± 2.204	100.00 ± 0.00
0.55	MBSC	6.067 ± 0.303	1.066 ± 0.000	2.341 ± 0.720	19.619 ± 0.000	100.00 ± 0.00
0.55	Fixed-2	5.972 ± 0.259	0.331 ± 0.070	2.341 ± 0.720	20.705 ± 3.433	100.00 ± 0.00
0.55	Fixed-3	8.807 ± 0.307	0.552 ± 0.115	4.629 ± 0.244	33.167 ± 1.782	100.00 ± 0.00
0.55	Rand-2/3	7.411 ± 0.405	0.353 ± 0.000	3.588 ± 0.318	30.128 ± 2.204	100.00 ± 0.00
0.70	MBSC	6.122 ± 0.245	1.066 ± 0.000	2.683 ± 0.881	19.619 ± 0.000	100.00 ± 0.00
0.70	Fixed-2	5.914 ± 0.372	0.331 ± 0.070	2.171 ± 0.540	20.705 ± 3.433	100.00 ± 0.00
0.70	Fixed-3	8.702 ± 0.384	0.574 ± 0.107	4.398 ± 0.399	32.152 ± 1.426	100.00 ± 0.00
0.70	Rand-2/3	7.389 ± 0.338	0.366 ± 0.028	3.423 ± 0.496	30.676 ± 0.422	100.00 ± 0.00
0.85	MBSC	6.005 ± 0.188	1.066 ± 0.000	2.000 ± 0.000	19.619 ± 0.000	100.00 ± 0.00
0.85	Fixed-2	5.835 ± 0.300	0.353 ± 0.000	2.000 ± 0.000	19.619 ± 0.000	100.00 ± 0.00
0.85	Fixed-3	8.593 ± 0.291	0.508 ± 0.115	4.629 ± 0.244	31.476 ± 0.000	100.00 ± 0.00
0.85	Rand-2/3	7.356 ± 0.247	0.353 ± 0.000	3.517 ± 0.364	29.281 ± 2.801	100.00 ± 0.00
1.00	MBSC	5.951 ± 0.164	1.066 ± 0.000	2.341 ± 0.720	19.619 ± 0.000	100.00 ± 0.00
1.00	Fixed-2	5.751 ± 0.253	0.331 ± 0.070	2.171 ± 0.540	19.619 ± 0.000	100.00 ± 0.00
1.00	Fixed-3	8.605 ± 0.129	0.508 ± 0.115	4.398 ± 0.399	32.490 ± 1.633	99.96 ± 0.12
1.00	Rand-2/3	7.325 ± 0.288	0.360 ± 0.021	3.424 ± 0.365	29.828 ± 2.048	100.00 ± 0.00
1.20	MBSC	5.923 ± 0.154	1.066 ± 0.000	2.341 ± 0.720	20.705 ± 3.433	99.75 ± 0.40
1.20	Fixed-2	5.881 ± 0.285	0.353 ± 0.000	2.171 ± 0.540	19.619 ± 0.000	100.00 ± 0.00
1.20	Fixed-3	8.713 ± 0.334	0.463 ± 0.094	4.552 ± 0.325	33.167 ± 1.782	99.10 ± 0.72
1.20	Rand-2/3	7.304 ± 0.288	0.353 ± 0.000	3.446 ± 0.515	29.281 ± 2.801	99.96 ± 0.12

Entries summarize ten run metrics as mean $\pm$ sample SD. The symbols $q_{.05}$, $q_{.50}$, and $q_{.95}$ denote the within-run 5th, 50th, and 95th percentiles of V/W among issued requests. Issued is the percentage of all cohort submissions that issue.

TABLE XXII
COHORT TRAFFIC AND CANDIDATE-VALIDATION COSTS PER SUBMISSION (NORMAL LINKS)

λ/C_{ref}	Method	Bytes/req.	Refs/req.	Parent fuel (10^3/req.)	Candidate CPU (ms/req.)
0.20	MBSC	7435.1 ± 44.3	2.412 ± 0.046	23098.93 ± 647.75	0.592 ± 0.016
0.20	Fixed-2	7040.0 ± 0.0	2.000 ± 0.000	19275.70 ± 616.46	0.494 ± 0.015
0.20	Fixed-3	8000.0 ± 0.0	3.000 ± 0.000	29114.83 ± 1218.63	0.746 ± 0.030
0.20	Rand-2/3	7519.2 ± 26.4	2.499 ± 0.028	23742.58 ± 955.04	0.609 ± 0.024
0.40	MBSC	7444.2 ± 41.3	2.421 ± 0.043	23208.59 ± 807.24	0.595 ± 0.020
0.40	Fixed-2	7040.0 ± 0.0	2.000 ± 0.000	19489.39 ± 804.37	0.500 ± 0.020
0.40	Fixed-3	8000.0 ± 0.0	3.000 ± 0.000	28764.28 ± 1390.68	0.737 ± 0.034
0.40	Rand-2/3	7519.2 ± 26.4	2.499 ± 0.028	23638.33 ± 852.13	0.606 ± 0.021
0.55	MBSC	7440.9 ± 39.3	2.418 ± 0.041	22930.01 ± 789.70	0.588 ± 0.020
0.55	Fixed-2	7040.0 ± 0.0	2.000 ± 0.000	19055.14 ± 813.24	0.489 ± 0.020
0.55	Fixed-3	8000.0 ± 0.0	3.000 ± 0.000	28620.56 ± 1495.38	0.734 ± 0.037
0.55	Rand-2/3	7519.2 ± 26.4	2.499 ± 0.028	23930.45 ± 1338.00	0.614 ± 0.033
0.70	MBSC	7441.2 ± 43.7	2.418 ± 0.046	22749.89 ± 831.68	0.584 ± 0.020
0.70	Fixed-2	7040.0 ± 0.0	2.000 ± 0.000	18855.26 ± 951.79	0.484 ± 0.024
0.70	Fixed-3	8000.0 ± 0.0	3.000 ± 0.000	28403.24 ± 925.81	0.729 ± 0.023
0.70	Rand-2/3	7519.2 ± 26.4	2.499 ± 0.028	23879.12 ± 1125.27	0.612 ± 0.028
0.85	MBSC	7486.6 ± 56.8	2.465 ± 0.059	22558.30 ± 730.06	0.579 ± 0.018
0.85	Fixed-2	7040.0 ± 0.0	2.000 ± 0.000	19070.39 ± 727.05	0.489 ± 0.018
0.85	Fixed-3	8000.0 ± 0.0	3.000 ± 0.000	28367.88 ± 1306.09	0.728 ± 0.032
0.85	Rand-2/3	7519.2 ± 26.4	2.499 ± 0.028	23637.17 ± 1020.93	0.606 ± 0.025
1.00	MBSC	7468.2 ± 46.5	2.453 ± 0.048	22355.11 ± 798.52	0.574 ± 0.020
1.00	Fixed-2	7040.0 ± 0.0	2.000 ± 0.000	18751.69 ± 947.61	0.481 ± 0.024
1.00	Fixed-3	8000.0 ± 0.0	3.000 ± 0.000	28468.86 ± 1460.32	0.730 ± 0.036
1.00	Rand-2/3	7519.2 ± 26.4	2.499 ± 0.028	23749.99 ± 849.55	0.609 ± 0.021
1.20	MBSC	7441.9 ± 77.6	2.470 ± 0.067	21712.13 ± 729.65	0.559 ± 0.018
1.20	Fixed-2	7040.0 ± 0.0	2.000 ± 0.000	18793.97 ± 926.50	0.482 ± 0.023
1.20	Fixed-3	7862.5 ± 128.6	2.948 ± 0.048	27611.83 ± 1337.32	0.709 ± 0.033
1.20	Rand-2/3	7514.6 ± 25.6	2.497 ± 0.026	23186.13 ± 816.63	0.595 ± 0.020

All entries are mean $\pm$ sample SD over ten runs, divided by all 512 cohort submissions. Refs are emitted references. Parent fuel includes accepted work of nonissued requests; candidate CPU includes all dispatched parent re-execution. No dispatched candidate fuel was unaccepted in these benign runs.

TABLE XXIII
COHORT TRAFFIC AND CANDIDATE-VALIDATION COSTS PER SUBMISSION (RESTRICTED LINKS)

λ/C_{ref}	Method	Bytes/req.	Refs/req.	Parent fuel (10^3/req.)	Candidate CPU (ms/req.)
0.20	MBSC	7435.1 ± 45.8	2.412 ± 0.048	23048.03 ± 866.80	0.591 ± 0.021
0.20	Fixed-2	7040.0 ± 0.0	2.000 ± 0.000	19660.61 ± 850.81	0.504 ± 0.021
0.20	Fixed-3	8000.0 ± 0.0	3.000 ± 0.000	28848.95 ± 1263.54	0.740 ± 0.031
0.20	Rand-2/3	7519.2 ± 26.4	2.499 ± 0.028	24098.22 ± 829.56	0.618 ± 0.020
0.40	MBSC	7435.2 ± 43.9	2.412 ± 0.046	22721.96 ± 626.51	0.583 ± 0.015
0.40	Fixed-2	7040.0 ± 0.0	2.000 ± 0.000	19085.75 ± 687.43	0.489 ± 0.017
0.40	Fixed-3	8000.0 ± 0.0	3.000 ± 0.000	28451.97 ± 1132.54	0.730 ± 0.028
0.40	Rand-2/3	7519.2 ± 26.4	2.499 ± 0.028	23772.66 ± 794.69	0.610 ± 0.020
0.55	MBSC	7446.7 ± 38.9	2.424 ± 0.040	22601.19 ± 592.76	0.580 ± 0.015
0.55	Fixed-2	7040.0 ± 0.0	2.000 ± 0.000	18877.25 ± 429.76	0.484 ± 0.011
0.55	Fixed-3	8000.0 ± 0.0	3.000 ± 0.000	28091.87 ± 1080.89	0.721 ± 0.027
0.55	Rand-2/3	7519.2 ± 26.4	2.499 ± 0.028	23453.28 ± 824.81	0.602 ± 0.020
0.70	MBSC	7469.6 ± 32.2	2.447 ± 0.034	22574.18 ± 697.13	0.580 ± 0.017
0.70	Fixed-2	7040.0 ± 0.0	2.000 ± 0.000	18731.69 ± 628.41	0.481 ± 0.016
0.70	Fixed-3	8000.0 ± 0.0	3.000 ± 0.000	27641.93 ± 1001.64	0.710 ± 0.025
0.70	Rand-2/3	7519.2 ± 26.4	2.499 ± 0.028	23339.70 ± 736.89	0.599 ± 0.018
0.85	MBSC	7498.1 ± 44.7	2.477 ± 0.047	22471.16 ± 703.39	0.577 ± 0.017
0.85	Fixed-2	7040.0 ± 0.0	2.000 ± 0.000	18647.47 ± 628.18	0.478 ± 0.016
0.85	Fixed-3	8000.0 ± 0.0	3.000 ± 0.000	27557.85 ± 910.87	0.707 ± 0.023
0.85	Rand-2/3	7519.2 ± 26.4	2.499 ± 0.028	23212.11 ± 779.89	0.596 ± 0.019
1.00	MBSC	7475.8 ± 64.4	2.454 ± 0.067	22279.16 ± 506.97	0.572 ± 0.013
1.00	Fixed-2	7040.0 ± 0.0	2.000 ± 0.000	18667.81 ± 543.96	0.479 ± 0.013
1.00	Fixed-3	7996.9 ± 9.9	2.999 ± 0.004	27621.01 ± 1043.92	0.709 ± 0.026
1.00	Rand-2/3	7519.2 ± 26.4	2.499 ± 0.028	23230.48 ± 868.43	0.596 ± 0.022
1.20	MBSC	7444.0 ± 31.0	2.434 ± 0.035	22271.21 ± 615.26	0.572 ± 0.015
1.20	Fixed-2	7040.0 ± 0.0	2.000 ± 0.000	18642.12 ± 521.65	0.478 ± 0.013
1.20	Fixed-3	7928.1 ± 57.6	2.973 ± 0.022	27545.52 ± 1089.33	0.707 ± 0.027
1.20	Rand-2/3	7516.1 ± 31.2	2.498 ± 0.029	23091.60 ± 896.56	0.593 ± 0.022

All entries are mean $\pm$ sample SD over ten runs, divided by all 512 cohort submissions. Refs are emitted references. Parent fuel includes accepted work of nonissued requests; candidate CPU includes all dispatched parent re-execution. No dispatched candidate fuel was unaccepted in these benign runs.

TABLE XXIV
TASK-CONDITIONAL COVERAGE SHORTFALLS AND PARENT-COUNT DISTRIBUTION (NORMAL LINKS)

λ/C_{ref}	Method	Alarm (%)	Access (%)	Energy (%)	P_2 (%)	P_3 (%)	$P_{\geq 4}$ (%)
0.20	MBSC	0.0 ± 0.0	0.0 ± 0.0	0.0 ± 0.0	81.4 ± 2.0	7.7 ± 1.6	11.0 ± 0.9
0.20	Fixed-2	0.0 ± 0.0	12.0 ± 3.3	42.7 ± 4.1	100.0 ± 0.0	0.0 ± 0.0	0.0 ± 0.0
0.20	Fixed-3	0.0 ± 0.0	3.9 ± 2.1	28.7 ± 1.5	0.0 ± 0.0	100.0 ± 0.0	0.0 ± 0.0
0.20	Rand-2/3	0.0 ± 0.0	8.6 ± 2.1	37.5 ± 4.3	50.1 ± 2.8	49.9 ± 2.8	0.0 ± 0.0
0.40	MBSC	0.0 ± 0.0	0.0 ± 0.0	0.0 ± 0.0	81.3 ± 2.1	7.3 ± 1.5	11.4 ± 1.1
0.40	Fixed-2	0.0 ± 0.0	11.6 ± 3.3	40.2 ± 3.2	100.0 ± 0.0	0.0 ± 0.0	0.0 ± 0.0
0.40	Fixed-3	0.0 ± 0.0	4.1 ± 1.8	29.2 ± 4.0	0.0 ± 0.0	100.0 ± 0.0	0.0 ± 0.0
0.40	Rand-2/3	0.0 ± 0.0	7.5 ± 2.3	35.5 ± 2.8	50.1 ± 2.8	49.9 ± 2.8	0.0 ± 0.0
0.55	MBSC	0.0 ± 0.0	0.0 ± 0.0	0.0 ± 0.0	81.6 ± 1.4	6.6 ± 0.5	11.8 ± 1.3
0.55	Fixed-2	0.0 ± 0.0	10.9 ± 1.9	44.9 ± 5.1	100.0 ± 0.0	0.0 ± 0.0	0.0 ± 0.0
0.55	Fixed-3	0.0 ± 0.0	3.3 ± 1.2	29.7 ± 3.2	0.0 ± 0.0	100.0 ± 0.0	0.0 ± 0.0
0.55	Rand-2/3	0.0 ± 0.0	7.6 ± 3.6	35.5 ± 4.0	50.1 ± 2.8	49.9 ± 2.8	0.0 ± 0.0
0.70	MBSC	0.0 ± 0.0	0.0 ± 0.0	0.0 ± 0.0	81.4 ± 1.9	7.5 ± 1.8	11.1 ± 1.2
0.70	Fixed-2	0.0 ± 0.0	11.9 ± 2.2	44.9 ± 4.6	100.0 ± 0.0	0.0 ± 0.0	0.0 ± 0.0
0.70	Fixed-3	0.0 ± 0.0	4.3 ± 1.6	28.1 ± 4.3	0.0 ± 0.0	100.0 ± 0.0	0.0 ± 0.0
0.70	Rand-2/3	0.0 ± 0.0	7.3 ± 2.1	37.1 ± 5.3	50.1 ± 2.8	49.9 ± 2.8	0.0 ± 0.0
0.85	MBSC	0.0 ± 0.0	0.0 ± 0.0	0.0 ± 0.0	79.9 ± 2.0	7.5 ± 1.3	12.6 ± 1.4
0.85	Fixed-2	0.0 ± 0.0	10.9 ± 2.5	41.7 ± 2.5	100.0 ± 0.0	0.0 ± 0.0	0.0 ± 0.0
0.85	Fixed-3	0.0 ± 0.0	3.8 ± 1.3	28.9 ± 4.0	0.0 ± 0.0	100.0 ± 0.0	0.0 ± 0.0
0.85	Rand-2/3	0.0 ± 0.0	6.9 ± 1.5	38.1 ± 4.7	50.1 ± 2.8	49.9 ± 2.8	0.0 ± 0.0
1.00	MBSC	0.0 ± 0.0	0.0 ± 0.0	0.0 ± 0.0	79.9 ± 1.6	8.0 ± 1.3	12.0 ± 1.4
1.00	Fixed-2	0.0 ± 0.0	10.9 ± 3.5	43.1 ± 2.8	100.0 ± 0.0	0.0 ± 0.0	0.0 ± 0.0
1.00	Fixed-3	0.0 ± 0.0	3.6 ± 2.1	30.3 ± 4.0	0.0 ± 0.0	100.0 ± 0.0	0.0 ± 0.0
1.00	Rand-2/3	0.0 ± 0.0	7.0 ± 2.8	36.1 ± 4.3	50.1 ± 2.8	49.9 ± 2.8	0.0 ± 0.0
1.20	MBSC	0.0 ± 0.0	0.0 ± 0.0	0.0 ± 0.0	78.5 ± 2.3	8.5 ± 1.6	13.0 ± 1.9
1.20	Fixed-2	0.0 ± 0.0	10.6 ± 3.5	45.6 ± 3.4	100.0 ± 0.0	0.0 ± 0.0	0.0 ± 0.0
1.20	Fixed-3	0.0 ± 0.0	4.2 ± 1.5	32.0 ± 4.9	0.0 ± 0.0	100.0 ± 0.0	0.0 ± 0.0
1.20	Rand-2/3	0.0 ± 0.0	8.0 ± 2.5	38.5 ± 5.5	50.1 ± 2.7	49.9 ± 2.7	0.0 ± 0.0

All entries are mean $\pm$ sample SD over ten runs. Alarm, Access, and Energy denote the threshold, access-rule, and squared-sum kernels; each column is the percentage with $V < W$ conditional on that own kernel and issuance. P_2, P_3, and $P_{\geq 4}$ are percentages of all issued requests having two, three, or at least four parents, respectively.

TABLE XXV
TASK-CONDITIONAL COVERAGE SHORTFALLS AND PARENT-COUNT DISTRIBUTION (RESTRICTED LINKS)

λ/C_{ref}	Method	Alarm (%)	Access (%)	Energy (%)	P_2 (%)	P_3 (%)	$P_{\geq 4}$ (%)
0.20	MBSC	0.0 ± 0.0	0.0 ± 0.0	0.0 ± 0.0	81.2 ± 1.9	8.1 ± 1.2	10.6 ± 1.2
0.20	Fixed-2	0.0 ± 0.0	10.7 ± 3.4	42.7 ± 3.0	100.0 ± 0.0	0.0 ± 0.0	0.0 ± 0.0
0.20	Fixed-3	0.0 ± 0.0	4.4 ± 1.9	27.2 ± 3.6	0.0 ± 0.0	100.0 ± 0.0	0.0 ± 0.0
0.20	Rand-2/3	0.0 ± 0.0	7.7 ± 2.6	35.2 ± 3.9	50.1 ± 2.8	49.9 ± 2.8	0.0 ± 0.0
0.40	MBSC	0.0 ± 0.0	0.0 ± 0.0	0.0 ± 0.0	81.8 ± 1.8	7.2 ± 1.2	11.0 ± 1.3
0.40	Fixed-2	0.0 ± 0.0	10.5 ± 2.7	43.7 ± 4.6	100.0 ± 0.0	0.0 ± 0.0	0.0 ± 0.0
0.40	Fixed-3	0.0 ± 0.0	3.6 ± 1.8	28.0 ± 5.8	0.0 ± 0.0	100.0 ± 0.0	0.0 ± 0.0
0.40	Rand-2/3	0.0 ± 0.0	7.9 ± 2.2	35.6 ± 4.7	50.1 ± 2.8	49.9 ± 2.8	0.0 ± 0.0
0.55	MBSC	0.0 ± 0.0	0.0 ± 0.0	0.0 ± 0.0	80.9 ± 1.6	7.6 ± 1.4	11.6 ± 1.0
0.55	Fixed-2	0.0 ± 0.0	11.9 ± 2.4	45.1 ± 3.0	100.0 ± 0.0	0.0 ± 0.0	0.0 ± 0.0
0.55	Fixed-3	0.0 ± 0.0	3.2 ± 1.6	31.0 ± 4.9	0.0 ± 0.0	100.0 ± 0.0	0.0 ± 0.0
0.55	Rand-2/3	0.0 ± 0.0	6.5 ± 1.7	37.5 ± 3.7	50.1 ± 2.8	49.9 ± 2.8	0.0 ± 0.0
0.70	MBSC	0.0 ± 0.0	0.0 ± 0.0	0.0 ± 0.0	80.5 ± 1.2	7.5 ± 1.0	12.0 ± 0.9
0.70	Fixed-2	0.0 ± 0.0	11.3 ± 2.7	45.6 ± 3.3	100.0 ± 0.0	0.0 ± 0.0	0.0 ± 0.0
0.70	Fixed-3	0.0 ± 0.0	4.0 ± 1.4	29.3 ± 3.5	0.0 ± 0.0	100.0 ± 0.0	0.0 ± 0.0
0.70	Rand-2/3	0.0 ± 0.0	8.2 ± 2.3	39.9 ± 4.1	50.1 ± 2.8	49.9 ± 2.8	0.0 ± 0.0
0.85	MBSC	0.0 ± 0.0	0.0 ± 0.0	0.0 ± 0.0	79.6 ± 1.7	7.7 ± 1.3	12.8 ± 1.4
0.85	Fixed-2	0.0 ± 0.0	11.5 ± 2.0	45.5 ± 2.0	100.0 ± 0.0	0.0 ± 0.0	0.0 ± 0.0
0.85	Fixed-3	0.0 ± 0.0	3.7 ± 1.8	33.3 ± 4.1	0.0 ± 0.0	100.0 ± 0.0	0.0 ± 0.0
0.85	Rand-2/3	0.0 ± 0.0	7.6 ± 2.4	38.7 ± 3.5	50.1 ± 2.8	49.9 ± 2.8	0.0 ± 0.0
1.00	MBSC	0.0 ± 0.0	0.0 ± 0.0	0.0 ± 0.0	79.6 ± 2.2	8.0 ± 1.6	12.4 ± 1.9
1.00	Fixed-2	0.0 ± 0.0	11.0 ± 2.3	44.3 ± 4.1	100.0 ± 0.0	0.0 ± 0.0	0.0 ± 0.0
1.00	Fixed-3	0.0 ± 0.0	4.1 ± 2.3	31.6 ± 5.8	0.0 ± 0.0	100.0 ± 0.0	0.0 ± 0.0
1.00	Rand-2/3	0.0 ± 0.0	7.3 ± 2.0	39.4 ± 4.4	50.1 ± 2.8	49.9 ± 2.8	0.0 ± 0.0
1.20	MBSC	0.0 ± 0.0	0.0 ± 0.0	0.0 ± 0.0	80.6 ± 1.5	7.3 ± 1.1	12.1 ± 1.4
1.20	Fixed-2	0.0 ± 0.0	11.0 ± 2.8	46.6 ± 3.1	100.0 ± 0.0	0.0 ± 0.0	0.0 ± 0.0
1.20	Fixed-3	0.0 ± 0.0	4.2 ± 2.4	31.8 ± 3.9	0.0 ± 0.0	100.0 ± 0.0	0.0 ± 0.0
1.20	Rand-2/3	0.0 ± 0.0	7.0 ± 2.1	40.2 ± 4.9	50.1 ± 2.8	49.9 ± 2.8	0.0 ± 0.0

All entries are mean $\pm$ sample SD over ten runs. Alarm, Access, and Energy denote the threshold, access-rule, and squared-sum kernels; each column is the percentage with $V < W$ conditional on that own kernel and issuance. P_2, P_3, and $P_{\geq 4}$ are percentages of all issued requests having two, three, or at least four parents, respectively.

APPENDIX VIII
PARENT-COUNT PARAMETER SENSITIVITY

A. Design

This extension examines how the parent-count parameter changes completion and resource demand. MBSC(k) denotes MBSC with minimum count $k_{min} = k$ and work coverage, whereas Fixed-k stops at exactly k valid parents. We added MBSC(3), MBSC(4), and Fixed-4 to the four configurations in Appendix VII. The three added configurations use the same seven offered loads, two networks, and ten seeds as that comparison, giving 420 new runs and 980 runs in total with its 560 existing runs. The seven configurations form 98 configuration/load/network cells, with $n = 10$ independent seeded runs per cell. Inputs are paired within seed, load, and network; reuse across configurations does not increase n. The task mixture, service demands, resource caps, 128-arrival warm-up, 512-request cohort, follow-up, and deadlines remain unchanged, with $c = 4$ (Table II). All new runs were retained, with no exclusions or replacements.

The extension was selected after examining the four-method results and is an exploratory parameter study. It does not establish a global optimum or provide independent validation of a selected parameter. Matching nominal k does not match realized parent counts, work, CPU service, or traffic, because MBSC may need additional parents to satisfy coverage. Figure 6 compares MBSC(k) with Fixed-k for $k = 2, 3, 4$. Tables XXVI and XXVII retain all seven configurations. Paired differences use the ten within-seed differences and pointwise 95% Student-t intervals with nine degrees of freedom. These intervals do not establish multiplicity-adjusted superiority or equivalence. Table XXX gives the numerical values of all fourteen MBSC(3) minus Fixed-3 contrasts shown in main-manuscript Fig. 3.

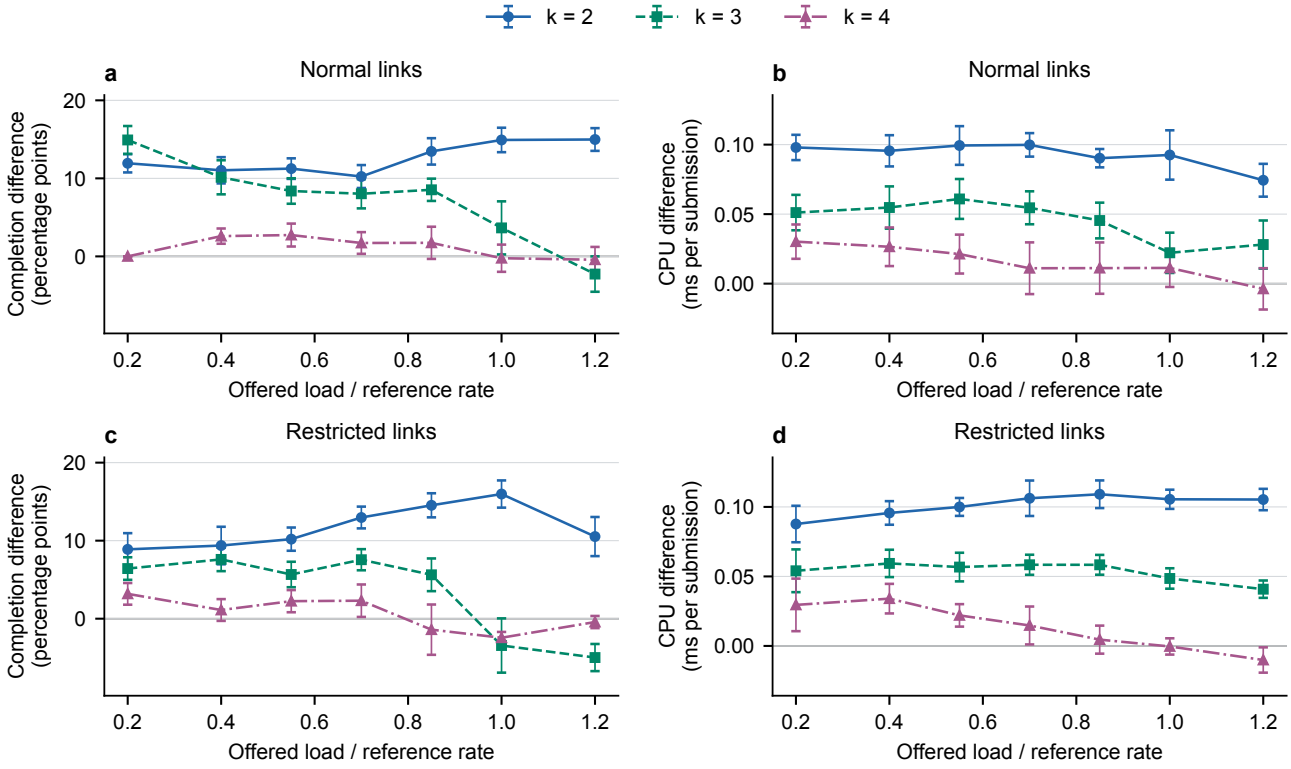


Fig. 6. Parent-count parameter sensitivity. Points show mean paired MBSC($\boldsymbol{k}$) minus Fixed-$\boldsymbol{k}$ differences over ten seeds; bars are pointwise 95% Student-$\boldsymbol{t}$ intervals. Panels (a,c) show timely completion differences in percentage points; (b,d) show cohort CPU differences in milliseconds per submitted request. Panels (a,b) use normal links and (c,d) restricted links. Each endpoint has 42 comparisons across three nominal counts and fourteen load/network settings. Lines connect sampled loads; zero denotes equal observed means.

B. Results and Scope

MBSC(3) had higher mean completion than MBSC(2) in thirteen of the fourteen settings, with 0.176–0.215 ms more cohort CPU service per submission in every setting. The highest restricted-link load reverses this completion ordering: at $1.20C_{\text{ref}}$, MBSC(2), MBSC(3), and MBSC(4) completed 47.40%, 24.92%, and 5.70% of submitted requests, respectively. The minimum parent count is therefore a load-dependent design parameter, and the smallest count serves the heaviest restricted-link load best.

At the same nominal count, MBSC(3) had higher mean completion than Fixed-3 in eleven settings, at an additional 0.022–0.061 ms of cohort CPU per submission in all settings. MBSC(4) had higher mean completion than Fixed-4 in eight settings, lower completion in five, and equal completion in one. Every issued MBSC request satisfied $V \geq W$, where V is accepted-parent fuel and W is own-task fuel. Fixed-4 still

left 6.17–7.69% of issued requests below this coverage level, averaged within each setting. These coverage statistics condition on issuance; nonissued requests remain in the completion and cost denominators.

Unlike the original four-method comparison, the added configurations incurred some dispatched candidate work whose result was not accepted. This work remains charged, as does work performed by requests that never issue. Tables XXVIII and XXIX report these costs alongside traffic, full-run totals, and deferrals. Full-run totals include warm-up and background work; cohort costs remain attributed to the request performing each operation. The finite warm-up, bootstrap, and follow-up interpretation in Appendix V-B therefore continues to apply. These results describe the tested parameter grid and workload; Appendix IX examines longer observation.

TABLE XXVI
PARENT-COUNT PARAMETER COMPARISON (NORMAL LINKS)

λ/C_{ref}	Configuration	F (%)	Cohort CPU (ms/req.)	$\overline{k}$	Issued (%)	$V < W$ (%)
0.20	MBSC(2)	26.95 ± 1.40	1.056 ± 0.024	2.412 ± 0.046	100.00 ± 0.00	0.00 ± 0.00
0.20	MBSC(3)	68.38 ± 1.55	1.262 ± 0.038	3.221 ± 0.029	100.00 ± 0.00	0.00 ± 0.00
0.20	MBSC(4)	100.00 ± 0.00	1.483 ± 0.046	4.110 ± 0.011	100.00 ± 0.00	0.00 ± 0.00
0.20	Fixed-2	15.02 ± 1.28	0.958 ± 0.022	2.000 ± 0.000	100.00 ± 0.00	18.28 ± 1.60
0.20	Fixed-3	53.46 ± 1.68	1.211 ± 0.036	3.000 ± 0.000	100.00 ± 0.00	10.96 ± 1.08
0.20	Fixed-4	100.00 ± 0.00	1.452 ± 0.046	4.000 ± 0.000	100.00 ± 0.00	6.37 ± 1.34
0.20	Rand-2/3	29.98 ± 1.67	1.073 ± 0.030	2.499 ± 0.028	100.00 ± 0.00	15.45 ± 1.64
0.40	MBSC(2)	26.11 ± 1.51	1.059 ± 0.027	2.421 ± 0.043	100.00 ± 0.00	0.00 ± 0.00
0.40	MBSC(3)	59.04 ± 2.62	1.256 ± 0.028	3.209 ± 0.044	100.00 ± 0.00	0.00 ± 0.00
0.40	MBSC(4)	98.11 ± 1.28	1.480 ± 0.040	4.110 ± 0.025	100.00 ± 0.00	0.00 ± 0.00
0.40	Fixed-2	15.08 ± 1.35	0.964 ± 0.026	2.000 ± 0.000	100.00 ± 0.00	17.32 ± 1.44
0.40	Fixed-3	48.91 ± 1.51	1.202 ± 0.042	3.000 ± 0.000	100.00 ± 0.00	11.11 ± 1.21
0.40	Fixed-4	95.51 ± 1.77	1.454 ± 0.051	4.000 ± 0.000	100.00 ± 0.00	6.52 ± 0.89
0.40	Rand-2/3	29.55 ± 1.93	1.070 ± 0.029	2.499 ± 0.028	100.00 ± 0.00	14.41 ± 1.06
0.55	MBSC(2)	26.56 ± 2.49	1.052 ± 0.027	2.418 ± 0.041	100.00 ± 0.00	0.00 ± 0.00
0.55	MBSC(3)	55.14 ± 3.46	1.259 ± 0.036	3.217 ± 0.027	100.00 ± 0.00	0.00 ± 0.00
0.55	MBSC(4)	85.47 ± 3.39	1.471 ± 0.044	4.120 ± 0.023	100.00 ± 0.00	0.00 ± 0.00
0.55	Fixed-2	15.31 ± 1.42	0.953 ± 0.027	2.000 ± 0.000	100.00 ± 0.00	18.63 ± 1.35
0.55	Fixed-3	46.76 ± 2.22	1.198 ± 0.045	3.000 ± 0.000	100.00 ± 0.00	11.05 ± 0.97
0.55	Fixed-4	82.73 ± 2.98	1.450 ± 0.045	4.000 ± 0.000	100.00 ± 0.00	6.43 ± 1.12
0.55	Rand-2/3	28.34 ± 2.13	1.078 ± 0.038	2.499 ± 0.028	100.00 ± 0.00	14.37 ± 1.82
0.70	MBSC(2)	25.64 ± 1.98	1.048 ± 0.027	2.418 ± 0.046	100.00 ± 0.00	0.00 ± 0.00
0.70	MBSC(3)	55.31 ± 3.72	1.247 ± 0.032	3.232 ± 0.034	100.00 ± 0.00	0.00 ± 0.00
0.70	MBSC(4)	82.85 ± 2.33	1.462 ± 0.039	4.105 ± 0.028	99.90 ± 0.10	0.00 ± 0.00
0.70	Fixed-2	15.41 ± 2.05	0.948 ± 0.031	2.000 ± 0.000	100.00 ± 0.00	18.96 ± 1.19
0.70	Fixed-3	47.29 ± 2.52	1.193 ± 0.028	3.000 ± 0.000	100.00 ± 0.00	10.82 ± 1.52
0.70	Fixed-4	81.13 ± 2.97	1.451 ± 0.048	4.000 ± 0.000	100.00 ± 0.00	6.27 ± 1.18
0.70	Rand-2/3	28.73 ± 2.08	1.076 ± 0.036	2.499 ± 0.028	100.00 ± 0.00	14.86 ± 1.70
0.85	MBSC(2)	30.16 ± 3.65	1.043 ± 0.026	2.465 ± 0.059	100.00 ± 0.00	0.00 ± 0.00
0.85	MBSC(3)	62.50 ± 3.35	1.237 ± 0.035	3.240 ± 0.030	99.96 ± 0.08	0.00 ± 0.00
0.85	MBSC(4)	77.70 ± 3.98	1.444 ± 0.044	4.105 ± 0.032	97.85 ± 1.37	0.00 ± 0.00
0.85	Fixed-2	16.70 ± 2.05	0.953 ± 0.025	2.000 ± 0.000	100.00 ± 0.00	17.60 ± 0.76
0.85	Fixed-3	53.96 ± 4.24	1.192 ± 0.038	3.000 ± 0.000	100.00 ± 0.00	10.96 ± 0.93
0.85	Fixed-4	75.96 ± 3.18	1.432 ± 0.046	4.000 ± 0.000	97.87 ± 1.68	6.53 ± 1.26
0.85	Rand-2/3	32.48 ± 3.13	1.070 ± 0.032	2.499 ± 0.028	100.00 ± 0.00	15.10 ± 1.31
1.00	MBSC(2)	35.64 ± 3.35	1.038 ± 0.028	2.457 ± 0.048	99.86 ± 0.19	0.00 ± 0.00
1.00	MBSC(3)	65.37 ± 4.26	1.216 ± 0.035	3.238 ± 0.037	98.98 ± 0.38	0.00 ± 0.00
1.00	MBSC(4)	50.18 ± 8.81	1.418 ± 0.039	4.097 ± 0.022	91.52 ± 3.21	0.00 ± 0.00
1.00	Fixed-2	20.72 ± 2.75	0.945 ± 0.028	2.000 ± 0.000	100.00 ± 0.00	18.11 ± 1.55
1.00	Fixed-3	61.72 ± 4.24	1.194 ± 0.041	3.000 ± 0.000	100.00 ± 0.00	11.37 ± 1.26
1.00	Fixed-4	50.41 ± 8.14	1.407 ± 0.041	4.000 ± 0.000	91.99 ± 3.29	6.52 ± 1.41
1.00	Rand-2/3	40.16 ± 4.14	1.073 ± 0.025	2.499 ± 0.028	100.00 ± 0.00	14.39 ± 1.61
1.20	MBSC(2)	43.57 ± 3.34	1.021 ± 0.024	2.493 ± 0.062	99.04 ± 0.42	0.00 ± 0.00
1.20	MBSC(3)	47.23 ± 6.20	1.197 ± 0.028	3.215 ± 0.036	96.05 ± 1.64	0.00 ± 0.00
1.20	MBSC(4)	20.27 ± 5.75	1.364 ± 0.040	4.080 ± 0.030	79.88 ± 4.53	0.00 ± 0.00
1.20	Fixed-2	28.59 ± 3.95	0.946 ± 0.030	2.000 ± 0.000	100.00 ± 0.00	18.77 ± 1.44
1.20	Fixed-3	49.49 ± 6.87	1.169 ± 0.037	3.000 ± 0.000	98.28 ± 1.61	12.11 ± 1.83
1.20	Fixed-4	20.70 ± 4.99	1.368 ± 0.040	4.000 ± 0.000	81.17 ± 4.15	6.84 ± 1.31
1.20	Rand-2/3	49.16 ± 3.91	1.059 ± 0.025	2.499 ± 0.027	99.94 ± 0.19	15.54 ± 1.30

Entries are mean $\pm$ sample SD over ten runs. F is timely completion among all 512 valid cohort submissions; CPU divides cohort-attributed service, including unsuccessful attempts, by all submissions. $\overline{k}$ is the mean parent count among issued requests; Issued uses all submissions, whereas $V < W$ uses issued requests only. V is accepted-parent fuel and W is own-task fuel. MBSC(k) has minimum parent count k and enforces $V \geq W$; Fixed-k issues exactly k parents. The arrival rate is λ and $C_{\mathrm{ref}} = 21923.1$ requests/s is the reference planning rate, not measured capacity.

TABLE XXVII
PARENT-COUNT PARAMETER COMPARISON (RESTRICTED LINKS)

λ/C_{ref}	Configuration	F (%)	Cohort CPU (ms/req.)	$\overline{k}$	Issued (%)	$V < W$ (%)
0.20	MBSC(2)	23.67 ± 2.51	1.057 ± 0.030	2.412 ± 0.048	100.00 ± 0.00	0.00 ± 0.00
0.20	MBSC(3)	46.64 ± 2.88	1.260 ± 0.033	3.225 ± 0.043	100.00 ± 0.00	0.00 ± 0.00
0.20	MBSC(4)	71.13 ± 2.35	1.482 ± 0.050	4.113 ± 0.021	100.00 ± 0.00	0.00 ± 0.00
0.20	Fixed-2	14.79 ± 1.60	0.969 ± 0.026	2.000 ± 0.000	100.00 ± 0.00	17.85 ± 1.55
0.20	Fixed-3	40.21 ± 2.52	1.206 ± 0.038	3.000 ± 0.000	100.00 ± 0.00	10.61 ± 1.53
0.20	Fixed-4	67.95 ± 1.93	1.453 ± 0.047	4.000 ± 0.000	100.00 ± 0.00	6.21 ± 1.04
0.20	Rand-2/3	26.27 ± 2.34	1.084 ± 0.028	2.499 ± 0.028	100.00 ± 0.00	14.36 ± 1.49
0.40	MBSC(2)	25.66 ± 3.47	1.055 ± 0.021	2.412 ± 0.046	100.00 ± 0.00	0.00 ± 0.00
0.40	MBSC(3)	48.18 ± 3.63	1.265 ± 0.031	3.240 ± 0.039	100.00 ± 0.00	0.00 ± 0.00
0.40	MBSC(4)	68.61 ± 2.70	1.492 ± 0.038	4.124 ± 0.022	100.00 ± 0.00	0.00 ± 0.00
0.40	Fixed-2	16.29 ± 1.20	0.960 ± 0.025	2.000 ± 0.000	100.00 ± 0.00	18.18 ± 1.59
0.40	Fixed-3	40.59 ± 2.12	1.206 ± 0.033	3.000 ± 0.000	100.00 ± 0.00	10.55 ± 1.87
0.40	Fixed-4	67.50 ± 2.73	1.458 ± 0.044	4.000 ± 0.000	100.00 ± 0.00	6.17 ± 1.20
0.40	Rand-2/3	28.30 ± 3.37	1.083 ± 0.026	2.499 ± 0.028	100.00 ± 0.00	14.51 ± 1.38
0.55	MBSC(2)	28.89 ± 2.47	1.060 ± 0.020	2.424 ± 0.040	100.00 ± 0.00	0.00 ± 0.00
0.55	MBSC(3)	52.60 ± 2.86	1.267 ± 0.029	3.234 ± 0.033	100.00 ± 0.00	0.00 ± 0.00
0.55	MBSC(4)	74.69 ± 1.72	1.486 ± 0.047	4.129 ± 0.040	100.00 ± 0.00	0.00 ± 0.00
0.55	Fixed-2	18.69 ± 2.82	0.960 ± 0.016	2.000 ± 0.000	100.00 ± 0.00	19.06 ± 1.73
0.55	Fixed-3	46.93 ± 1.83	1.210 ± 0.032	3.000 ± 0.000	100.00 ± 0.00	11.54 ± 1.89
0.55	Fixed-4	72.44 ± 1.47	1.464 ± 0.051	4.000 ± 0.000	100.00 ± 0.00	7.42 ± 1.39
0.55	Rand-2/3	31.97 ± 2.56	1.083 ± 0.023	2.499 ± 0.028	100.00 ± 0.00	14.77 ± 1.15
0.70	MBSC(2)	34.12 ± 1.94	1.068 ± 0.023	2.447 ± 0.034	100.00 ± 0.00	0.00 ± 0.00
0.70	MBSC(3)	61.54 ± 1.79	1.273 ± 0.032	3.241 ± 0.055	99.98 ± 0.06	0.00 ± 0.00
0.70	MBSC(4)	79.22 ± 4.87	1.494 ± 0.045	4.125 ± 0.034	99.43 ± 1.08	0.00 ± 0.00
0.70	Fixed-2	21.15 ± 1.41	0.962 ± 0.020	2.000 ± 0.000	100.00 ± 0.00	19.08 ± 1.42
0.70	Fixed-3	53.98 ± 2.15	1.215 ± 0.030	3.000 ± 0.000	100.00 ± 0.00	11.19 ± 1.23
0.70	Fixed-4	76.91 ± 4.56	1.480 ± 0.052	4.000 ± 0.000	99.75 ± 0.80	7.20 ± 1.54
0.70	Rand-2/3	37.36 ± 3.02	1.090 ± 0.020	2.499 ± 0.028	100.00 ± 0.00	16.11 ± 1.29
0.85	MBSC(2)	40.04 ± 3.15	1.076 ± 0.025	2.477 ± 0.047	100.00 ± 0.00	0.00 ± 0.00
0.85	MBSC(3)	67.85 ± 3.19	1.289 ± 0.029	3.276 ± 0.063	99.90 ± 0.14	0.00 ± 0.00
0.85	MBSC(4)	50.29 ± 8.54	1.479 ± 0.037	4.119 ± 0.023	94.98 ± 3.24	0.00 ± 0.00
0.85	Fixed-2	25.51 ± 2.36	0.967 ± 0.022	2.000 ± 0.000	100.00 ± 0.00	19.10 ± 1.02
0.85	Fixed-3	62.23 ± 2.67	1.231 ± 0.026	3.000 ± 0.000	100.00 ± 0.00	12.42 ± 1.25
0.85	Fixed-4	51.70 ± 7.64	1.475 ± 0.042	4.000 ± 0.000	96.78 ± 2.93	7.44 ± 1.73
0.85	Rand-2/3	43.91 ± 3.34	1.098 ± 0.024	2.499 ± 0.028	100.00 ± 0.00	15.53 ± 1.36
1.00	MBSC(2)	47.07 ± 3.55	1.077 ± 0.018	2.454 ± 0.067	100.00 ± 0.00	0.00 ± 0.00
1.00	MBSC(3)	57.48 ± 5.08	1.293 ± 0.024	3.254 ± 0.031	99.39 ± 0.44	0.00 ± 0.00
1.00	MBSC(4)	18.59 ± 4.97	1.458 ± 0.037	4.109 ± 0.023	86.88 ± 3.35	0.00 ± 0.00
1.00	Fixed-2	31.09 ± 2.22	0.972 ± 0.019	2.000 ± 0.000	100.00 ± 0.00	18.52 ± 1.13
1.00	Fixed-3	60.92 ± 4.85	1.244 ± 0.030	3.000 ± 0.000	99.96 ± 0.12	11.94 ± 1.99
1.00	Fixed-4	21.04 ± 4.40	1.459 ± 0.038	4.000 ± 0.000	89.59 ± 3.21	7.69 ± 1.48
1.00	Rand-2/3	50.25 ± 2.97	1.107 ± 0.025	2.499 ± 0.028	100.00 ± 0.00	15.70 ± 1.64
1.20	MBSC(2)	47.40 ± 2.96	1.077 ± 0.021	2.441 ± 0.041	99.75 ± 0.40	0.00 ± 0.00
1.20	MBSC(3)	24.92 ± 4.45	1.278 ± 0.032	3.238 ± 0.039	96.74 ± 1.75	0.00 ± 0.00
1.20	MBSC(4)	5.70 ± 1.81	1.401 ± 0.039	4.111 ± 0.026	76.15 ± 3.67	0.00 ± 0.00
1.20	Fixed-2	36.88 ± 1.80	0.972 ± 0.019	2.000 ± 0.000	100.00 ± 0.00	19.32 ± 1.12
1.20	Fixed-3	29.90 ± 4.69	1.237 ± 0.031	3.000 ± 0.000	99.10 ± 0.72	12.06 ± 1.10
1.20	Fixed-4	6.13 ± 2.53	1.411 ± 0.037	4.000 ± 0.000	79.04 ± 3.90	7.51 ± 1.21
1.20	Rand-2/3	47.40 ± 3.47	1.104 ± 0.027	2.499 ± 0.028	99.96 ± 0.12	15.81 ± 1.56

Entries are mean $\pm$ sample SD over ten runs. F is timely completion among all 512 valid cohort submissions; CPU divides cohort-attributed service, including unsuccessful attempts, by all submissions. $\overline{k}$ is the mean parent count among issued requests; Issued uses all submissions, whereas $V < W$ uses issued requests only. V is accepted-parent fuel and W is own-task fuel. MBSC(k) has minimum parent count k and enforces $V \geq W$; Fixed-k issues exactly k parents. The arrival rate is λ and $C_{\mathrm{ref}} = 21923.1$ requests/s is the reference planning rate, not measured capacity.

TABLE XXVIII
ADDITIONAL-CONFIGURATION COSTS AND DEFERRALS (NORMAL LINKS)

λ/C_{ref}	Configuration	Cohort bytes (kB/req.)	Full CPU (s)	Full bytes (MiB)	Deferred (%)	Unaccepted fuel (10^3/req.)	Nonissued CPU (μs/req.)
0.20	MBSC(3)	8.212 ± 0.028	1.904 ± 0.038	11.930 ± 0.244	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00
0.20	MBSC(4)	9.066 ± 0.011	2.241 ± 0.053	13.158 ± 0.259	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00
0.20	Fixed-4	8.960 ± 0.000	2.199 ± 0.047	12.997 ± 0.255	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00
0.40	MBSC(3)	8.201 ± 0.042	2.996 ± 0.076	18.734 ± 0.367	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00
0.40	MBSC(4)	9.066 ± 0.024	3.528 ± 0.085	20.669 ± 0.408	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00
0.40	Fixed-4	8.960 ± 0.000	3.461 ± 0.093	20.416 ± 0.381	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00
0.55	MBSC(3)	8.208 ± 0.026	3.818 ± 0.071	23.864 ± 0.434	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00
0.55	MBSC(4)	9.075 ± 0.022	4.476 ± 0.090	26.348 ± 0.492	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00
0.55	Fixed-4	8.960 ± 0.000	4.400 ± 0.087	26.027 ± 0.484	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00
0.70	MBSC(3)	8.223 ± 0.033	4.630 ± 0.084	28.987 ± 0.508	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00
0.70	MBSC(4)	9.052 ± 0.028	5.260 ± 0.054	28.531 ± 1.087	0.10 ± 0.10	0.00 ± 0.00	1.16 ± 1.24
0.70	Fixed-4	8.960 ± 0.000	5.214 ± 0.064	28.373 ± 0.760	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00
0.85	MBSC(3)	8.227 ± 0.032	5.227 ± 0.035	30.533 ± 1.358	0.04 ± 0.08	0.00 ± 0.00	0.51 ± 1.10
0.85	MBSC(4)	8.866 ± 0.135	5.513 ± 0.038	20.504 ± 1.656	2.15 ± 1.37	0.00 ± 0.00	27.20 ± 17.20
0.85	Fixed-4	8.769 ± 0.151	5.516 ± 0.040	19.867 ± 1.467	2.13 ± 1.68	0.00 ± 0.00	26.14 ± 22.20
1.00	MBSC(3)	8.145 ± 0.052	5.409 ± 0.038	24.175 ± 1.643	1.02 ± 0.38	0.00 ± 0.00	11.99 ± 4.40
1.00	MBSC(4)	8.286 ± 0.300	5.490 ± 0.287	13.498 ± 2.084	8.48 ± 3.21	26.37 ± 33.47	104.73 ± 40.66
1.00	Fixed-4	8.242 ± 0.295	5.491 ± 0.375	12.670 ± 2.019	8.01 ± 3.29	12.90 ± 22.24	97.14 ± 37.66
1.20	MBSC(3)	7.882 ± 0.149	5.462 ± 0.046	14.967 ± 1.490	3.95 ± 1.64	5.27 ± 16.68	40.54 ± 16.41
1.20	MBSC(4)	7.219 ± 0.413	4.116 ± 0.125	5.942 ± 0.547	20.12 ± 4.53	28.38 ± 27.84	234.08 ± 49.92
1.20	Fixed-4	7.273 ± 0.372	4.128 ± 0.113	5.899 ± 0.496	18.83 ± 4.15	37.75 ± 34.34	223.94 ± 48.72

Entries are mean $\pm$ sample SD over ten runs. Cohort bytes, unaccepted fuel, and nonissued CPU divide their cohort totals by all 512 submissions. Unaccepted fuel is dispatched candidate fuel minus accepted-parent fuel; nonissued CPU is service charged to requests that never issue. Deferred is the percentage of all cohort submissions that never issue. Full CPU and bytes are totals over warm-up, cohort, background, and drain activity. One kB is 10^3 bytes and one MiB is 2^{20} bytes. Fuel is the pinned runtime's metering unit.

TABLE XXIX
ADDITIONAL-CONFIGURATION COSTS AND DEFERRALS (RESTRICTED LINKS)

λ/C_{ref}	Configuration	Cohort bytes (kB/req.)	Full CPU (s)	Full bytes (MiB)	Deferred (%)	Unaccepted fuel (10^3/req.)	Nonissued CPU (μs/req.)
0.20	MBSC(3)	8.216 ± 0.041	1.914 ± 0.037	11.916 ± 0.251	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00
0.20	MBSC(4)	9.069 ± 0.020	2.240 ± 0.039	13.161 ± 0.271	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00
0.20	Fixed-4	8.960 ± 0.000	2.203 ± 0.043	12.997 ± 0.255	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00
0.40	MBSC(3)	8.231 ± 0.037	3.009 ± 0.068	18.732 ± 0.363	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00
0.40	MBSC(4)	9.079 ± 0.021	3.543 ± 0.088	20.680 ± 0.395	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00
0.40	Fixed-4	8.960 ± 0.000	3.474 ± 0.080	20.416 ± 0.381	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00
0.55	MBSC(3)	8.225 ± 0.032	3.826 ± 0.065	23.883 ± 0.449	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00
0.55	MBSC(4)	9.084 ± 0.038	4.489 ± 0.075	26.348 ± 0.520	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00
0.55	Fixed-4	8.960 ± 0.000	4.409 ± 0.082	26.027 ± 0.484	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00
0.70	MBSC(3)	8.229 ± 0.054	4.637 ± 0.081	28.924 ± 0.490	0.02 ± 0.06	0.00 ± 0.00	0.20 ± 0.63
0.70	MBSC(4)	9.028 ± 0.104	5.282 ± 0.059	27.975 ± 1.140	0.57 ± 1.08	0.00 ± 0.00	7.64 ± 14.06
0.70	Fixed-4	8.937 ± 0.072	5.236 ± 0.065	28.244 ± 1.094	0.25 ± 0.80	0.00 ± 0.00	3.38 ± 10.69
0.85	MBSC(3)	8.257 ± 0.060	5.266 ± 0.030	30.141 ± 1.217	0.10 ± 0.14	0.00 ± 0.00	1.21 ± 1.71
0.85	MBSC(4)	8.619 ± 0.296	5.461 ± 0.039	18.703 ± 2.185	5.02 ± 3.24	9.37 ± 19.95	60.90 ± 37.41
0.85	Fixed-4	8.671 ± 0.262	5.472 ± 0.031	18.635 ± 1.733	3.22 ± 2.93	4.10 ± 12.95	40.30 ± 36.39
1.00	MBSC(3)	8.194 ± 0.050	5.425 ± 0.039	23.104 ± 1.536	0.61 ± 0.44	5.27 ± 13.07	7.35 ± 5.47
1.00	MBSC(4)	7.875 ± 0.309	5.249 ± 0.271	10.532 ± 2.454	13.12 ± 3.35	10.82 ± 20.33	153.30 ± 35.34
1.00	Fixed-4	8.027 ± 0.288	5.220 ± 0.244	10.068 ± 2.159	10.41 ± 3.21	18.74 ± 29.43	126.23 ± 39.81
1.20	MBSC(3)	7.960 ± 0.135	5.420 ± 0.094	14.699 ± 1.525	3.26 ± 1.75	0.00 ± 0.00	34.19 ± 16.60
1.20	MBSC(4)	6.905 ± 0.339	4.223 ± 0.121	5.253 ± 0.238	23.85 ± 3.67	65.24 ± 45.05	259.73 ± 43.63
1.20	Fixed-4	7.082 ± 0.350	4.363 ± 0.201	5.423 ± 0.444	20.96 ± 3.90	34.56 ± 33.19	233.67 ± 45.52

Entries are mean $\pm$ sample SD over ten runs. Cohort bytes, unaccepted fuel, and nonissued CPU divide their cohort totals by all 512 submissions. Unaccepted fuel is dispatched candidate fuel minus accepted-parent fuel; nonissued CPU is service charged to requests that never issue. Deferred is the percentage of all cohort submissions that never issue. Full CPU and bytes are totals over warm-up, cohort, background, and drain activity. One kB is 10^3 bytes and one MiB is 2^{20} bytes. Fuel is the pinned runtime's metering unit.

TABLE XXX
MBSC(3) MINUS FIXED-3 IN ALL FOURTEEN EARLY-COHORT SETTINGS

Offered load (C_{ref})	Completion difference (percentage points)	95% interval (percentage points)	CPU difference (ms/submission)	95% interval (ms/submission)
		Normal links		
0.20	14.9219	[13.1348, 16.7090]	0.05112	[0.03838, 0.06386]
0.40	10.1367	[7.9528, 12.3206]	0.05476	[0.03957, 0.06995]
0.55	8.3789	[6.7463, 10.0115]	0.06092	[0.04657, 0.07526]
0.70	8.0273	[6.1535, 9.9012]	0.05456	[0.04270, 0.06641]
0.85	8.5352	[7.1072, 9.9631]	0.04540	[0.03256, 0.05825]
1.00	3.6523	[0.2509, 7.0538]	0.02217	[0.00768, 0.03666]
1.20	−2.2656	[−4.5341, 0.0028]	0.02812	[0.01082, 0.04542]
		Restricted links		
0.20	6.4258	[4.9806, 7.8710]	0.05406	[0.03868, 0.06945]
0.40	7.5977	[6.0936, 9.1017]	0.05936	[0.04953, 0.06920]
0.55	5.6641	[4.0320, 7.2961]	0.05674	[0.04644, 0.06703]
0.70	7.5586	[6.2055, 8.9117]	0.05839	[0.05117, 0.06561]
0.85	5.6250	[3.5300, 7.7200]	0.05836	[0.05123, 0.06549]
1.00	−3.4375	[−6.9153, 0.0403]	0.04853	[0.04118, 0.05587]
1.20	−4.9805	[−6.7196, −3.2413]	0.04083	[0.03459, 0.04707]

Entries are means and pointwise descriptive 95% Student-t intervals across ten seed-paired runs, with nine degrees of freedom. Differences are MBSC(3) minus Fixed-3. Completion and CPU use all submitted cohort requests as their denominator. Positive values indicate higher completion or higher CPU demand for MBSC(3), respectively. Values are rounded for display; the negative-mean completion intervals at normal-link load 1.20 and restricted-link load 1.00 include zero. Later-window results are reported in Appendix IX.

APPENDIX IX
OBSERVATION-TIME SENSITIVITY

A. Continuous Trajectories and Observation Windows

This experiment examined whether completion and resource demand depended on when observation began and how long it lasted. It retained MBSC with $k_{min} = 2, 3, 4$, Fixed-2, Fixed-3, Fixed-4, and Rand-2/3, with $c = 4$ for all seven configurations. The catalog, equal task mixture, 21 serial node queues, resource caps, network models, 50 ms pre-issuance deadline, 200 ms request deadline, and 300 ms approval window were unchanged. The two settings were $0.20C_{ref}$ with normal links and $1.20C_{ref}$ with restricted links. These corners were selected from the preceding evaluation to examine low contention and the most demanding offered-load/network combination; they do not establish time robustness across all fourteen original environments.

Each setting used the ten original comparison seeds 2026101801–2026101810 for every configuration, giving 140 complete trajectories. The observation times and analysis windows were fixed before these new trajectories were run. Earlier results using these seeds were already known, so this is a paired sensitivity analysis rather than an independent test set. Within a seed and setting, all configurations shared request times, issuers, tasks, and keyed link-delay draws. The Rand-2/3 target used its separate keyed namespace. All planned trajectories were retained.

Each trajectory began at time zero with the same 42 bootstrap fixtures and generated requests continuously through 1.2 s. Measurement windows ended no later than 1.0 s, leaving at least 200 ms of continuing background input after their last possible arrival; all remaining events were then drained. An observation boundary neither stopped arrivals nor reset queues, candidate pools, fixtures, known messages, or random streams. Here, warm-up denotes the time from initialization to the selected observation start, without changing system actions.

Ten nonoverlapping 100 ms windows covered $[0, 1)$ s. The six primary windows were the Cartesian product of start times $w \in \{0.1, 0.5, 0.8\}$ s and durations $d \in \{0.1, 0.2\}$ s. Each window included requests created in $[w, w+d)$; their final outcomes were read after the full trajectory drained. Comparing starts at fixed d isolates the observation-start contrast, while comparing durations at fixed w isolates the duration contrast. The later 200 ms windows $[0.6, 0.8)$ and $[0.8, 1.0)$ s supplied a separate drift comparison. These ten atomic, six primary, and two drift views give 18 views per trajectory, with overlap between some views. They are repeated observations, not 18 independent experiments.

B. Endpoints, Monitoring, and Replication

For each window, the completion fraction divided valid requests reaching c within 200 ms of creation by all submitted requests created in that window. Issued fractions used the same denominator. Deferred, unconfirmed, and late requests remained in the counts. CPU service and bytes per submission also used all window submissions, including unsuccessful ones. They followed the request-attribution rule: own execution, selected-parent execution, and resulting-message admission were charged to the requesting child, including work dispatched after the observation window ended. Background children validating measured targets retained their own costs. Thus these window-attributed costs are distinct from full-trajectory costs, which include all generated requests and drain activity. The latter were also retained and normalized by all generated requests. Initial fixture provisioning was separate, whereas every later re-execution was charged.

Read-only monitoring sampled the trajectory every 10 ms through 1.2 s and once after drain. It inserted no service events, consumed no random draws, and did not change the simulated clock. Queue and pool statistics were averaged across nodes and then over the samples in the indicated clock window.

Eligible-pool size excluded expired or locally rejected entries, even when lazy cleanup left them in storage. Approval supply was counted when a new eligible child reference was actually observed at its target's issuer. The reported prethreshold credit rate counted such credits only while the generated target had fewer than c approvals; bootstrap targets were excluded. This rate concerns all generated targets observed during the clock window, irrespective of their creation cohort. It is not a certified honest-opportunity intensity.

Published messages, bytes, approval credits, and dispatched CPU service were cumulative counters whose window increments were analyzed. In particular, CPU was charged in full at dispatch, as in the original model; its increment is dispatched service demand, not exact busy time or physical CPU utilization. Busy-node snapshots were retained separately. Growth of cumulative approval or known-message metadata was not treated as evidence of nonstationarity.

Each complete seeded trajectory supplied one value for each metric and window, so $n = 10$ per configuration and setting. Tables of window outcomes report mean $\pm$ sample SD. Differences between observation conditions were first formed within the same seed; their means and pointwise descriptive 95% Student-t intervals use nine degrees of freedom and $t_{0.975,9} \approx 2.262$. Requests and overlapping windows were not pooled as independent replicates. No superiority, equivalence, or stationarity test was performed, and an interval containing zero does not certify a steady state.

Work coverage and successful-only delay remain conditional metrics. A run with no issued request has undefined issued-only coverage statistics, not zero coverage shortfall; a run with no timely completion has undefined successful-only delay. A ten-run mean of either conditional metric is reported as unavailable if any run lacks that metric, with the available run count retained. Completion, issuance, and cost per submission remain defined when a window contains submissions but none succeeds. For high-load restricted links in $[0.8, 1.0)$ s, issued-only coverage and successful-only delay were available in only two of ten runs for MBSC with $k_{min} = 3$, one of ten for Fixed-3, and none for MBSC with $k_{min} = 4$ or Fixed-4. Their ten-run conditional aggregates are therefore unavailable rather than zero.

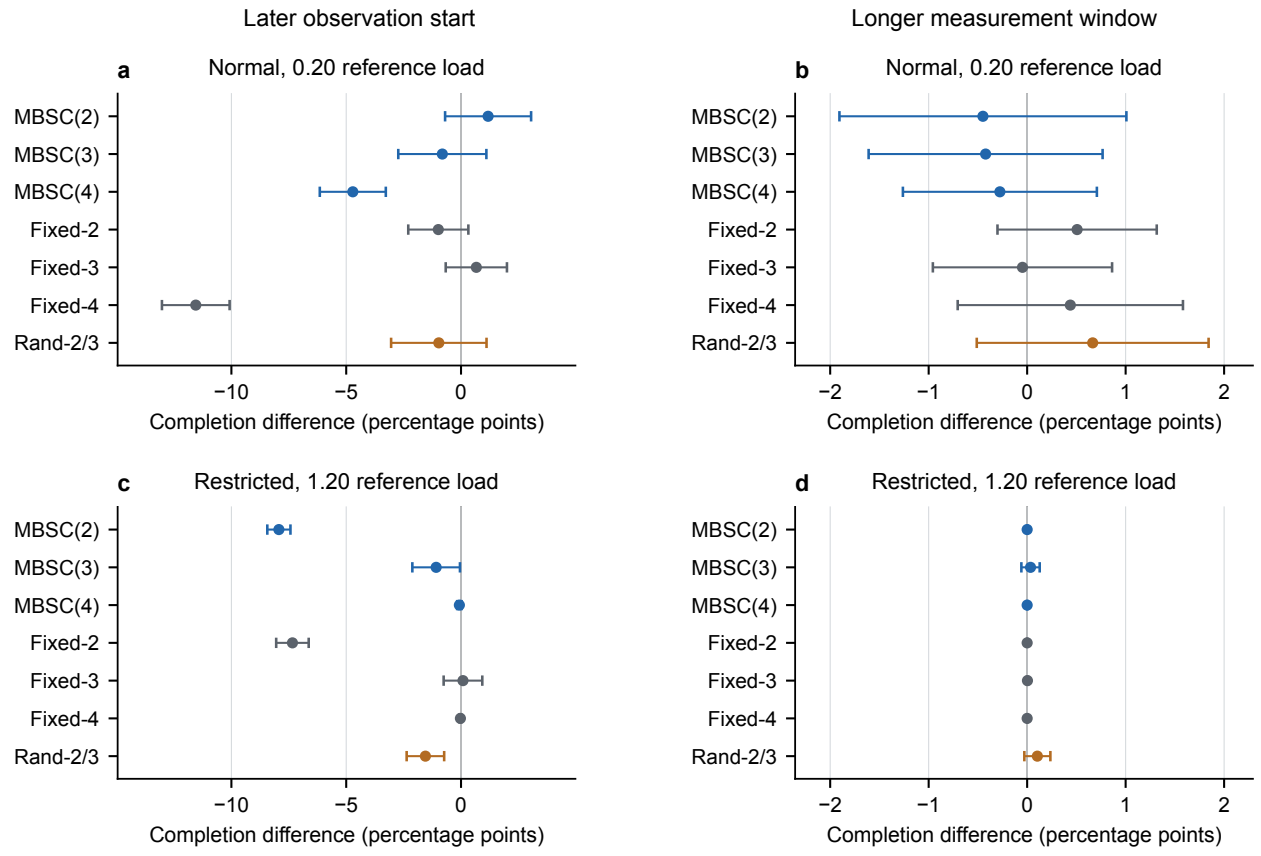


Fig. 7. Observation-time sensitivity of completion for all seven configurations in the two selected settings. The start-time contrast is $w = 0.8$ minus 0.1 s with $d = 0.1$ s; the duration contrast is $d = 0.2$ minus 0.1 s with $w = 0.8$ s. Points show means of ten within-seed differences and bars show pointwise descriptive 95% Student-t intervals. Differences are in percentage points. The windows share continuous trajectories; intervals containing zero do not establish equivalence or stationarity.

C. Observed Sensitivity and Operating Boundaries

Figure 7 shows two completion contrasts from the predefined window grid, while Tables XXXI and XXXII retain all six primary windows for every configuration. At high load with restricted links, MBSC with $k_{min} = 2$ completed 7.932% of requests in $[0.1, 0.2)$ s, but 0.011% in $[0.5, 0.6)$ s and none in $[0.8, 0.9)$ s. Fixed-2 likewise fell from 7.341% to 0.004% and then zero. In $[0.8, 0.9)$ s, mean completion was 0.604% for MBSC with $k_{min} = 3$, 0.383% for Fixed-3, and 0.027% for Rand-2/3; MBSC with $k_{min} = 4$ and Fixed-4 had no issued requests in that window. No tested configuration therefore sustains timely completion at this load and network setting, which marks the overload boundary of the simulated system.

Measurement duration also affected the earlier window. At high load with restricted links and $w = 0.1$ s, increasing d from 0.1 to 0.2 s changed MBSC ($k_{min} = 2$) completion from 7.932% to 6.009%, a within-seed mean difference of -1.923 percentage points (95% interval -2.168 to -1.678). The near-zero later-window contrasts in Fig. 7 therefore do not imply insensitivity to duration throughout the trajectory.

The low-load normal-link setting also changed with observation time. MBSC with $k_{min} = 4$ and Fixed-4 both completed every request in $[0.1, 0.2)$ s, but their corresponding means in $[0.8, 0.9)$ s were 95.287% and 88.449%, respectively. MBSC with $k_{min} = 4$ kept the higher completion in the later window. The start and duration contrasts are kept separate; extending a window also includes a later part of the same trajectory and does not supply additional independent replications.

A smaller late queue reflected reduced issuance rather than recovery. For high-load restricted links, MBSC with $k_{min} = 2$ had sampled mean queues of 195.118 and 15.996 operations per node in $[0.1, 0.2)$ and $[0.8, 0.9)$ s, respectively, while its completion fell to zero. Its issued fraction fell from 65.004% to 14.098%. MBSC with $k_{min} = 4$ and Fixed-4 had zero sampled eligible-pool means and no issuance in the later window, despite small queues. These state observations are consistent with coupled issuance, candidate supply, expiry, local removal, and prior congestion; they do not isolate the causal contribution of each process.

Tables XXXIII and XXXIV report the later-minus-earlier 200 ms contrasts. Near-zero drift can accompany near-zero completion and cannot be interpreted as satisfactory steady operation. Conversely, completion alone can hide state drift: at low load on normal links, MBSC with $k_{min} = 4$ had an eligible-pool increase of 7.919 entries per node (95% interval 4.040 to 11.797) between these blocks. The 1.2 s experiment thus maps the sensitivity and operating boundaries of the seven configurations.

TABLE XXXI
OBSERVATION-TIME OUTCOMES: LOW LOAD ($\mathbf{0.20}C_{ref}$), NORMAL LINKS

Configuration	w (s)	d (s)	Completion (%)	Issued (%)	CPU/submission (ms)
MBSC ($k_{min} = 2$)	0.1	0.1	23.623 ± 1.963	100.000 ± 0.000	1.0530 ± 0.0323
MBSC ($k_{min} = 2$)	0.1	0.2	24.461 ± 1.238	100.000 ± 0.000	1.0534 ± 0.0302
MBSC ($k_{min} = 2$)	0.5	0.1	23.298 ± 1.245	100.000 ± 0.000	1.0501 ± 0.0167
MBSC ($k_{min} = 2$)	0.5	0.2	23.503 ± 1.430	100.000 ± 0.000	1.0524 ± 0.0119
MBSC ($k_{min} = 2$)	0.8	0.1	24.802 ± 2.702	100.000 ± 0.000	1.0468 ± 0.0232
MBSC ($k_{min} = 2$)	0.8	0.2	24.353 ± 1.044	100.000 ± 0.000	1.0560 ± 0.0140
MBSC ($k_{min} = 3$)	0.1	0.1	52.899 ± 1.630	100.000 ± 0.000	1.2530 ± 0.0409
MBSC ($k_{min} = 3$)	0.1	0.2	52.785 ± 1.016	100.000 ± 0.000	1.2505 ± 0.0382
MBSC ($k_{min} = 3$)	0.5	0.1	52.052 ± 1.823	100.000 ± 0.000	1.2467 ± 0.0254
MBSC ($k_{min} = 3$)	0.5	0.2	52.123 ± 1.420	100.000 ± 0.000	1.2473 ± 0.0251
MBSC ($k_{min} = 3$)	0.8	0.1	52.085 ± 2.283	100.000 ± 0.000	1.2496 ± 0.0413
MBSC ($k_{min} = 3$)	0.8	0.2	51.663 ± 1.842	100.000 ± 0.000	1.2524 ± 0.0166
MBSC ($k_{min} = 4$)	0.1	0.1	100.000 ± 0.000	100.000 ± 0.000	1.4759 ± 0.0571
MBSC ($k_{min} = 4$)	0.1	0.2	100.000 ± 0.000	100.000 ± 0.000	1.4718 ± 0.0481
MBSC ($k_{min} = 4$)	0.5	0.1	99.266 ± 1.000	100.000 ± 0.000	1.4556 ± 0.0291
MBSC ($k_{min} = 4$)	0.5	0.2	97.912 ± 1.770	100.000 ± 0.000	1.4677 ± 0.0262
MBSC ($k_{min} = 4$)	0.8	0.1	95.287 ± 2.010	100.000 ± 0.000	1.4649 ± 0.0523
MBSC ($k_{min} = 4$)	0.8	0.2	95.010 ± 1.362	100.000 ± 0.000	1.4754 ± 0.0152
Fixed-2	0.1	0.1	14.952 ± 1.095	100.000 ± 0.000	0.9589 ± 0.0293
Fixed-2	0.1	0.2	14.435 ± 0.721	100.000 ± 0.000	0.9550 ± 0.0253
Fixed-2	0.5	0.1	14.240 ± 1.042	100.000 ± 0.000	0.9494 ± 0.0204
Fixed-2	0.5	0.2	13.748 ± 1.257	100.000 ± 0.000	0.9503 ± 0.0183
Fixed-2	0.8	0.1	13.963 ± 1.434	100.000 ± 0.000	0.9466 ± 0.0274
Fixed-2	0.8	0.2	14.470 ± 0.575	100.000 ± 0.000	0.9530 ± 0.0115
Fixed-3	0.1	0.1	42.840 ± 1.417	100.000 ± 0.000	1.1990 ± 0.0453
Fixed-3	0.1	0.2	43.093 ± 0.570	100.000 ± 0.000	1.1993 ± 0.0424
Fixed-3	0.5	0.1	43.408 ± 1.733	100.000 ± 0.000	1.1859 ± 0.0359
Fixed-3	0.5	0.2	43.361 ± 1.314	100.000 ± 0.000	1.1946 ± 0.0286
Fixed-3	0.8	0.1	43.506 ± 1.148	100.000 ± 0.000	1.1908 ± 0.0433
Fixed-3	0.8	0.2	43.459 ± 1.107	100.000 ± 0.000	1.2014 ± 0.0149

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TABLE XXXI
OBSERVATION-TIME OUTCOMES: LOW LOAD ($\mathbf{0.20}C_{ref}$), NORMAL LINKS (CONTINUED)

Configuration	w (s)	d (s)	Completion (%)	Issued (%)	CPU/submission (ms)
Fixed-4	0.1	0.1	100.000 ± 0.000	100.000 ± 0.000	1.4448 ± 0.0539
Fixed-4	0.1	0.2	99.754 ± 0.249	100.000 ± 0.000	1.4441 ± 0.0459
Fixed-4	0.5	0.1	89.117 ± 2.683	100.000 ± 0.000	1.4261 ± 0.0410
Fixed-4	0.5	0.2	88.699 ± 1.227	100.000 ± 0.000	1.4382 ± 0.0266
Fixed-4	0.8	0.1	88.449 ± 2.060	100.000 ± 0.000	1.4361 ± 0.0587
Fixed-4	0.8	0.2	88.887 ± 1.324	100.000 ± 0.000	1.4454 ± 0.0190
Rand-2/3	0.1	0.1	27.068 ± 1.942	100.000 ± 0.000	1.0742 ± 0.0320
Rand-2/3	0.1	0.2	26.484 ± 1.085	100.000 ± 0.000	1.0689 ± 0.0331
Rand-2/3	0.5	0.1	26.444 ± 1.954	100.000 ± 0.000	1.0690 ± 0.0261
Rand-2/3	0.5	0.2	26.542 ± 1.491	100.000 ± 0.000	1.0720 ± 0.0165
Rand-2/3	0.8	0.1	26.101 ± 1.200	100.000 ± 0.000	1.0666 ± 0.0375
Rand-2/3	0.8	0.2	26.766 ± 1.240	100.000 ± 0.000	1.0778 ± 0.0122

Entries are mean $\pm$ sample SD over ten complete trajectories. Each window contains requests created in $[w, w + d)$, with their full deadline follow-up. Completion and issuance divide by all window submissions. CPU is request-attributed dispatched service divided by all submissions, including failures; it excludes background children's work attributed to those children. Values are rounded, so a displayed 0.000% alone does not imply an exact zero count.

TABLE XXXII
OBSERVATION-TIME OUTCOMES: HIGH LOAD ($\mathbf{1.20}C_{ref}$), RESTRICTED LINKS

Configuration	w (s)	d (s)	Completion (%)	Issued (%)	CPU/submission (ms)
MBSC ($k_{min} = 2$)	0.1	0.1	7.932 ± 0.704	65.004 ± 5.039	0.8134 ± 0.0172
MBSC ($k_{min} = 2$)	0.1	0.2	6.009 ± 0.607	66.805 ± 3.804	0.7996 ± 0.0149
MBSC ($k_{min} = 2$)	0.5	0.1	0.011 ± 0.025	66.711 ± 3.454	0.7426 ± 0.0223
MBSC ($k_{min} = 2$)	0.5	0.2	0.008 ± 0.013	59.137 ± 2.480	0.6875 ± 0.0178
MBSC ($k_{min} = 2$)	0.8	0.1	0.000 ± 0.000	14.098 ± 4.540	0.3525 ± 0.0358
MBSC ($k_{min} = 2$)	0.8	0.2	0.000 ± 0.000	8.664 ± 2.734	0.3118 ± 0.0229
MBSC ($k_{min} = 3$)	0.1	0.1	1.687 ± 0.943	4.059 ± 2.621	0.7833 ± 0.0164
MBSC ($k_{min} = 3$)	0.1	0.2	1.973 ± 0.771	10.409 ± 2.923	0.7395 ± 0.0431
MBSC ($k_{min} = 3$)	0.5	0.1	0.963 ± 1.905	3.425 ± 4.215	0.3343 ± 0.1022
MBSC ($k_{min} = 3$)	0.5	0.2	1.011 ± 2.069	3.165 ± 5.111	0.3181 ± 0.1083
MBSC ($k_{min} = 3$)	0.8	0.1	0.604 ± 1.360	1.889 ± 4.008	0.2992 ± 0.1145
MBSC ($k_{min} = 3$)	0.8	0.2	0.638 ± 1.395	1.883 ± 4.004	0.2917 ± 0.1000
MBSC ($k_{min} = 4$)	0.1	0.1	0.072 ± 0.133	0.798 ± 0.688	0.5435 ± 0.0369
MBSC ($k_{min} = 4$)	0.1	0.2	0.036 ± 0.067	0.399 ± 0.344	0.3950 ± 0.0201
MBSC ($k_{min} = 4$)	0.5	0.1	0.000 ± 0.000	0.000 ± 0.000	0.2409 ± 0.0025
MBSC ($k_{min} = 4$)	0.5	0.2	0.000 ± 0.000	0.000 ± 0.000	0.2419 ± 0.0022
MBSC ($k_{min} = 4$)	0.8	0.1	0.000 ± 0.000	0.000 ± 0.000	0.2438 ± 0.0050
MBSC ($k_{min} = 4$)	0.8	0.2	0.000 ± 0.000	0.000 ± 0.000	0.2437 ± 0.0033
Fixed-2	0.1	0.1	7.341 ± 0.989	80.794 ± 3.115	0.8266 ± 0.0134
Fixed-2	0.1	0.2	5.522 ± 0.846	79.301 ± 2.635	0.8113 ± 0.0148
Fixed-2	0.5	0.1	0.004 ± 0.012	78.256 ± 2.142	0.7929 ± 0.0165
Fixed-2	0.5	0.2	0.002 ± 0.006	76.296 ± 1.662	0.7839 ± 0.0103
Fixed-2	0.8	0.1	0.000 ± 0.000	36.505 ± 6.375	0.5277 ± 0.0464
Fixed-2	0.8	0.2	0.000 ± 0.000	27.176 ± 4.790	0.4600 ± 0.0347
Fixed-3	0.1	0.1	0.298 ± 0.464	2.543 ± 2.138	0.7829 ± 0.0132
Fixed-3	0.1	0.2	0.749 ± 0.543	5.307 ± 3.596	0.6814 ± 0.0883
Fixed-3	0.5	0.1	0.297 ± 0.902	1.558 ± 3.361	0.2886 ± 0.0871
Fixed-3	0.5	0.2	0.319 ± 0.988	1.263 ± 3.176	0.2779 ± 0.0802
Fixed-3	0.8	0.1	0.383 ± 1.210	0.784 ± 2.480	0.2641 ± 0.0667
Fixed-3	0.8	0.2	0.386 ± 1.219	0.961 ± 3.040	0.2670 ± 0.0754
Fixed-4	0.1	0.1	0.027 ± 0.062	0.525 ± 0.521	0.5919 ± 0.0661
Fixed-4	0.1	0.2	0.013 ± 0.030	0.272 ± 0.271	0.4212 ± 0.0381
Fixed-4	0.5	0.1	0.000 ± 0.000	0.000 ± 0.000	0.2409 ± 0.0025
Fixed-4	0.5	0.2	0.000 ± 0.000	0.000 ± 0.000	0.2419 ± 0.0022
Fixed-4	0.8	0.1	0.000 ± 0.000	0.000 ± 0.000	0.2438 ± 0.0050
Fixed-4	0.8	0.2	0.000 ± 0.000	0.000 ± 0.000	0.2437 ± 0.0033

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TABLE XXXII
OBSERVATION-TIME OUTCOMES: HIGH LOAD ($\mathbf{1.20}C_{ref}$), RESTRICTED LINKS (CONTINUED)

Configuration	w (s)	d (s)	Completion (%)	Issued (%)	CPU/submission (ms)
Rand-2/3	0.1	0.1	1.578 ± 1.175	27.999 ± 6.576	0.8001 ± 0.0140
Rand-2/3	0.1	0.2	1.267 ± 0.785	26.100 ± 6.160	0.7982 ± 0.0171
Rand-2/3	0.5	0.1	0.121 ± 0.103	20.496 ± 3.589	0.6806 ± 0.0448
Rand-2/3	0.5	0.2	0.107 ± 0.096	19.835 ± 3.206	0.6371 ± 0.0487
Rand-2/3	0.8	0.1	0.027 ± 0.048	9.220 ± 2.543	0.4207 ± 0.0519
Rand-2/3	0.8	0.2	0.130 ± 0.194	7.722 ± 1.626	0.4008 ± 0.0435

Entries are mean $\pm$ sample SD over ten complete trajectories. Each window contains requests created in $[w, w + d)$, with their full deadline follow-up. Completion and issuance divide by all window submissions. CPU is request-attributed dispatched service divided by all submissions, including failures; it excludes background children's work attributed to those children. Values are rounded, so a displayed 0.000% alone does not imply an exact zero count.

TABLE XXXIII
LATE-BLOCK DIFFERENCES IN COMPLETION AND ISSUANCE

Setting	Configuration	ΔF (percentage points)	ΔI (percentage points)
Low/normal	MBSC ($k_{min} = 2$)	0.899 ± 0.628	0.000 ± 0.000
Low/normal	MBSC ($k_{min} = 3$)	-0.836 ± 1.548	0.000 ± 0.000
Low/normal	MBSC ($k_{min} = 4$)	-1.487 ± 1.827	0.000 ± 0.000
Low/normal	Fixed-2	0.586 ± 1.165	0.000 ± 0.000
Low/normal	Fixed-3	0.147 ± 1.007	0.000 ± 0.000
Low/normal	Fixed-4	0.288 ± 1.233	0.000 ± 0.000
Low/normal	Rand-2/3	0.462 ± 0.869	0.000 ± 0.000
High/restricted	MBSC ($k_{min} = 2$)	-0.002 ± 0.004	-31.804 ± 2.540
High/restricted	MBSC ($k_{min} = 3$)	-0.408 ± 0.639	-0.885 ± 1.329
High/restricted	MBSC ($k_{min} = 4$)	0.000 ± 0.000	0.000 ± 0.000
High/restricted	Fixed-2	0.000 ± 0.000	-40.215 ± 3.028
High/restricted	Fixed-3	0.098 ± 0.221	0.105 ± 0.238
High/restricted	Fixed-4	0.000 ± 0.000	0.000 ± 0.000
High/restricted	Rand-2/3	0.067 ± 0.155	-8.450 ± 2.487

Each entry is the mean within-seed difference $[0.8, 1.0) - [0.6, 0.8)$ s $\pm$ the half-width of its pointwise descriptive 95% Student-t interval ($n = 10$, df$= 9$), not sample SD. F is timely completion and I is issuance, each divided by all submissions in its respective creation window.

TABLE XXXIV
LATE-BLOCK DIFFERENCES IN SAMPLED STATE AND OBSERVED APPROVAL SUPPLY

Setting	Configuration	ΔQ (operations/node)	ΔP_e (entries/node)	Δa_o (credits/s)
Low/normal	MBSC ($k_{min} = 2$)	0.007 ± 0.057	0.037 ± 0.043	269.5 ± 462.6
Low/normal	MBSC ($k_{min} = 3$)	0.007 ± 0.075	0.046 ± 0.067	322.0 ± 615.5
Low/normal	MBSC ($k_{min} = 4$)	0.029 ± 0.097	7.919 ± 3.878	400.0 ± 689.1
Low/normal	Fixed-2	0.023 ± 0.036	0.006 ± 0.017	227.0 ± 376.7
Low/normal	Fixed-3	0.005 ± 0.070	-0.040 ± 0.123	318.0 ± 562.3
Low/normal	Fixed-4	0.016 ± 0.066	0.155 ± 0.312	436.5 ± 649.2
Low/normal	Rand-2/3	0.006 ± 0.054	-0.017 ± 0.056	293.5 ± 462.5
High/restricted	MBSC ($k_{min} = 2$)	-80.710 ± 5.522	-73.192 ± 6.329	-9493.5 ± 1472.8
High/restricted	MBSC ($k_{min} = 3$)	-0.505 ± 1.208	-0.002 ± 0.004	-506.0 ± 617.7
High/restricted	MBSC ($k_{min} = 4$)	0.020 ± 0.015	0.000 ± 0.000	0.0 ± 0.0
High/restricted	Fixed-2	-109.153 ± 6.620	-96.850 ± 5.283	-8272.5 ± 1333.7
High/restricted	Fixed-3	-0.397 ± 0.927	0.000 ± 0.000	-164.0 ± 196.8
High/restricted	Fixed-4	0.020 ± 0.015	0.000 ± 0.000	0.0 ± 0.0
High/restricted	Rand-2/3	-42.849 ± 6.434	-48.480 ± 7.177	-3355.5 ± 865.0

Entries are later-minus-earlier differences for $[0.8, 1.0)$ and $[0.6, 0.8)$ s, expressed as mean $\pm$ pointwise 95% interval half-width over ten within-seed differences. Q and P_e are sampled means across nodes and 10 ms observations of queued operations and eligible pool entries. a_o is the increment per second of issuer-observed prethreshold credits for all generated targets, excluding bootstrap targets; it is not restricted to the creation cohort and is not a certified opportunity intensity. These sampled-state and clock-window rates have a different attribution scope from the cohort outcomes above.

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